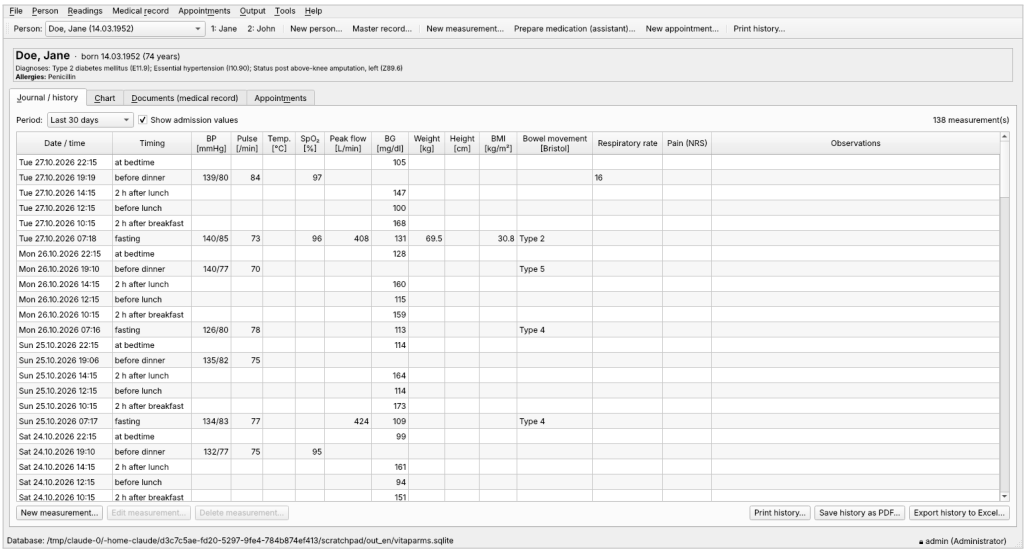
VitaParms

Vital signs, medication and medical record

User Manual

Your personal health diary,
step by step

Version 0.22.0
for Windows, macOS and Linux
English edition (user interface in English and German)



Michael Bergmann

October 2026

VitaParms – User Manual
User manual for program version 0.22.0, as of October 2026. English edition.

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Web: https://mbergmann-sh.de GitHub: https://github.com/mbergmann-sh

Software: VitaParms is free software (open source) under the MIT licence and is provided free of charge. The full licence text and the licences of the third-party components used can be found in Appendix B.

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Not a medical device. VitaParms is a personal diary for health-related data. It makes no diagnoses, gives no treatment recommendations and does not replace medical or pharmaceutical advice. Use is at your own risk. In an emergency call **112** (Europe), **999** (UK) or **911** (USA/Canada).

Liability: The program and this manual have been prepared with great care. Nevertheless, errors cannot be ruled out. The author and contributors accept no responsibility for correctness, completeness or currency, in particular not for the automatic classification of medicines or for content from the internet (see Appendix B).

Sample data: All persons, addresses, telephone numbers, readings, diagnoses and appointments shown in the figures are fictitious. Any resemblance to real persons is coincidental. Medicines with the suffix ‘Sample’ are fictitious; the other products shown serve only as examples and are not a recommendation.

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Typesetting: LaTeX with KOMA-Script; typefaces Palatino and Helvetica. Screenshots: VitaParms 0.22.0 on Linux (Qt 6, style ‘Fusion’), English user interface.

Preface

Anyone who cares for a relative or lives with a chronic illness will know the situation: blood pressure and blood glucose are written down in a notebook, the medication plan hangs on the fridge, doctors' letters are kept in a folder – and at the next appointment, the one report the doctor asks about is missing. VitaParms grew out of exactly this situation.

The idea was simple: a diary for vital signs that keeps several people apart, shows trends clearly and can be printed on A4 paper. While working on it, everything that turned out to be needed in everyday life was added – a medication plan with regular, as-needed and emergency medication, an assistant that reads aloud while you prepare the tablets, a digital medical record for reports and letters, and an appointment planner with reminders. And because this is about health data, the entire database is encrypted.

One thing mattered to me from the start: VitaParms is *not* a medical device. It calculates, sorts and reminds, but it does not assess anything. Whether a reading needs treatment is for doctors to decide; what the package leaflet says is what counts. The program is meant to save you paperwork and help you have the right documents at hand at your next appointment – no more and no less.

This manual guides you through the program step by step: from the first login and entering readings to sending an encrypted medical record. You do not have to read it from cover to cover. Chapters 1 to 6 are enough to get started; you can look up everything else when you need it.

VitaParms was developed in Germany. Its user interface is available in English and German, and some features – the PZN for looking up package leaflets, or the classification of controlled narcotics under the German Narcotics Act – are tailored to the German healthcare system. This manual points this out wherever it matters.

VitaParms is free software and costs nothing. If you find a bug, have an idea or would like to translate the program into your language, I look forward to hearing from you.

Michael Bergmann
October 2026

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1 Welcome to VitaParms

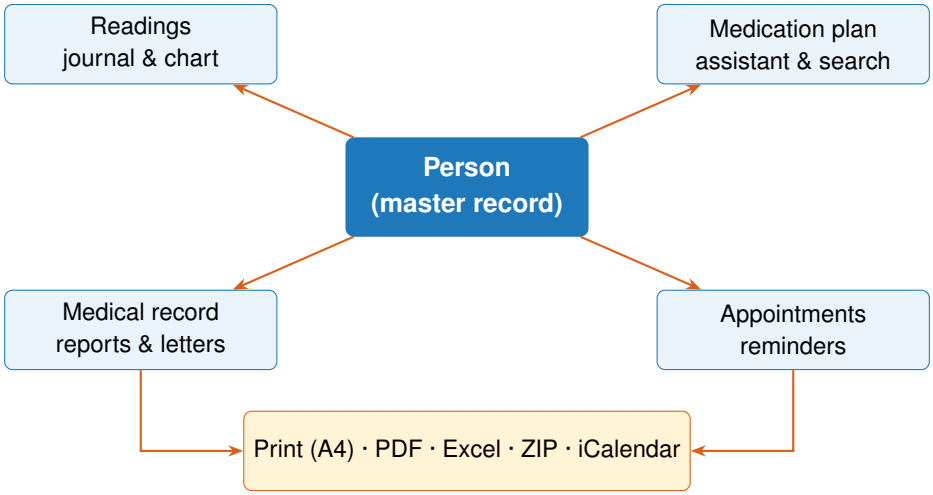
VitaParms is a diary for health-related data. For every person you create, it records what comes up in everyday health and care: readings such as blood pressure, pulse or blood glucose, the current medication plan, doctors' letters and reports, and upcoming appointments. From the collected readings the program produces history tables and colour-coded trend charts that you can print on A4 paper, save as PDF or export to Excel.

1.1 Who is VitaParms for?

- **For yourself**, if you measure your readings regularly – for example with high blood pressure or diabetes – and want to show your doctor a clean history.
- **For family carers** who record readings for one or more people, prepare medication and keep track of medical appointments.
- **For small teams**, for example in a shared household or in home care, who work together on one database – each person with their own login account.

Not a medical device: VitaParms makes no diagnoses and does not assess readings. It replaces neither medical treatment nor the advice of your pharmacist. Never rely on the program alone when making decisions about medication. If you have acute symptoms, call the emergency number (**112** in Europe, **999** in the UK, **911** in the USA and Canada) or your out-of-hours medical service.

1.2 What VitaParms can do – an overview



Master record Personal details, diagnoses, allergies, emergency contact, information on missing limbs and the vital signs on admission (Chapter 5).

Readings Pulse, blood pressure, temperature, oxygen saturation, peak flow, blood glucose, weight and height with an automatically calculated BMI; any number of measurements per day, daily profiles (Chapter 6).

Journal and chart All readings by date, newest first; trend charts per day, week and month; printing, PDF and Excel (Chapter 7).

Medication plan Divided into regular, as-needed and emergency medication plus ‘Other’, with intake schedule, package leaflet and a search for controlled narcotics, painkillers and blood thinners (Chapter 8).

Prepare medication An assistant reads out the medication that is due and logs who prepared it and when (Chapter 9).

Medical record Reports, lab results and letters as PDF, searchable and ready to be sent as an encrypted ZIP archive (Chapter 10).

Appointments Planner with recurring series, reminders and export to calendar apps (Chapter 11).

Security Encrypted database, login accounts with roles, automatic lock, backups (Chapter 12).

1.3 How this manual is organised

The chapters follow the path you take when you first work with VitaParms: installation, first start, a tour of the main window, then creating a person and entering the first

measurement. After that come the tools for output, medication, the medical record and appointments. The appendix contains all keyboard shortcuts, the legal background, help with problems and a glossary of technical terms.

Many sections contain step-by-step instructions. It is best to follow them directly at your computer.

Conventions used in this manual

Shown as	Meaning
File › Create backup...	Menu command: choose *Create backup...* in the *File* menu
Ctrl + M	Keyboard shortcut: press the keys together (on macOS use Cmd instead of Ctrl)
Save	Button in a window
Chart	Tab in a window
Timing	Field or label in a window
`vitaparms.sqlite`	File or folder name
1	Step in a set of instructions

Note: The figures show VitaParms on Linux with the 'Fusion' style and the English user interface. On Windows and macOS, windows and buttons look slightly different; layout and labels are the same.

Tip: The program is also available with a German user interface (Section 13.1 (p. 93)). This manual uses the English labels. Dates are shown in the day.month.year format throughout the program.

2 Installation

VitaParms runs on Windows, macOS and Linux. It needs neither an internet connection nor a server; all data is kept in a single encrypted file on your computer. The program only goes online when you look up a package leaflet yourself or have a reminder sent by e-mail.

2.1 System requirements

Operating system	Windows 10/11, macOS 12 or later, a current Linux distribution (64-bit)
Screen	1280 × 800 pixels or more recommended
Disk space	about 100 MB for the program, plus your data (documents in the medical record take up the most space)
Printer	optional, any printer with A4 paper
Speech output	optional, for the 'Prepare medication' assistant: a voice of the operating system in your language

2.2 Installing the program

Windows

1. Start `VitaParms-0.22.0-Setup.exe` and choose the language of the setup wizard.
2. Read the licence and disclaimer and confirm with *I accept the agreement*.
3. Keep the destination folder `C:\VitaParms` or choose another one.
4. Choose the components – the program and the user manuals (for F1) – and whether a desktop shortcut should be created.
5. Click Install. At the end you can start VitaParms or the demo (Section 3.5 (p. 12)) straight away.

The Start menu then contains *VitaParms*, *VitaParms Demo*, the user manual and the uninstaller. **Updating:** simply run the setup of a newer version. It detects the existing installation and asks whether to update or uninstall; updates go into the existing folder. **Uninstalling** via the Start menu or *Settings* → *Apps*. Your databases, key files and settings are kept in both cases.

Alternatively there is a second installer, `...-Setup-IFW.exe`, built with the *Qt Installer Framework*. It offers the same components and also installs to `C:\VitaParms`; changes and uninstallation are then handled by `VitaParms-Wartung.exe`. Without installing, it is also enough to unpack a ZIP archive with VitaParms into a folder and start `VitaParms.exe`.

macOS

1. Open the file `VitaParms.dmg` and drag VitaParms into the *Applications* folder.
2. When you start it for the first time, macOS asks whether you want to open the program downloaded from the internet. Confirm with Open.

Linux

Use your distribution's package if available, or build the program from source (Section 2.3 (p. 6)). For speech output, `speech-dispatcher` should be installed with voices for your language, on Debian and Ubuntu for example with the package `speech-dispatcher-espeak-ng`.

2.3 Building the program from source

VitaParms is written in C++ with the Qt 6 framework. To build the program yourself you need Qt 6.4 or later with the modules *Widgets*, *PrintSupport*, *UiTools* and the *Linguist* tools. Also recommended are *Qt PDF* (preview and full-text search in the medical record), *Qt TextToSpeech* (reading aloud) and *Qt Multimedia* (signal tones). If these optional modules are missing, the program still builds; the corresponding features are then limited.

```
qmake6 VitaParms.pro
make -j
```

On Debian and Ubuntu, install the required packages with:

```
sudo apt install qt6-base-dev qt6-tools-dev qt6-tools-dev-tools \
     qt6-l10n-tools qt6-pdf-dev qt6-speech-dev qt6-multimedia-dev \
     qt6-declarative-dev qmake6
```

The encrypting database (SQLite3 Multiple Ciphers) is included with the source code and is built along with it; no further libraries are required. On macOS, `macdeployqt` bundles the Qt libraries with the program.

On **Windows** the build does the rest itself: after each build, `windeployqt` copies the required Qt libraries next to `VitaParms.exe`. A release build also fills the installation directory `installer\install_src` and – if installed – builds both installers with *Inno Setup* and the *Qt Installer Framework* in `installer\Output`. Details are in `installer\README.md`.

2.4 Where VitaParms stores its data

File	Contents
`vitaparms.sqlite`	The encrypted database with all persons, readings, medication, documents and appointments. The default location is the application data folder (see below); with File › New database… you can choose any other location.
`vitaparms.sqlite.keys`	The key file with the login accounts. It always belongs to the database and must be copied along with it when making backups.
Forms folder	Your customised input forms (Section 13.5 (p. 95)).

The application data folder is located at

- Windows: `%APPDATA%\VitaParms\VitaParms`
- macOS: `~/Library/Application Support/VitaParms/VitaParms`
- Linux: `~/.local/share/VitaParms/VitaParms`

The full path of the open database is shown at the bottom left of the main window's status bar. You can open a different database with File › Open database… or when starting the program with the parameter `--database <file>`.

> **Caution:** Never copy or move the database while VitaParms is running, and never separate the database from its key file `.keys`. Use the backup function to make copies (Section 12.2 (p. 90)).

3 Getting started

When you start it for the first time, VitaParms sets up a new, encrypted database. To do this you create an administrator account and receive a recovery key. This only takes a few minutes – but do take the time to store the key safely.

> **Getting Started:** The quick guide **Getting Started** shows the most important tasks with examples using the demo database – readings, charts, medication, stock, appointments and appointment series, tablet. You open it at the first start, in the login window (even without logging in) and at any time via Help › Getting Started.

3.1 Step by step: setting up the database

1. **Choose the language.** At the very first start VitaParms asks for the language in both languages: Deutsch or English (Fig. 3.1). You can change it later under Tools › Settings... . The user manual and 'Getting Started' always open in the chosen language.



Fig. 3.1: The language choice at the very first start

2. **Confirm the notice.** First you see the welcome notice (Fig. 3.2). It reminds you that VitaParms is not a medical device. Click Understood. If *Do not show this notice again* is ticked, it will not appear again; you can always find it under Help › Disclaimer... .

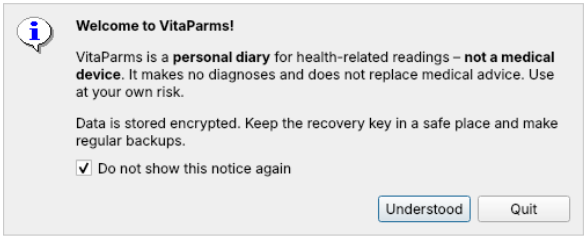


Fig. 3.2: The welcome notice at the first start

3 **Set up.** If there is no database yet, VitaParms offers four ways (Fig. 3.3): Create new database 1, Import database... 2 (Section 12.2 (p. 90)), Getting Started 3 opens the quick guide and Try the demo 4 starts the demo in a second window (Section 3.5 (p. 12)). The last two leave the window open. For your own database click Create new database.

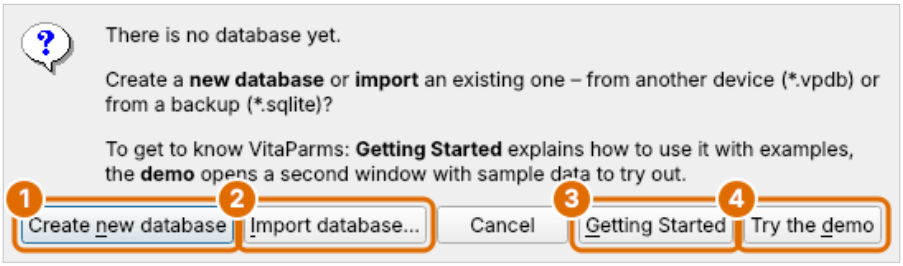


Fig. 3.3: At the first start: new database, import, quick guide or demo

4 **Create the administrator account.** VitaParms reports *New encrypted database* and asks for the first account (Fig. 3.4). The account name *admin* is suggested; you can change it, for example to your first name. Choose a password with at least eight characters and repeat it. Click Save.

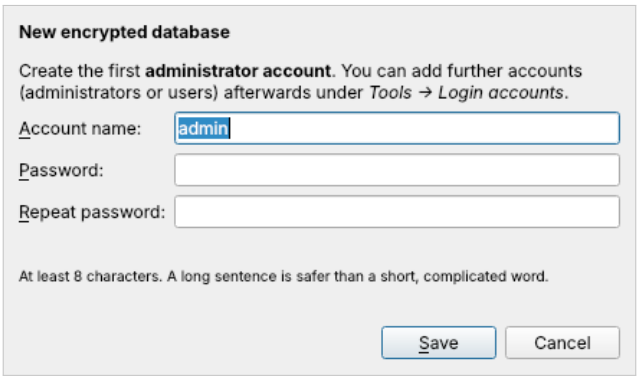


Fig. 3.4: The first account is always an administrator

5 **Secure the recovery key.** VitaParms now shows the recovery key – a long sequence of letters and digits (Fig. 3.5). With this key the database can be opened even if

all passwords have been forgotten or the key file has been lost. Choose Print… or Save as file…, tick *I have stored the recovery key safely* and click Done.

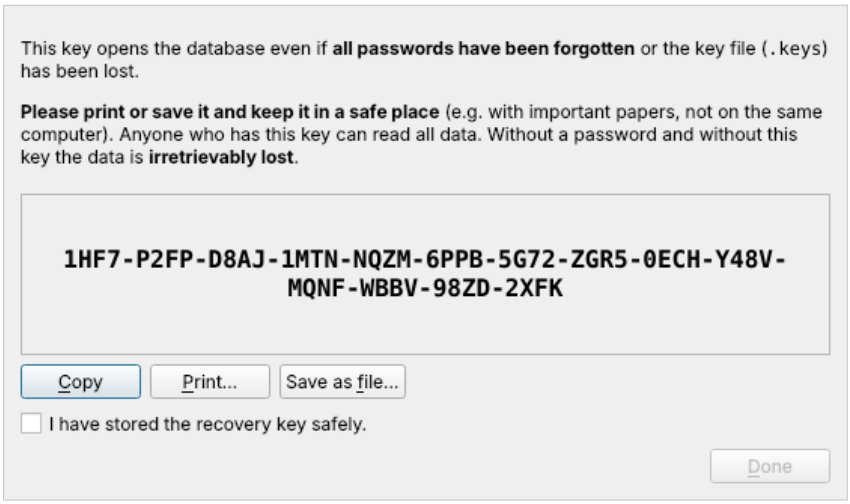


Fig. 3.5: The recovery key – please print it and keep it in a safe place

6 **Done.** The main window opens. It is still empty because there is no person yet. Chapter 5 shows how to create the first person.

No password and no key – no data: The encryption is real: there is no back door. Anyone who forgets their password and can no longer find the recovery key cannot open their data again – not even the author can help. Therefore keep the key away from the computer, for example printed out with your important papers. Conversely: whoever has the key can read all data. It does not belong under the keyboard.

3.2 Logging in

Each time you start the program, VitaParms asks for account and password (Fig. 3.6). Once you have entered them correctly, the database is unlocked automatically; you never need to know a separate database key.

Use File » Log out / lock or Ctrl + L to lock VitaParms when you leave the computer briefly or hand it over to someone else. The main window disappears, the database is closed and the login window appears – now another account can log in as well. If you like, the program locks itself after a configurable period of inactivity (Section 13.2 (p. 94)).

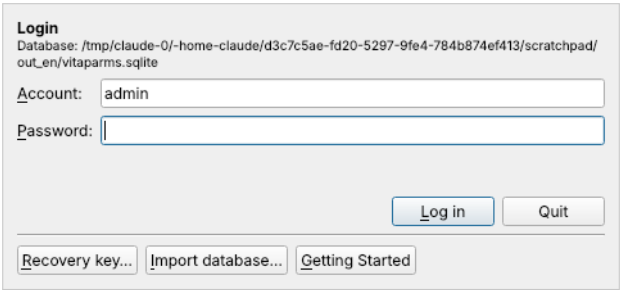


Fig. 3.6: Logging in

3.3 Forgotten your password?

- **A user has forgotten their password:** an administrator sets a new one under Tools › Login accounts… with Set password… (Section 12.1 (p. 89)).
- **Database from another device:** in the login window (and at the first start) Import database… takes over a `.vpdb` transfer file or a backup (Section 12.2 (p. 90)).
- **All administrators have forgotten their passwords:** click Recovery key… in the login window and enter the key. Upper and lower case, spaces and hyphens do not matter; mixing up O and 0 or I and 1 is also tolerated. Then choose a new password for an existing administrator account.
- **The key file `.keys` has been lost:** VitaParms notices this at start-up and offers recovery. With the recovery key you set up the accounts again.

3.4 An older, unencrypted database

If you used a very early version of VitaParms whose database was not yet encrypted, the program detects this when opening it and offers to encrypt the database. You then create the first administrator account as described above.

3.5 Trying it out with the demo database

If you would like to get to know VitaParms first without entering your own data, choose Help › Start demo database… – or Try the demo at the first start. After a short explanation, a **second program window** opens with the fictitious person *Erika Mustermann* (Fig. 3.7):

- 365 days of readings up to today: blood pressure and pulse morning and evening, fasting blood glucose and regular daily profiles, weight, peak flow, occasionally temperature and oxygen saturation – with a visible treatment success for blood pressure, a mild hypo and a respiratory infection with fever;

- a medication plan with regular, as-needed and emergency medication, an archived medicine and a weekly vitamin D dose;
- appointments for today, tomorrow and the coming weeks, a physiotherapy series, the ‘Prepare medication’ log and four reports in the medical record.

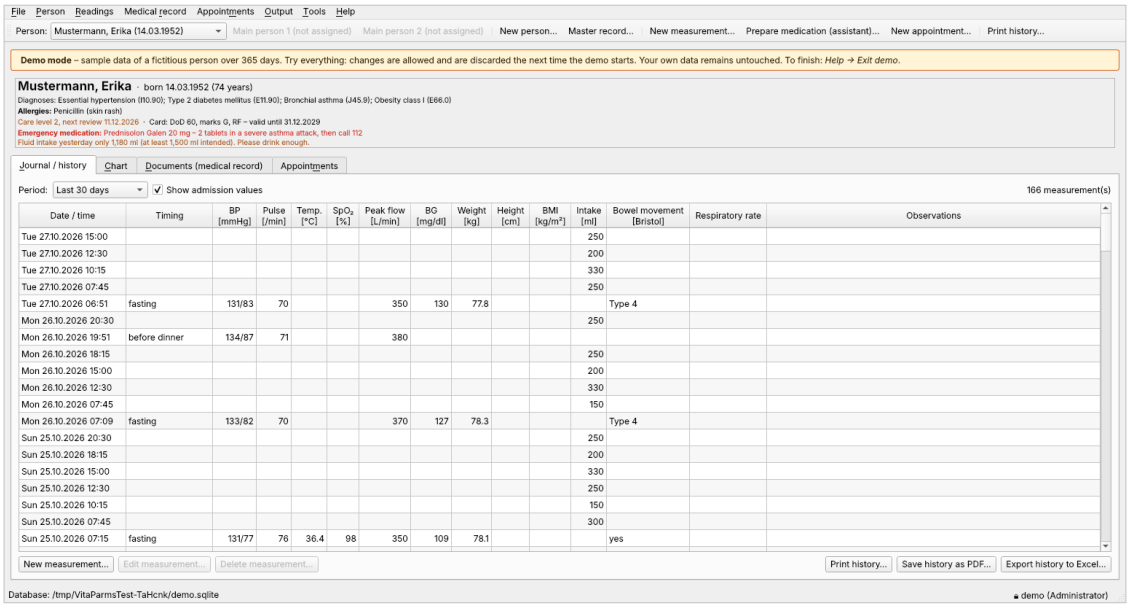


Fig. 3.7: The demo: a yellow band indicates demo mode

The demo has its own encrypted database and its own settings; it takes over language and appearance from your settings. You are logged in automatically (account `demo`, password `demo1234`). Try everything – take readings, change medication, print, delete: the demo is recreated every time it starts, and your own data remains untouched. Help › Exit demo closes the demo window. On Windows the installer also creates a *VitaParms Demo* shortcut in the Start menu.

4 The main window

After logging in you see the main window. Everything VitaParms can do is reached from here – via the menus, the toolbar or the tabs. Fig. 4.1 shows the window with a sample person.

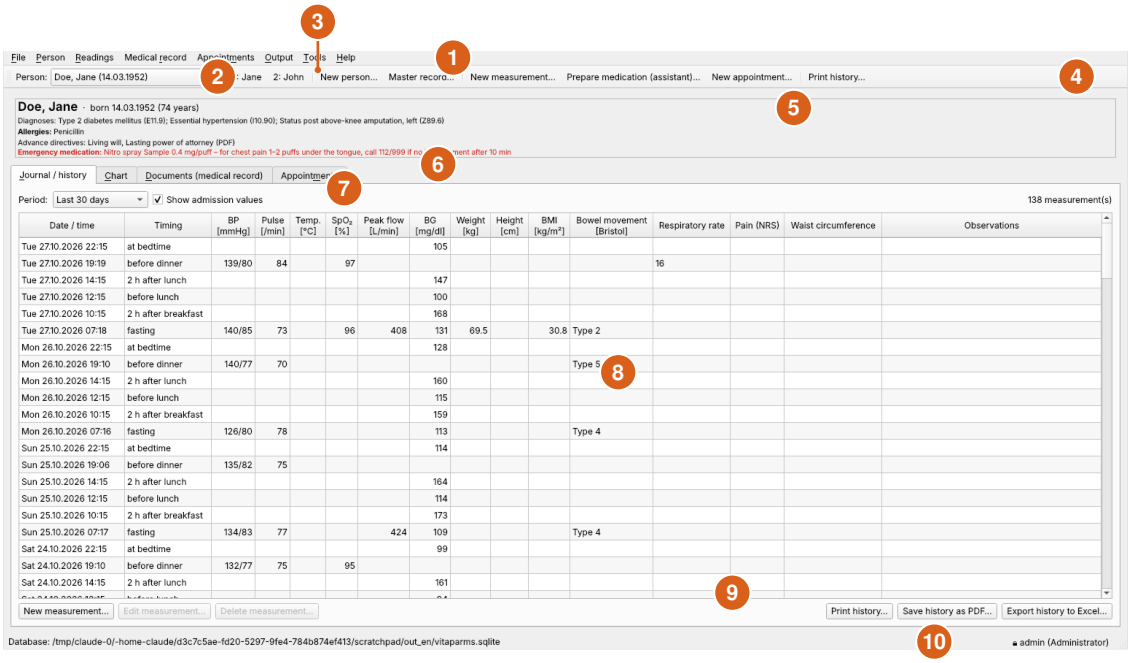


Fig. 4.1: The main window with the journal of a sample person

1. **Menu bar** with all commands, arranged by topic (overview in Table 4.1).
2. **Person selection:** here you choose the person whose data is shown. Everything you enter afterwards belongs to this person.
3. **Main persons 1 and 2:** quick selection for the two people you look after most often – also with Ctrl + 1 and Ctrl + 2.
4. **Toolbar** with the most frequent commands: new person, master record, new measurement, prepare medication, new appointment, print.
5. **Header** with name, date of birth and age, diagnoses and allergies. If emergency medication has been entered, it is shown here in red – so it is visible immediately in an emergency.

6. **Tabs** for the four work areas: Journal / history, Chart, Documents (medical record) and Appointments.
7. **Filters** of the current tab, here the period of the journal.
8. **Work area**, here the table of all measurements.
9. **Buttons** for the current tab: create and edit measurements, print, save as PDF, export to Excel.
10. **Status bar** with the path of the open database (left) and the logged-in account with its role (right).

4.1 The menus at a glance

Table 4.1: The menus of the main window

File	New database, Open database, Create backup, Restore backup / import, Export database, Update website (Chapter 14), Synchronise database, Sync with tablet via USB (Chapter 15), Log out / lock, Quit
Person	New person, Master record, Delete person, Prepare medication (assistant), Medication stock and orders (Section 8.8 (p. 62)), Search medications (controlled narcotics, painkillers, blood thinners, blood pressure, cortisone, free text), Medication preparation log, Print (master record, medication list, medication list for the pharmacy, reminders), set and call main persons
Readings	New measurement, Edit measurement, Delete measurement, BG day profile (section 6.4)
Medical record	Add documents, Pack record / send by e-mail, Print document list, Show medical record
Appointments	New appointment, Show appointment planner, Check reminders now, Appointments today and tomorrow, Export appointments (iCalendar)
Output	Print, Print preview, Save as PDF, Export to Excel, Print master record, Print medication plan, Medication list for the pharmacy
Tools	Settings, Customize forms, Login accounts, Website
Help	User manual (F1), Getting Started, Start demo database (Section 3.5 (p. 12)), Disclaimer, About VitaParms, About Qt

The commands in the *Output* menu always refer to the tab that is currently visible: if the journal is open, the history table is printed; if the chart is open, the trend charts are printed.

4.2 Help: the manual inside the program

This manual is built into VitaParms: press F1 anywhere in the program or choose Help › User manual (Fig. 4.2). VitaParms opens the manual in the language set for the program.



Fig. 4.2: The 'Help' menu with the user manual at the top

Windows The manual appears in *Windows Help* – with contents, keyword index and full-text search in a window of its own.

macOS and Linux The manual opens as a PDF in the default program for PDF files, such as Preview, Okular or Evince.

The files are located in the `docs` subfolder of the program folder:

- `VitaParms-Manual.pdf` (English) and `VitaParms-Handbuch.pdf` (German),
- on Windows also `VitaParms-Manual.chm` and `VitaParms-Handbuch.chm`.

If the Windows Help file is missing, VitaParms uses the PDF; if the file for the selected language is missing, it uses the other language. If no manual is found at all, a message names the folder where it is expected.

4.3 Working with several people

VitaParms manages any number of people. Each person has their own master record and their own readings, medication, documents and appointments. You can always see which person you are working on in the header (5).

Setting main persons

1. Choose the desired person in the person selection (2).
2. Choose Person » Set current person as main person 1 (or 2).
3. From now on you switch directly to this person with Ctrl + 1 or Ctrl + 2 or with the buttons (3).

The first person you create automatically becomes main person 1. Both slots can also be assigned under Tools » Settings…

> **Note:** In VitaParms, the person whose data is recorded is called a 'person'. This is different from the *login accounts* used by the people who operate the program (Section 12.1 (p. 89)). A carer with one login account can therefore look after the data of several persons.

5 Creating a person: the master record

The master record contains everything about a person that rarely changes: personal details, diagnoses, allergies, the medication plan, information about the body and the vital signs on admission. It is divided into four tabs.

5.1 Step by step: the first person

1. Choose Person › New person… , press Ctrl + Shift + N or click New person… in the toolbar.
2. On the **Person** tab, fill in at least the *Last name* – it is the only required field. First name, date of birth and sex are also useful because they appear on every printout (Fig. 5.1).
3. Under *Further details: contacts*, enter health insurance, GP practice and an emergency contact (Section 5.2). These details appear on the printed master record.
4. Switch to the **Diagnoses** tab and enter the diagnoses – one per line, ideally with the ICD-10 code (Fig. 5.2). Below follow *Allergies / intolerances* and free-text *Remarks*.
5. Click Save. The person appears in the person selection; diagnoses and allergies are shown in the header of the main window.

You can open the master record of an existing person at any time with Person › Master record… or Ctrl + E. Print it with Person › Print › Master record… or Output › Print master record… (Section 11.8 (p. 86)).

5.2 Contacts: health insurance, doctors, hospitals, emergency contact

On the right of the **Person** tab there is the list *Further details: contacts*: health insurance, GP practice, specialists, hospitals, home care service, pharmacy, emergency contact and other contacts – as many as you need. The list scrolls; the search field above it filters immediately by name, speciality, town, telephone number or any other entry.

1. Click Add… below the list. If health insurance or GP practice are still missing, VitaParms suggests that type, otherwise ‘Specialist’.

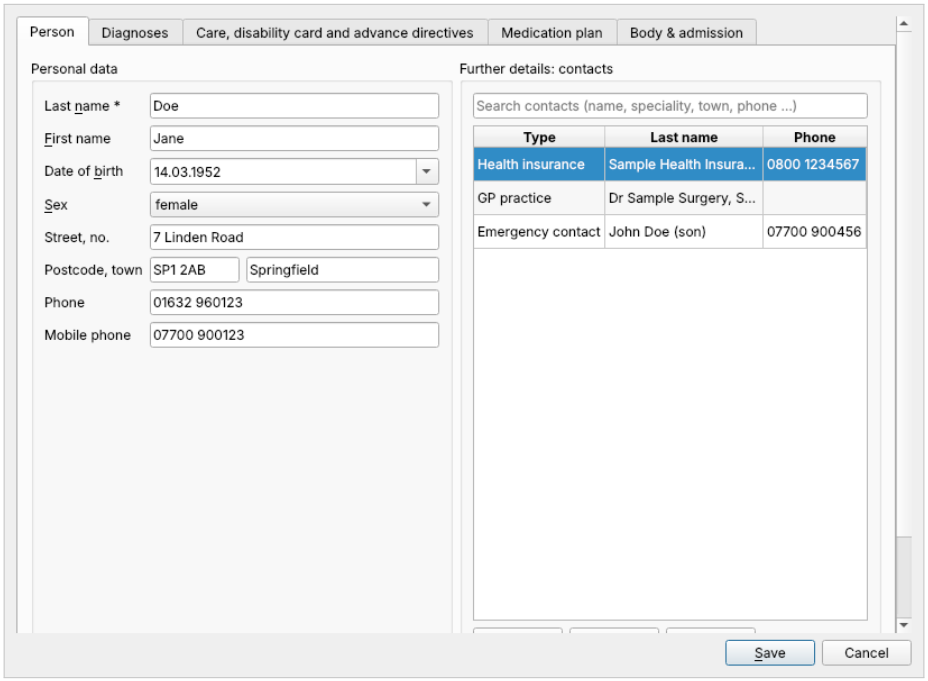


Fig. 5.1: Master record, ‘Person’ tab

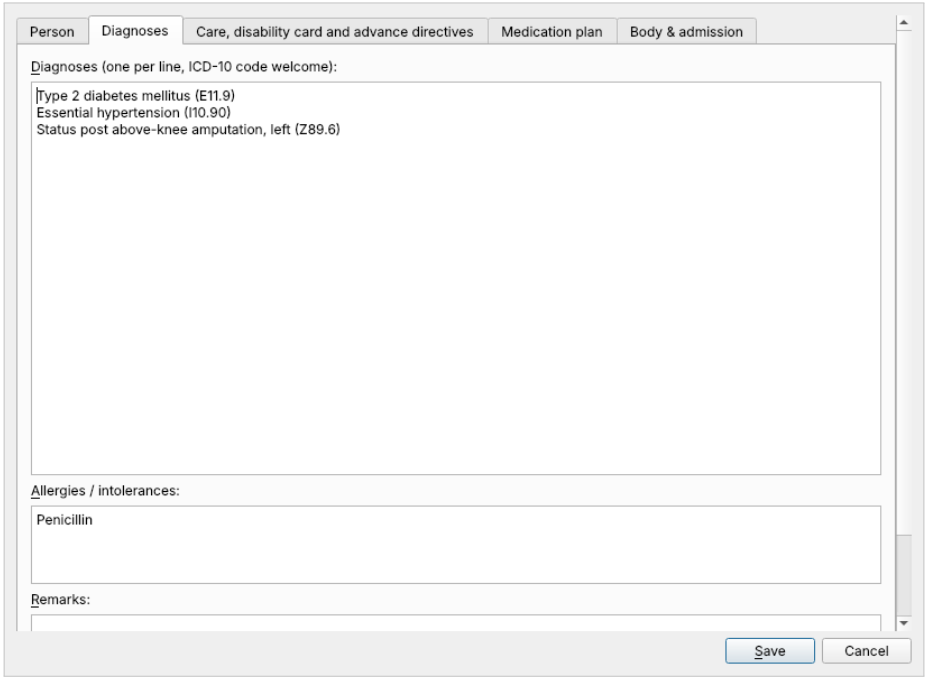


Fig. 5.2: Master record, ‘Diagnoses’ tab with allergies and remarks

2. Choose the *Type* and enter name, address, *Telephone, Mobile, E-mail* and *Website* (Fig. 5.3). Specialists and hospitals also have *Speciality* or *Department / ward,* the health insurance the *Insurance number,* the emergency contact the *Relationship* (e.g. husband). The ↗ buttons next to e-mail and website write an e-mail or open the website.

3. Click Apply and then Save in the master record.

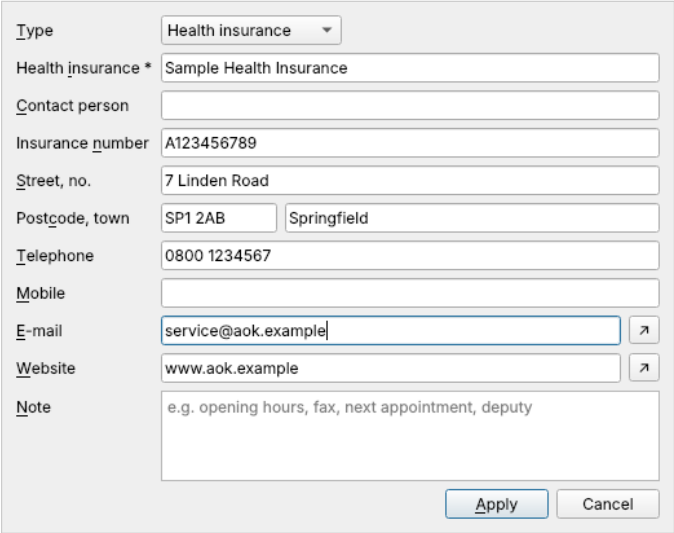


Fig. 5.3: A contact: health insurance with insurance number

A double-click on an entry (or Edit…) opens it for changing, Remove deletes it after asking. The contacts appear on the printed master record; health insurance with insurance number and GP practice also on the medication list for the pharmacy and on the order list. The website and the tablet show the contacts too; they are edited on the computer.

> **Note:** Earlier versions only had the fields *Health insurance, GP practice, Practice phone* and *Emergency contact.* VitaParms takes them over into the contact list automatically when they are first opened – the emergency contact split into name, relationship and telephone number where possible. You can still add your own custom fields as described in Section 13.5 (p. 95).

5.3 The medication plan

The third tab, **Medication plan**, is so extensive that it has a chapter of its own: Chapter 8.

5.4 Body and admission

The **Body & admission** tab (Fig. 5.4) contains six things:

Biological sex Reference tables – such as predicted peak flow – depend on the biological sex. The *For reference values* field automatically takes 'male' or 'female' from the *Sex* field on the **Person** tab; change it only if the biological sex differs or that field says 'diverse' or is empty. A difference also appears on the printed master record.

Limbs If a person is missing limbs or parts of them, select here for arm and leg, left and right, what is missing – from *hand* or *foot* to *entire arm* or *entire leg*. VitaParms calculates the missing proportion of the body from this and takes it into account for the BMI (Section 6.10 (p. 37)).

Peak flow The *Personal best* is the highest peak flow during a symptom-free period, agreed with the doctor. If it is entered, warnings are based on it, otherwise on the predicted value for sex, age and height (section 6.5). Below, VitaParms shows today's predicted value and the limits for yellow and red (Fig. 5.5).

Coagulation (INR) The doctor sets the *Target range*, usually 2.0–3.0, with mechanical heart valves often 2.5–3.5. If nothing is entered, VitaParms assesses INR values against the default 2.0–3.0 (section 6.7).

Fluid intake and weight *Fluid intake at least* and *Fluid intake at most* (only with a prescribed restriction, e.g. heart failure or dialysis) in ml per day, and *Report weight gain from*: with heart failure a note is usual if weight rises by more than 2 kg in three days (section 6.8). 'off' or empty switches the notes off (Fig. 5.6).

Vital signs on admission The readings taken on admission – for example when care began or when you started recording. Edit admission values… opens the usual measurement form (Chapter 6). In the journal, the admission values can be shown or hidden with *Show admission values*.

> **Note:** If you change the details on limbs later, VitaParms asks when saving whether the BMI of the measurements already stored should be recalculated.

5.5 Care level and severe disability card

The **Care, disability card and advance directives** tab (Fig. 5.7) records the German long-term care level (*Pflegegrad*) and severe disability card (*Schwerbehindertenausweis*):

Care level *no care level* or 1 to 5. With a care level you can enter the *Next review* by the medical service if the notice states a date.

Degree of disability in steps of ten from 20 to 100 (*no card* = 0).

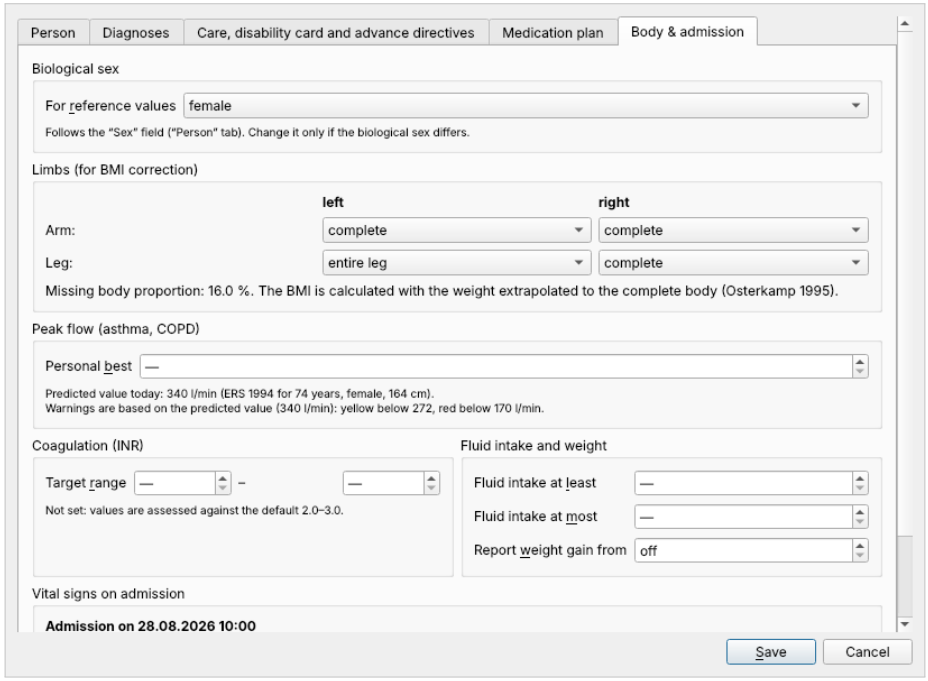


Fig. 5.4: Master record, 'Body & admission' tab: the left leg is missing (16.0 % of body weight)

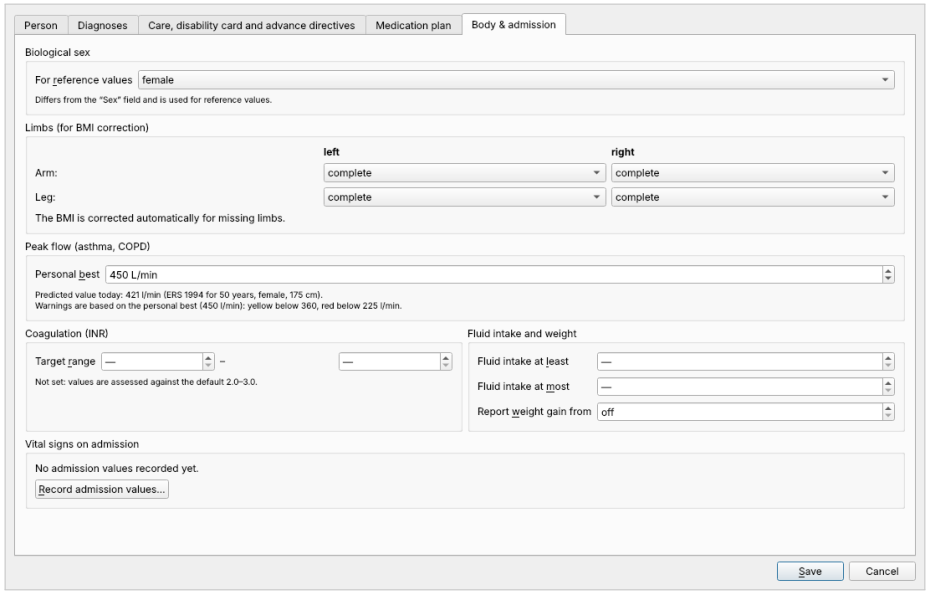


Fig. 5.5: 'Body & admission' tab: biological sex (here differing from 'diverse') and peak flow with personal best

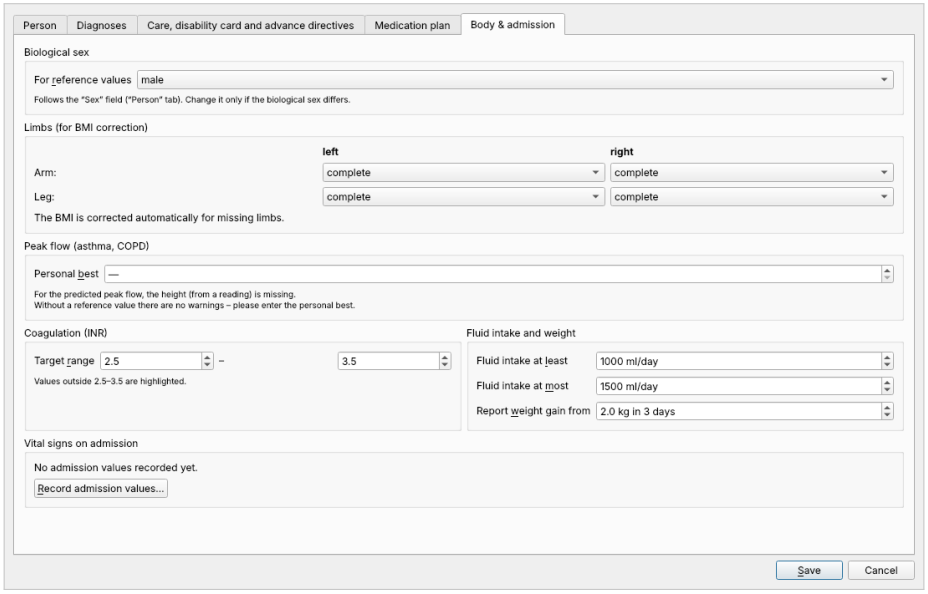


Fig. 5.6: INR target range, fluid limits and weight check in the master record

Marks as on the card: **G** (considerably impaired mobility), **aG** (exceptionally impaired walking), **B** (escort), **H** (helpless), **Bl** (blind), **Gl** (deaf), **TBl** (deafblind), **RF** (reduced broadcasting fee) and **1. Kl.** (first class). Hovering over a mark shows its meaning.

Valid until the expiry date of the card, or *unlimited*.

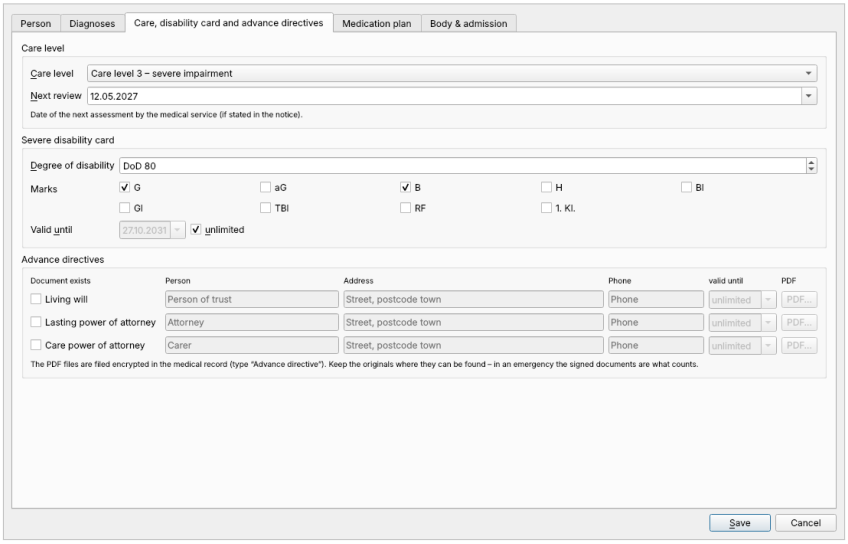


Fig. 5.7: Care level and severe disability card

Both appear in the header of the main window, on the printed master record, on the website and in the tablet app. If the review is due in less than two months or the card expires in less than three months, VitaParms highlights the entry in orange.

Advance directives

The same tab records which advance directives exist (Fig. 5.8) – in an emergency one of the first questions of paramedics and hospitals:

Document exists tick *Living will* (*Patientenverfügung*), *Lasting power of attorney* (*Vorsorgevollmacht*) and/or *Care power of attorney* (*Betreuungsvollmacht*).

Person, address, phone the person named – the attorney for the powers of attorney, the appointed carer for the care power of attorney, and for the living will a person of trust or the place where the original is kept.

valid until the expiry date if the document is limited in time; otherwise *unlimited*.

PDF PDF… stores a scanned copy as PDF. It is filed encrypted in the medical record (type 'Advance directive') and can be shown, replaced or detached via the same button (PDF ✓ indicates a stored file).

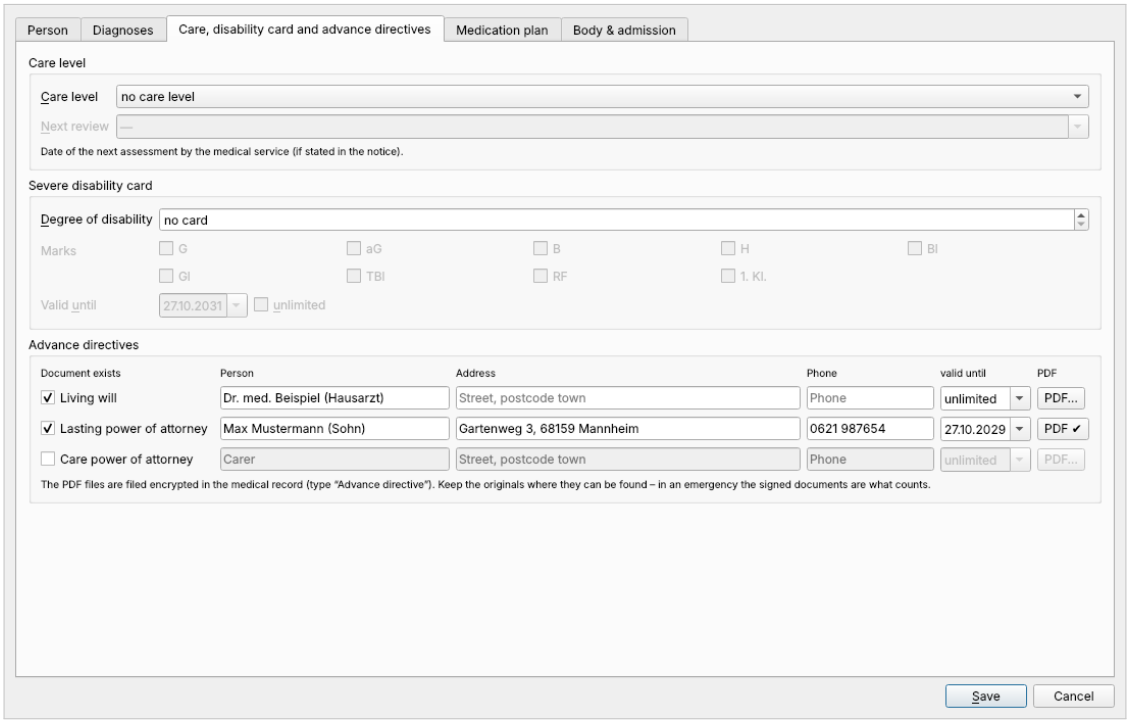


Fig. 5.8: Advance directives in the master record

The entries appear in the header of the main window, on the printed master record, on the website and in the tablet app – the PDF itself stays on your own devices like the whole medical record. If a time-limited document expires in less than three months, the entry is shown in orange.

> **Note:** In an emergency the signed originals are what counts. Keep them where they can be found and note for the living will where the original is kept.

5.6 Deleting a person

Person » Delete person... removes the selected person together with all readings, medication, documents and appointments. Because this cannot be undone, only administrators may delete persons, and VitaParms asks for confirmation first. If in doubt, make a backup beforehand (Section 12.2 (p. 90)).

6 Entering readings

At the heart of VitaParms is the journal: a running list of all measurements of a person. Any number of measurements per day is possible – for example a complete blood glucose daily profile. You do not have to record every value each time; empty fields simply stay empty.

6.1 Step by step: entering a measurement

1. Choose the person in the person selection.

2. Choose Readings » New measurement… or press Ctrl + M (alternatively Ctrl + N). The measurement form opens (Fig. 6.1). Its header shows last name, first name and date of birth from the master record – so mix-ups are ruled out.

3. *Date / time* is preset to the current time. If you measured earlier, change it; Now sets it back to the current time.

4. Choose the *Timing*, for example *fasting* or *2 h after breakfast*. It makes daily profiles comparable.

5. Enter the measured values. Fields showing '—' count as *not measured*. Use the Tab key to move to the next field.

6. Note under *Observations* anything important about the measurement: how the person feels, special circumstances, measuring conditions.

7. Click Save. If you want to enter the next measurement straight away – for example when transferring readings from a paper notebook – choose Save & next measurement. The form then stays open and is cleared for the next measurement.

The form adapts to the height of the screen: if the screen is smaller than the form, it can be scrolled with the mouse wheel or the scroll bar – the Save and Cancel buttons always stay visible.

> **Peak flow:** Peak flow (peak expiratory flow, PEF) is measured mainly in asthma and COPD. Enter the best of three attempts. On the **Chart** tab, peak flow has a panel of its own, which you switch on under *Panels*. How VitaParms rates the value and warns you is described in section 6.5.

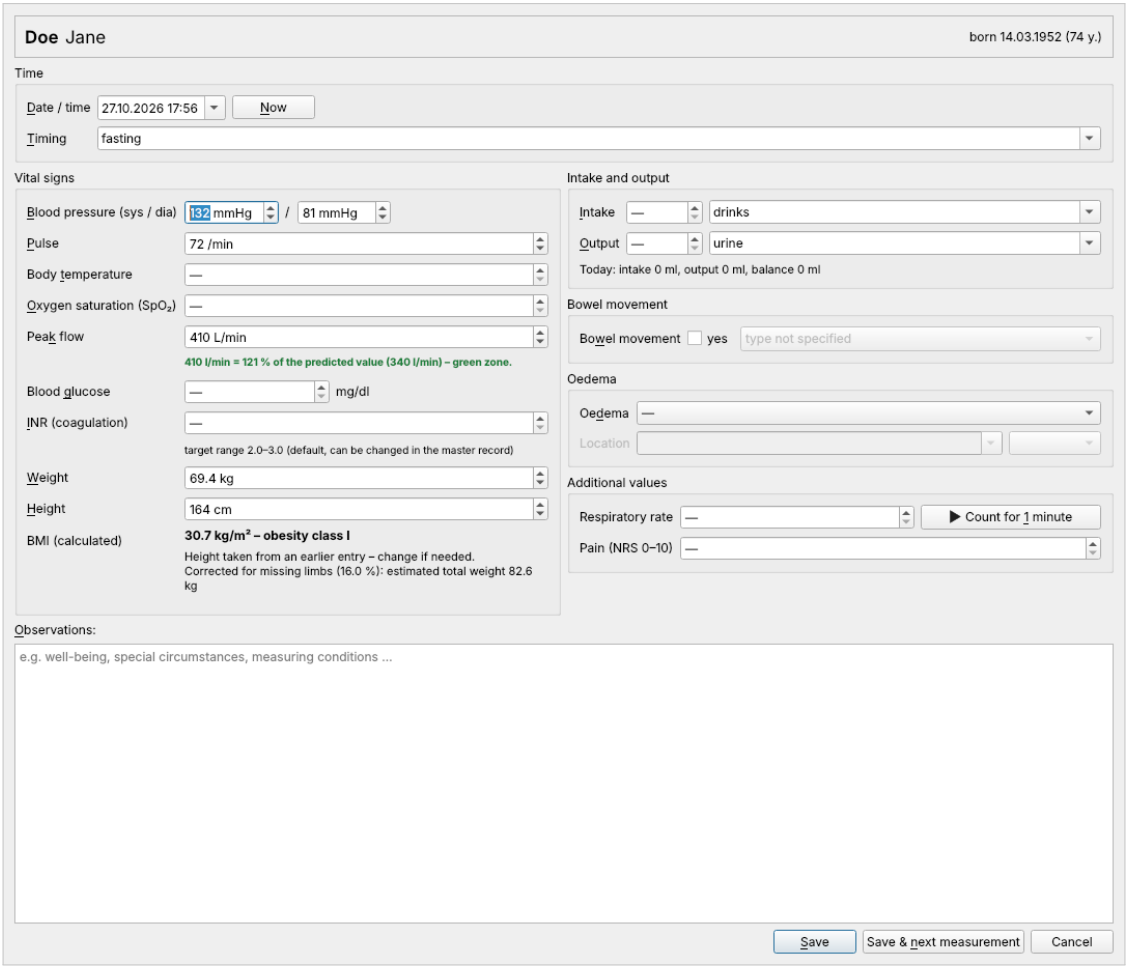


Fig. 6.1: The form for a measurement: vital signs on the left; intake and output, bowel movement, oedema and extra values on the right

6.2 The readings in detail

Reading	Abbr.	Unit	Note
Blood pressure	BP	mmHg	systolic / diastolic
Pulse	Pulse	/min	beats per minute
Body temperature	Temp.	°C	
Oxygen saturation	SpO_2	%	peripheral measurement with a finger clip
Peak flow	PEF	L/min	peak expiratory flow, measured with a peak flow meter; best of three attempts
Blood glucose	BG	mg/dl or mmol/l	unit selectable in the settings
Weight		kg	
Height		cm	preset with the known height (admission or last recorded) for a new reading; can be changed
BMI		kg/m^2	calculated, not entered
INR			coagulation value, section 6.7
Intake / output		ml	with type (drinks, infusion … / urine …), section 6.8
Oedema		grade 0–4	with site and side, section 6.9
Respiratory rate		/min	example of a custom field
Pain	NRS	0–10	numeric rating scale, custom field
Bowel movement		type 1–7	tick *yes*, then the type on the Bristol scale; section 6.6

You can adapt the list of timings for daily profiles to your daily routine in the settings (Section 13.3 (p. 94)).

Counting breaths with the minute timer

Next to the *Respiratory rate* field there is a Count for 1 minute button (Alt + 1). It starts a timer for exactly one minute:

1. Click Count for 1 minute. A short tone signals the start.
2. Count the breaths – preferably the rising of the chest – without the person noticing; otherwise they involuntarily breathe differently. The button shows the remaining seconds (Fig. 6.2).
3. After one minute a double tone sounds. The cursor is in the *Respiratory rate* field; type in the number you counted.

Clicking again while counting cancels. If you use the *Respiratory rate* field in your own form (Section 13.5 (p. 95)), the button appears there as well.

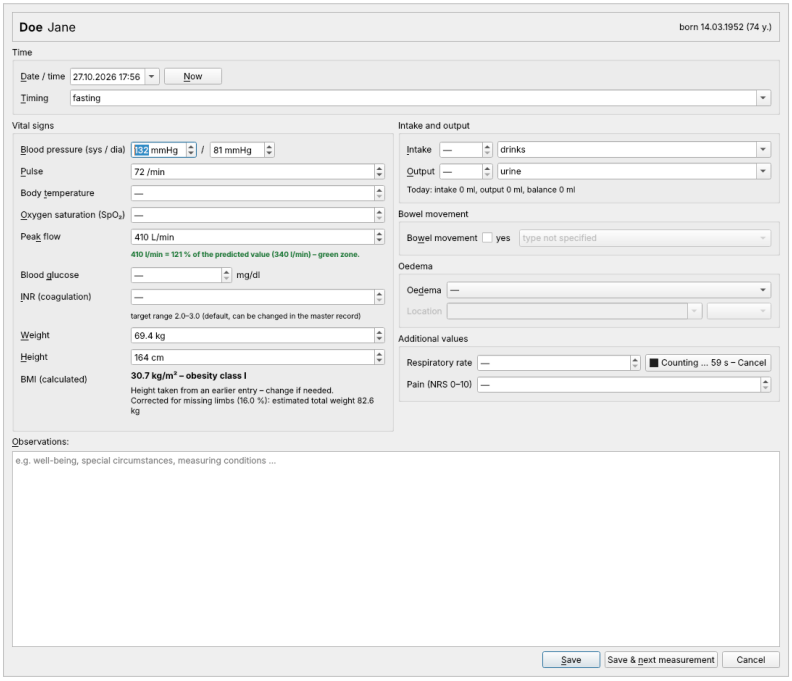


Fig. 6.2: The minute timer counting down

6.3 Editing and deleting measurements

Select the measurement in the journal and press F2 (or double-click the row) to edit it. Del deletes the selected measurement after a confirmation. Both commands are also available in the *Readings* menu and as buttons below the table.

6.4 BG day profile

For a blood glucose day profile you measure at fixed times on one day. Readings › BG day profile… sets the reminders for it (Fig. 6.3):

1. Select the person, then Readings › BG day profile….
2. Choose the *Day* (suggested: today, after the last time point tomorrow) and check the times of the seven time points: fasting before breakfast, two hours after breakfast, before lunch, two hours after lunch, before dinner, two hours after dinner and before bedtime. Untick the ones you do not need.
3. Start day profile. VitaParms creates an appointment with a reminder for each time point and remembers the times for next time.

4 At the time, the reminder appears (Fig. 6.4). Record measurement… opens the entry window with the subject ‘BG day profile’ and the time as well as the matching time point; after saving, the appointment is marked as done.

Stop day profile deletes the remaining open reminders of the selected day. Because the reminders are ordinary appointments, they appear in the appointment planner and reach the tablet when synchronising. The day profile itself is shown under **Chart** (day view).

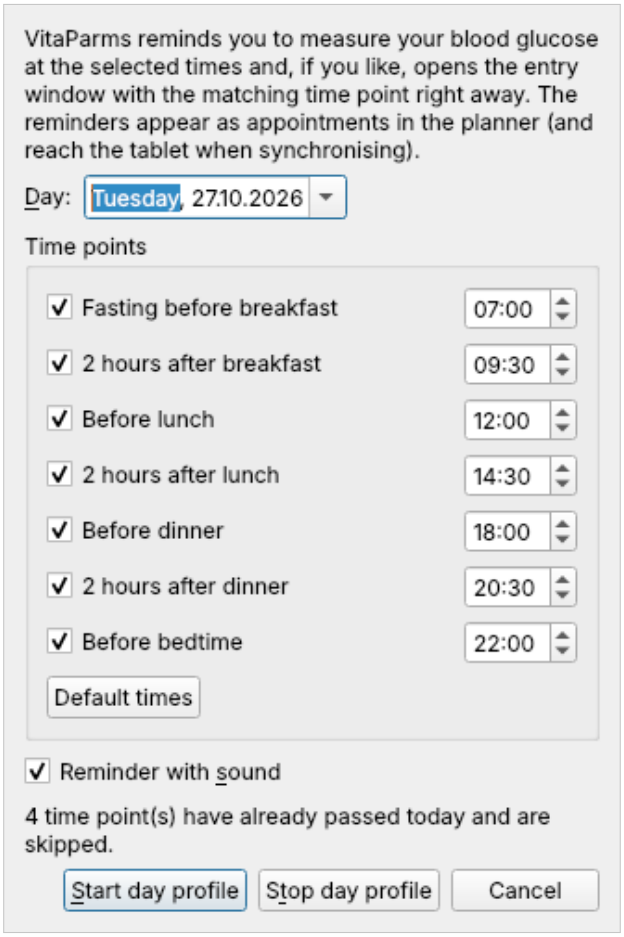


Fig. 6.3: Setting up a BG day profile

6.5 Peak flow: predicted value and warnings

VitaParms compares every peak flow value with a *reference value* and assigns it to a zone of the familiar traffic-light scheme:

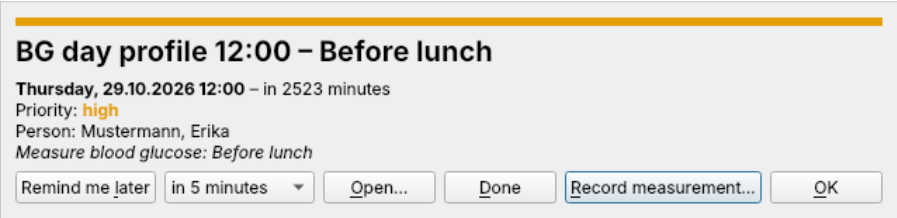


Fig. 6.4: Reminder for a measurement of the day profile

Zone	Share of the reference value	VitaParms's note
green	80 % or more	none
yellow	50 to 79 %	'Caution' – follow the asthma action plan
red	below 50 %	'Warning' – emergency medication, medical help, call 112

The reference value is the *personal best* if it is entered in the master record (section 5.4) – this is what asthma guidelines recommend, because the personal best reflects your own lungs. Otherwise VitaParms uses the *predicted value* from the EGKS/ERS 1994 reference table. It depends on

- the **biological sex** (master record, **Body & admission** tab; it follows the *Sex* field and is changed only if it differs or that field says 'diverse'),
- the **age** on the day of the reading (from the date of birth) and
- the **height** (from the reading or the most recent one).

Between the table rows (5, 10, 15 … 80 years) and columns (135, 140 … 195 cm) VitaParms interpolates linearly; above 80 years the row for 80 applies. For children under 5 and heights outside 135 to 195 cm the table gives no value – the personal best is needed then. If something is missing, VitaParms says what below the input field.

When entering a reading, the reference value and the limits for yellow and red appear below the *Peak flow* field; as soon as you enter a value, the percentage and the zone are shown (Fig. 6.5). If the saved reading is in the yellow or red zone, a window points it out again (Fig. 6.6). In the history such values are highlighted in orange or red (red also in bold); the tooltip gives the percentage, the note and the reference. The printed master record shows the personal best and the predicted value.

Important: VitaParms is not a medical device. These notes do not replace the asthma action plan agreed with your doctor; its limits and instructions take precedence. The predicted value is an average of healthy people – your personal best is the better measure.

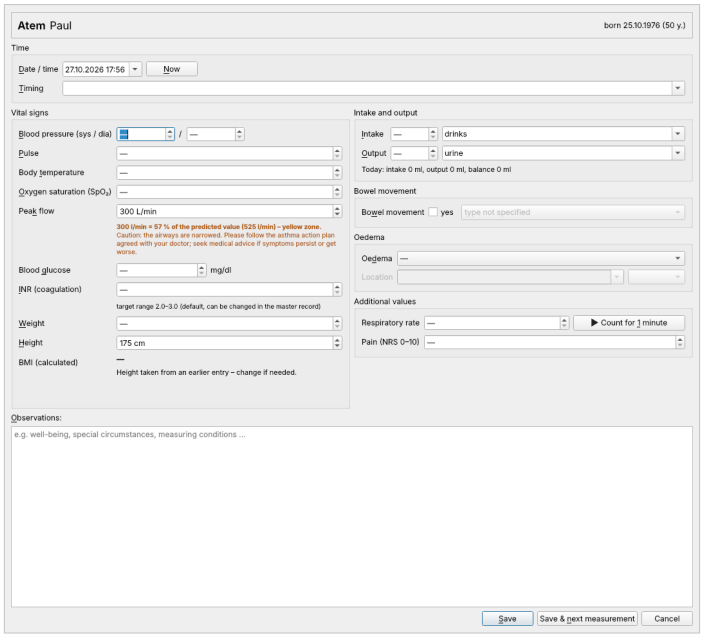


Fig. 6.5: Peak flow in the yellow zone: note below the field

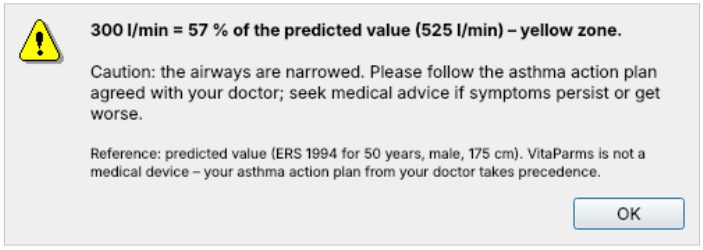


Fig. 6.6: Note after saving a value in the yellow zone

6.6 Bowel movements on the Bristol scale

In care, bowel movements are among the most important observations. The reading form has the row *Bowel movement* for them (Fig. 6.7):

1. Tick *yes* if the person had a bowel movement. Only then is the list next to it enabled.
2. Choose the type on the *Bristol stool scale* (see table) – or leave it at *type not specified*.
3. For the bowel movement log you can enter *Amount*, *Colour*, *Admixtures / odour* and *Accompanying symptoms* below. The lists offer suggestions; you may also type your own text, several admixtures separated by commas.
4. Save. Record several bowel movements on one day as separate readings.

Type	Description	Meaning
1	separate hard lumps, hard to pass	hard – suggests constipation
2	sausage-shaped but lumpy	hard – suggests constipation
3	sausage-shaped with cracks on the surface	normal
4	sausage- or snake-like, smooth and soft	normal
5	soft blobs with clear-cut edges	soft – suggests diarrhoea
6	mushy, fluffy pieces with ragged edges	soft – suggests diarrhoea
7	watery, no solid pieces	liquid – suggests diarrhoea

Note after three days without a bowel movement. If no bowel movement has been recorded for three days or more, the header of the main window shows the orange line 'Bowel movement: none recorded for … days'; its tooltip names the last day and advises watching for complaints, drinking enough and seeking medical advice if constipation persists (Fig. 6.8). The reading form reminds you of it below the *Bowel movement* row as well. The note only appears if bowel movements were recorded at all in the last 30 days – if you do not log them, you are not bothered. The day without an entry counts: if the last bowel movement was on Monday, the note appears on Thursday.

Black or bloody stool. If you choose *black (tarry)* or *red / bloody* as the colour or enter blood as an admixture, VitaParms points out after saving that this should be checked by a doctor promptly – unless it is explained by iron tablets, charcoal or food such as beetroot. In the journal the entry is marked with '▲'.

The *Bowel movements (frequency)* panel on the **Chart** tab shows the frequency (Section 7.2 (p. 42)); the bowel movement log is described in Section 7.4 (p. 44).

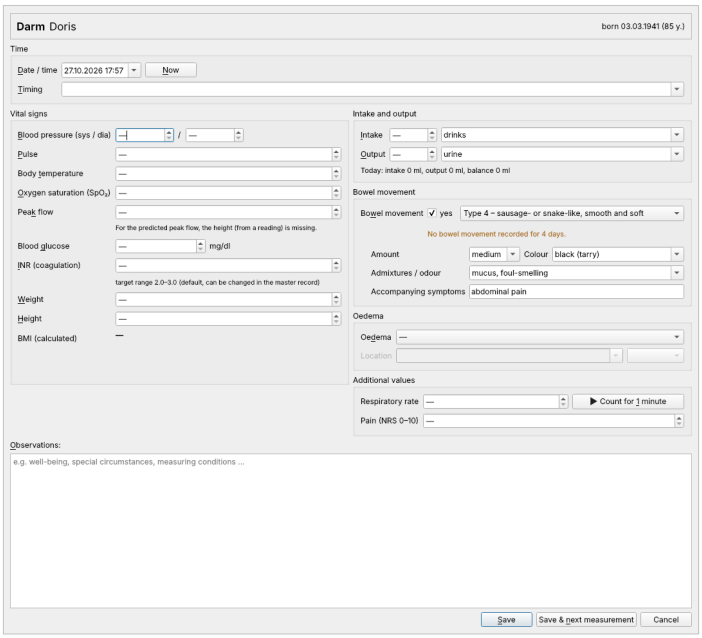


Fig. 6.7: Recording a bowel movement: Bristol type and details for the log

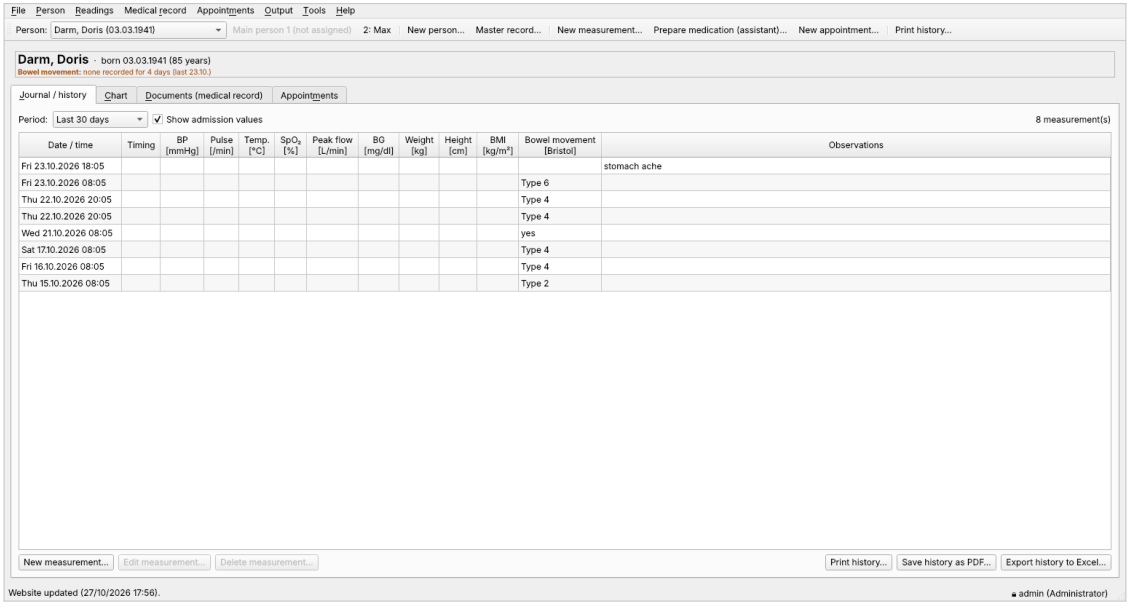


Fig. 6.8: Note in the header of the main window after four days without a bowel movement

Important: VitaParms is not a medical device. The Bristol scale only describes the form of the stool; the notes do not replace a medical assessment.

6.7 Coagulation: INR

If you take vitamin K antagonists such as warfarin or phenprocoumon, your INR is checked regularly or you test it yourself. Enter the value in the *INR (coagulation)* field. Right below, VitaParms shows whether it is within the target range from the master record (Section 5.4 (p. 22)); without an entry, the default 2.0–3.0 applies.

Value	Display	Note after saving
within target range	green	none
below target range	orange	clotting less inhibited (risk of thrombosis)
above target range	orange	increased bleeding risk
5.0 and above	red, bold	call your doctor at once; with bleeding call the emergency number

A value outside the target range from the last 14 days also appears in the header of the main window. On the **Chart** tab the *Coagulation (INR)* panel shows the target range as a green band. Change the dose only as instructed by your doctor.

6.8 Intake and output (fluid balance)

At the top right of the reading form, enter under *Intake* what was drunk or given (drinks, soup, tube feed, infusion …) and under *Output* what was passed (urine in a measuring jug, vomit, drain …) – each in ml and with its type. For a drinking log, record each glass as a reading of its own; Save & next measurement helps. Below, VitaParms shows the day's balance including the amount just entered.

If fluid limits are set in the master record, the header of the main window shows a note when less than the minimum was drunk the day before or more than the prescribed maximum today. With the *weight check* switched on, VitaParms reports a gain of more than the set amount within three days – an early sign of fluid retention in heart failure. After saving, one window sums up all notes (Fig. 6.9).

6.9 Oedema

Under *Oedema*, choose the grade by the depth of the pit: press with a finger for a few seconds, for example on the shin, and assess the pit.

Grade	Finding
0 – none	no pit after finger pressure
1 – mild	pit up to 2 mm, disappears immediately
2 – moderate	pit up to 4 mm, disappears after 10–15 seconds
3 – marked	pit up to 6 mm, lasts up to a minute
4 – severe	pit 8 mm or deeper, lasts 2–5 minutes

From grade 1, choose the *Location* (ankles/feet, lower legs, entire leg, hands/arms, face/eyelids …) and the side. VitaParms points out in particular

- swelling of **one** leg – it can indicate a thrombosis,
- swelling of the **face** or eyelids,
- grades 3 and 4 and an increase of two grades over the last entry.

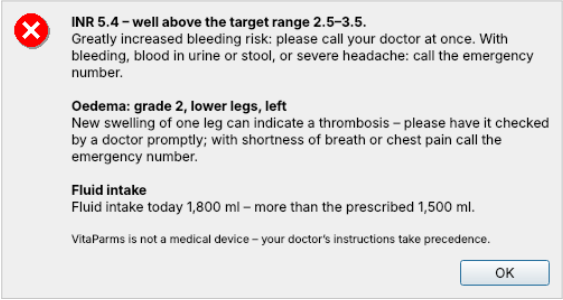


Fig. 6.9: Notes after saving: INR well above range, one-sided oedema, fluid intake above the restriction

Important: VitaParms is not a medical device. The notes do not replace a medical assessment; your doctor's instructions take precedence.

6.10 The BMI – including missing limbs

The body mass index is calculated from weight and height:

$$\mathrm{BMI} = \frac{\text{weight in kg}}{(\text{height in m})^2}$$

If a person is missing limbs, this value would be too low, because the measured weight is less than that of a complete body of the same height. VitaParms therefore first extrapolates the weight to the complete body:

$$W_{\mathrm{corr}} = \frac{W}{1 - p/100}, \qquad \mathrm{BMI} = \frac{W_{\mathrm{corr}}}{(\text{height in m})^2}$$

Here p is the proportion of the missing limbs in body weight, in per cent. The proportions come from a widely used table by Osterkamp[1]:

Arm	Proportion	Leg	Proportion
hand	0.7 %	foot	1.5 %
forearm with hand	2.3 %	lower leg with foot	5.9 %
entire arm	5.0 %	entire leg	16.0 %

Example: Jane Doe is missing her left leg ($p = 16.0$). She weighs 69.4 kg and is 164 cm tall. The extrapolated weight is $69.4/0.84 = 82.6$ kg, so the BMI is $82.6/1.64^2 = 30.7\,\mathrm{kg/m^2}$. Without the correction the BMI would be only 25.8. The measurement form shows the calculation below the BMI (Fig. 6.1).

> **Note:** The BMI is a rough guide. The classification as underweight, normal weight or obesity that VitaParms shows next to the value follows the usual WHO categories for adults and takes neither age nor muscle mass into account. Assess it together with your doctor.

[1] L. K. Osterkamp: Current perspective on assessment of human body proportions of relevance to amputees. *J Am Diet Assoc* 1995; 95: 215–218.

7 Journal, chart and output

Collected readings are only useful if you can see them at a glance. VitaParms offers two views for this – the table in the journal and the trend charts on the **Chart** tab – and outputs both on paper, as PDF and as an Excel workbook.

7.1 The journal

The **Journal / history** tab lists all measurements of the selected person, **newest first** (Fig. 4.1 on p. 15). Each reading has its own column; custom fields such as respiratory rate or pain automatically appear as further columns.

Use *Period* to limit the display: *All measurements*, the last 7, 30 or 90 days, the last year or a custom period (*From – to*). The number of measurements shown appears at the top right.

7.2 Trend charts

The **Chart** tab draws the readings as curves (Fig. 7.1).

1. Under *View*, choose the period: *Daily profile*, *Weekly trend*, *Monthly trend*, *Yearly trend* or *Custom period* with any start and end date. VitaParms remembers your choice.

2. Use the arrow buttons next to the date to move to the previous or next day, week, month or year. Today jumps back to the current period.

3. Under *Panels*, choose which charts are shown: blood pressure & pulse, body temperature, oxygen saturation, peak flow, blood glucose, coagulation (INR), weight, BMI, intake and output, oedema and bowel movements (frequency). If the window is not tall enough for all selected panels, the chart scrolls vertically – every panel keeps a readable minimum height.

4. Point the mouse at a data point to see the exact value and time.

Each curve has its own colour *and* its own marker shape (circle, square, triangle and so on). This keeps the curves distinguishable on a black-and-white printout and for people with colour vision deficiency. The legend is shown above each chart.

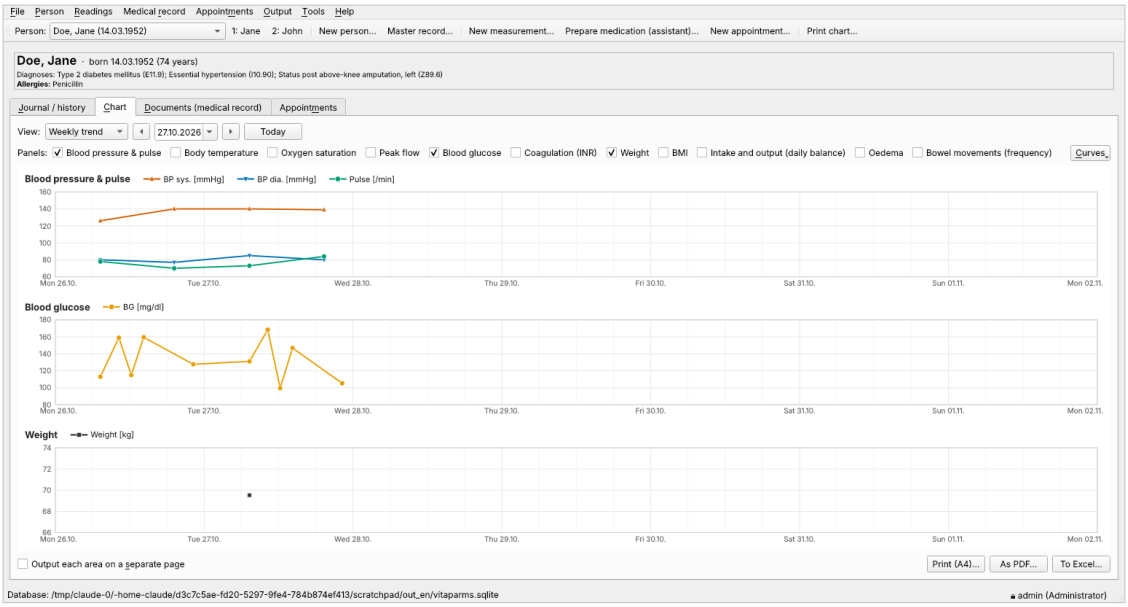


Fig. 7.1: Weekly trend with blood pressure, pulse, blood glucose and weight

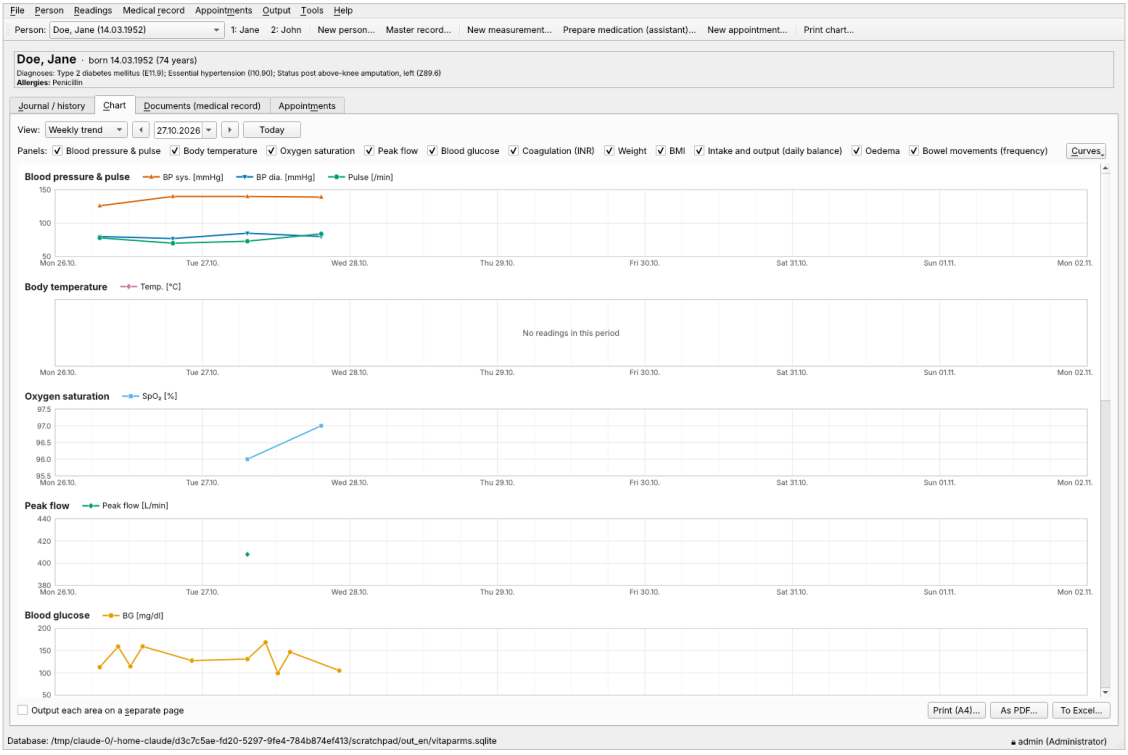


Fig. 7.2: All seven panels: the chart scrolls vertically and every panel keeps its minimum height

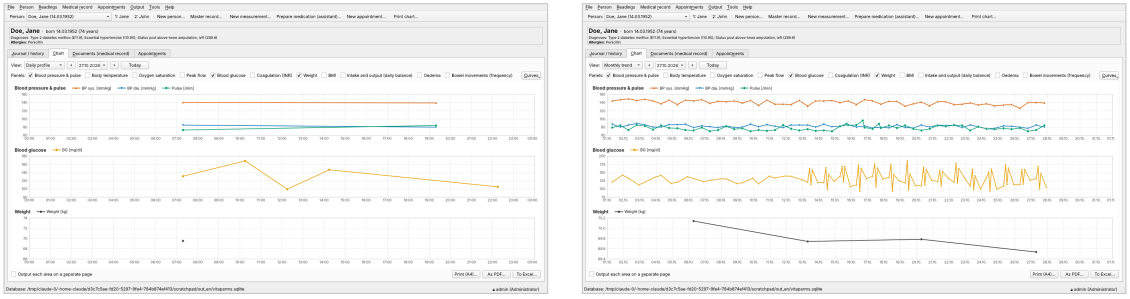


Fig. 7.3: Daily profile (left) with the blood glucose profile, and monthly trend (right)

Yearly trend and long periods

The *yearly trend* shows a calendar year. For periods longer than about two months, VitaParms combines the readings of a day into one **daily value**: the curve connects the daily means, and a vertical line in the same colour reaches from the lowest to the highest value of the day (Fig. 7.4). This keeps even a year with several readings a day clear, while daily fluctuations – for blood glucose, say – remain visible. The tooltip gives the mean, the range and the number of readings of the day. Gaps of more than a week without readings are not bridged. The statistics (minimum, maximum, mean) always refer to the individual readings.

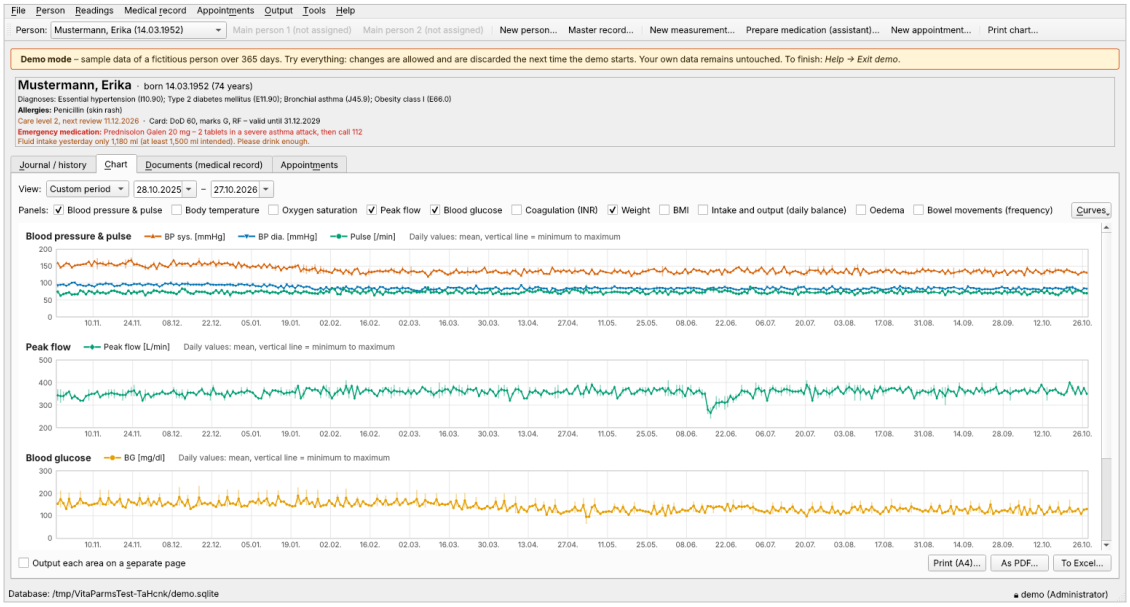
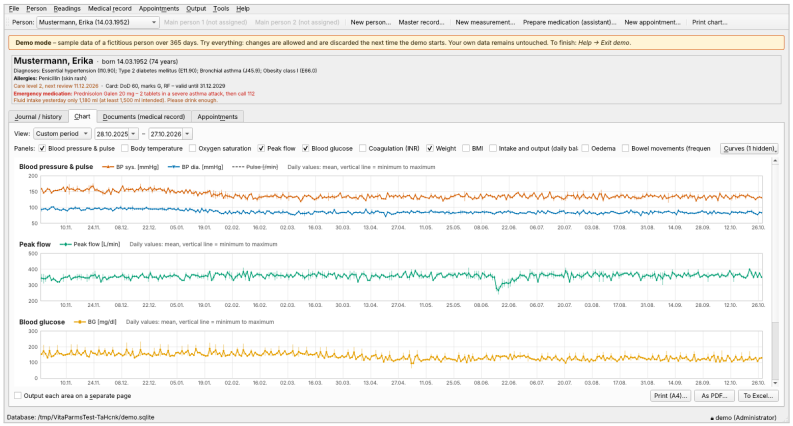


Fig. 7.4: One year as a custom period: daily values with range (demo database)

Showing and hiding individual curves

Besides the panels, every single curve can be shown or hidden – for example the pulse in the *Blood pressure & pulse* panel if only the blood pressure matters:

- **Click the legend entry** above the chart. The curve disappears, its entry stays in grey and struck through; a second click shows it again.
- Or open Curves next to the panels and tick or untick the curves you want (Fig. 7.5). *Show all curves* cancels all hiding. The button shows how many curves are currently hidden.



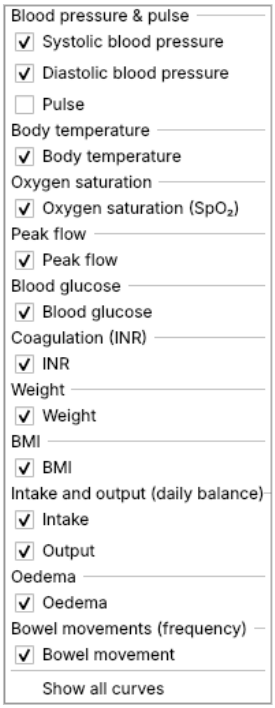


Fig. 7.5: Pulse hidden (left) and the ‘Curves’ menu (right)

Hidden curves are also left out of the printout, the PDF and the Excel export; the axes adapt to the visible curves. The setting is kept until you change it.

Bowel movements: frequency

The *Bowel movements (frequency)* panel does not draw a curve but a bar for each day: its height is the number of bowel movements, the segments are coloured by the Bristol scale – orange for hard (types 1–2), green for normal (types 3–4), blue for soft to liquid (types 5–7), grey for not specified. Periods of three or more days without a bowel movement are shaded and labelled (Fig. 7.6); a period that began before the selected range is counted correctly. In the daily view each bowel movement is a marker at its time, its height shows the type. In the key figures below the printed chart, bowel movements show the number of movements, the smallest, largest and mean Bristol type, ‘on … of … days’ and the longest period without.

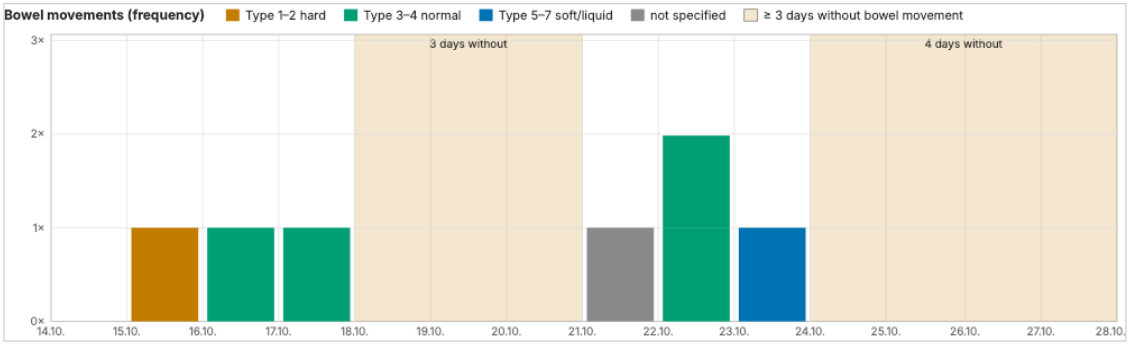


Fig. 7.6: Bowel movements: a bar per day, three and four days without shaded

INR, intake and output, oedema

The *Coagulation (INR)* panel shades the target range from the master record in green. *Intake and output (daily balance)* shows two bars per day – blue for intake, orange for output – and the balance as a diamond; minimum and maximum fluid intake are dashed lines (Fig. 7.7). In the daily view intake and output rise as steps over the day, so you can see how much had been drunk by when. *Oedema* shows the grade on a fixed scale from 0 to 4. The key figures state how many INR values were within the target range and give daily totals for intake and output.

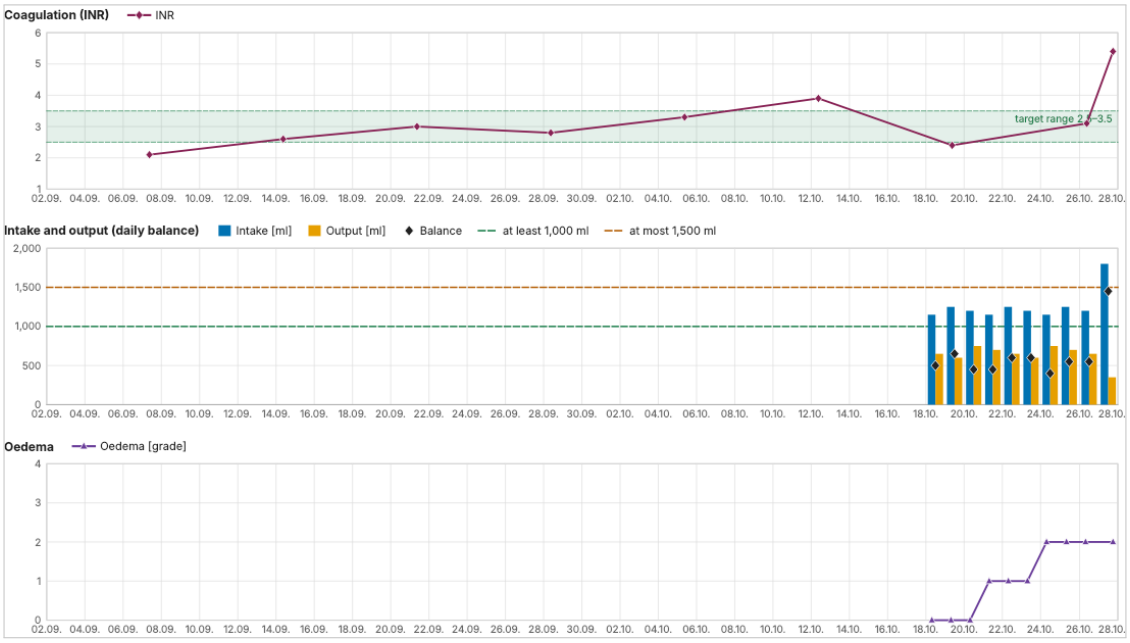


Fig. 7.7: INR with target range 2.5–3.5, daily balance with fluid limits, oedema

7.3 Printing, PDF and Excel

All output refers to the visible tab and the selected period.

Output	Menu	Shortcut	Result
Print	*Output* → *Print…*	Ctrl + P	printout on A4
Preview	*Output* → *Print preview…*	Ctrl + Shift + P	page view before printing
PDF	*Output* → *Save as PDF…*		PDF file, A4
Excel	*Output* → *Export to Excel…*	Ctrl + Shift + E	`.xlsx` file

The **history** is printed as a table in landscape format, the **chart** in portrait format (Fig. 7.9). VitaParms adds a header with name, date of birth and period as well as page numbers.

If *Output each area on a separate page* below the chart is ticked, each panel gets an A4 sheet of its own with its own legend and statistics – the curves become much larger and can be passed on individually (Fig. 7.8). This applies to printing, print preview and PDF.

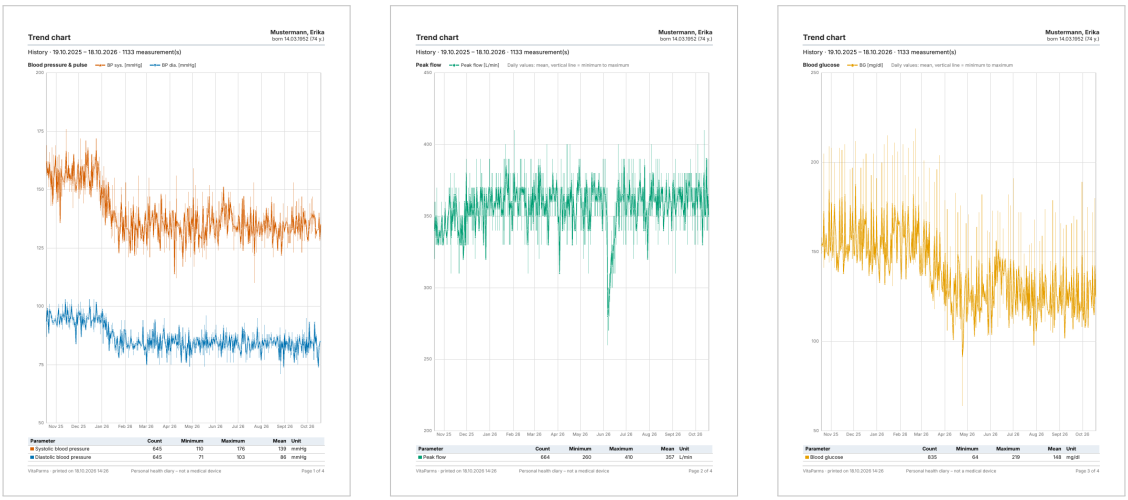


Fig. 7.8: Yearly trend with one panel per page: blood pressure, peak flow, blood glucose

An **Excel export** from the journal produces a worksheet *History* with all measurements shown. The export from the **Chart** tab produces three worksheets: *Chart* with the trend charts as an image, *Statistics* with minimum, maximum and mean per reading, and *Readings* with the raw data of the period. The files can be opened with Microsoft Excel, LibreOffice Calc and other spreadsheet programs.

7.4 Bowel movement log, daily glucose profiles and fluid balance

VitaParms compiles three logs from the recorded readings by itself – you do not have to keep anything separately. All are in the *Output* menu and apply to the selected person.

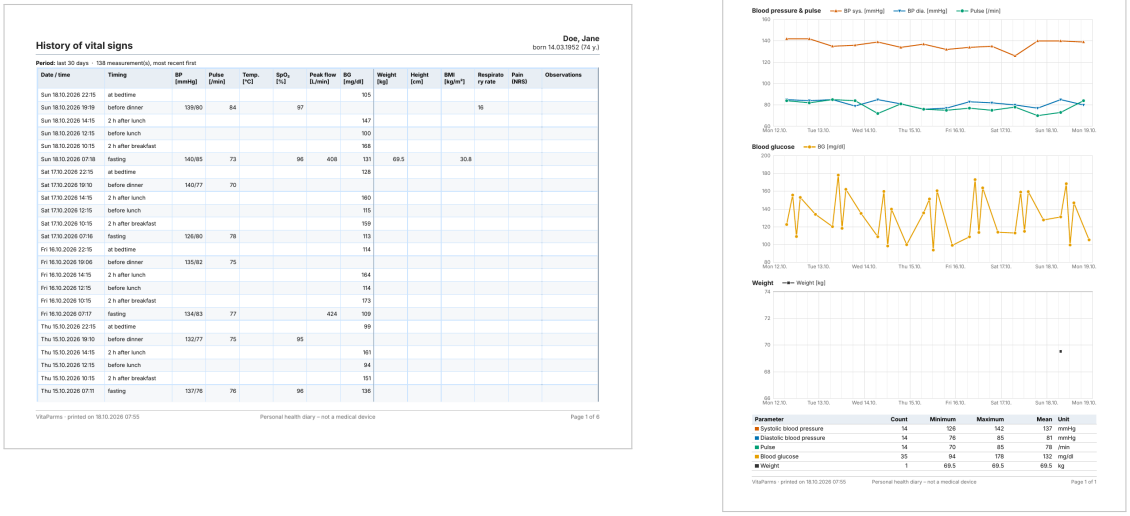


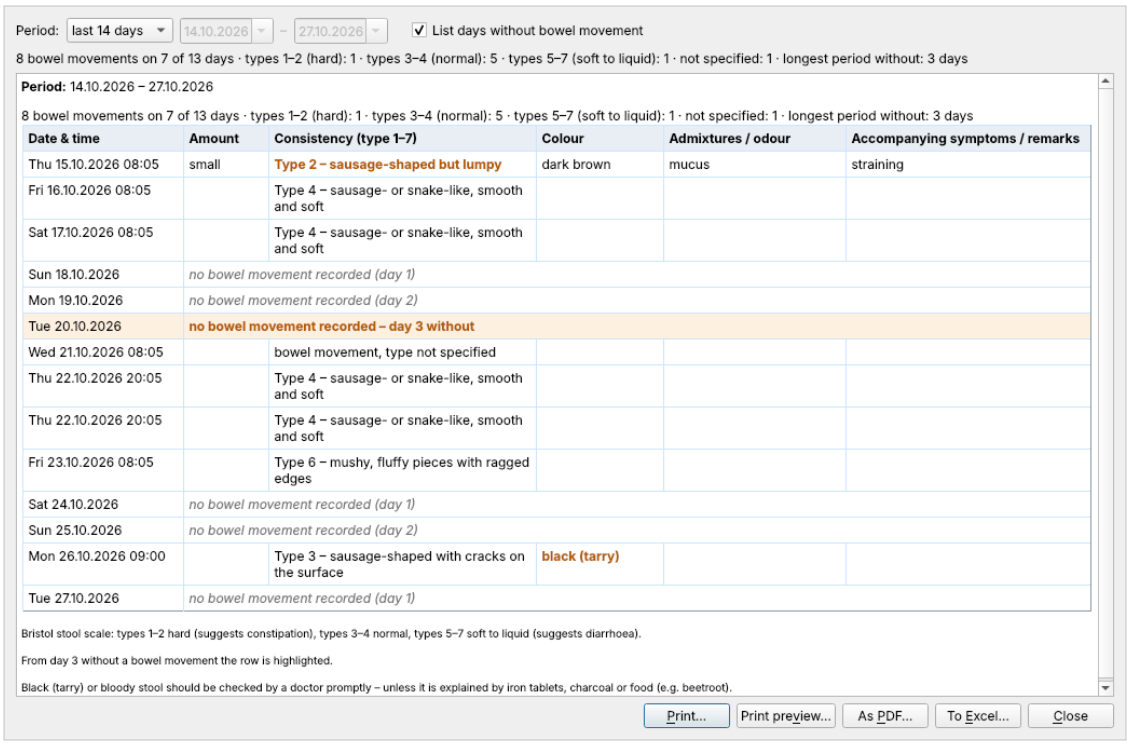
Fig. 7.9: Printout of the history (landscape) and the weekly chart (portrait)

Output » **Bowel movement log…** lists all bowel movements of the period by date and time with the columns *Date & time*, *Amount*, *Consistency (type 1–7)*, *Colour*, *Admixtures / odour* and *Accompanying symptoms / remarks* (Fig. 7.10). If *List days without bowel movement* is ticked, every day without an entry is listed too, with how many days it has been; from the third day on the row is highlighted. A line above sums up: number of bowel movements, on how many days, the distribution on the Bristol scale and the longest period without.

Output » **Daily glucose profiles…** assigns the blood glucose values of each day to the time points of the daily glucose profile (Section 6.4 (p. 30)): one row per day, one column per time point, plus *other values* (e.g. at night) and the daily mean (Fig. 7.11). The timing of the reading decides; values without timing go to the nearest time point (up to 75 minutes away). The time of day is in brackets. Below the table are the mean and count per column; values below 70 or above 250 mg/dl are highlighted. *Days with at least … values* decides which days count as a profile (default 4; with 1, every day with a glucose value appears).

Output » **Fluid balance…** lists every intake and output with time, amount, type and remark; each day ends with a total row with the balance, and the end shows the average per day (Fig. 7.12). Daily totals outside the fluid limits from the master record are highlighted.

For all three, choose the period (7, 14, 30 or 90 days or *From – to*) and output the log with Print…, Print preview…, As PDF… (A4 landscape) or To Excel….



Period: last 14 days 14.10.2026 – 27.10.2026 ☑ List days without bowel movement

8 bowel movements on 7 of 13 days · types 1–2 (hard): 1 · types 3–4 (normal): 5 · types 5–7 (soft to liquid): 1 · not specified: 1 · longest period without: 3 days

Period: 14.10.2026 – 27.10.2026

8 bowel movements on 7 of 13 days · types 1–2 (hard): 1 · types 3–4 (normal): 5 · types 5–7 (soft to liquid): 1 · not specified: 1 · longest period without: 3 days

Date & time	Amount	Consistency (type 1–7)	Colour	Admixtures / odour	Accompanying symptoms / remarks
Thu 15.10.2026 08:05	small	**Type 2 – sausage-shaped but lumpy**	dark brown	mucus	straining
Fri 16.10.2026 08:05		Type 4 – sausage- or snake-like, smooth and soft			
Sat 17.10.2026 08:05		Type 4 – sausage- or snake-like, smooth and soft			
Sun 18.10.2026	*no bowel movement recorded (day 1)*				
Mon 19.10.2026	*no bowel movement recorded (day 2)*				
Tue 20.10.2026	**no bowel movement recorded – day 3 without**				
Wed 21.10.2026 08:05		bowel movement, type not specified			
Thu 22.10.2026 20:05		Type 4 – sausage- or snake-like, smooth and soft			
Thu 22.10.2026 20:05		Type 4 – sausage- or snake-like, smooth and soft			
Fri 23.10.2026 08:05		Type 6 – mushy, fluffy pieces with ragged edges			
Sat 24.10.2026	*no bowel movement recorded (day 1)*				
Sun 25.10.2026	*no bowel movement recorded (day 2)*				
Mon 26.10.2026 09:00		Type 3 – sausage-shaped with cracks on the surface	**black (tarry)**		
Tue 27.10.2026	*no bowel movement recorded (day 1)*				

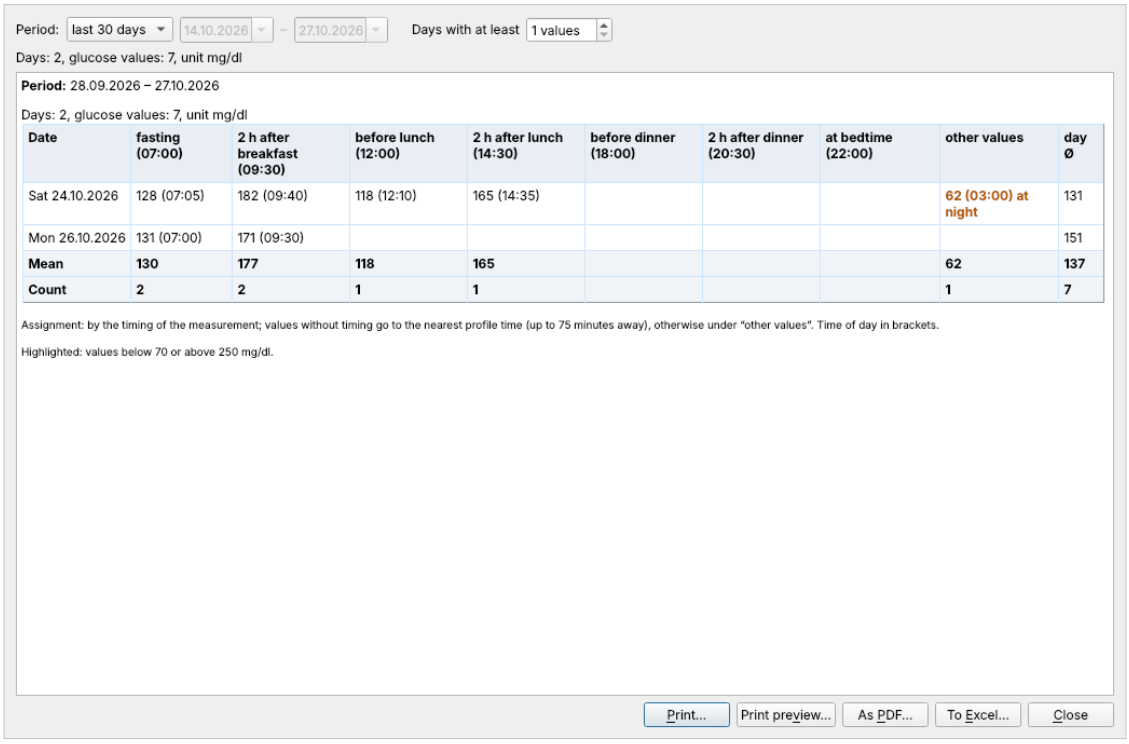
Bristol stool scale: types 1–2 hard (suggests constipation), types 3–4 normal, types 5–7 soft to liquid (suggests diarrhoea).

From day 3 without a bowel movement the row is highlighted.

Black (tarry) or bloody stool should be checked by a doctor promptly – unless it is explained by iron tablets, charcoal or food (e.g. beetroot).

Print... Print preview... As PDF... To Excel... Close

Fig. 7.10: Bowel movement log of the last 14 days



Period: last 30 days 14.10.2026 – 27.10.2026 Days with at least 1 values

Days: 2, glucose values: 7, unit mg/dl

Period: 28.09.2026 – 27.10.2026

Days: 2, glucose values: 7, unit mg/dl

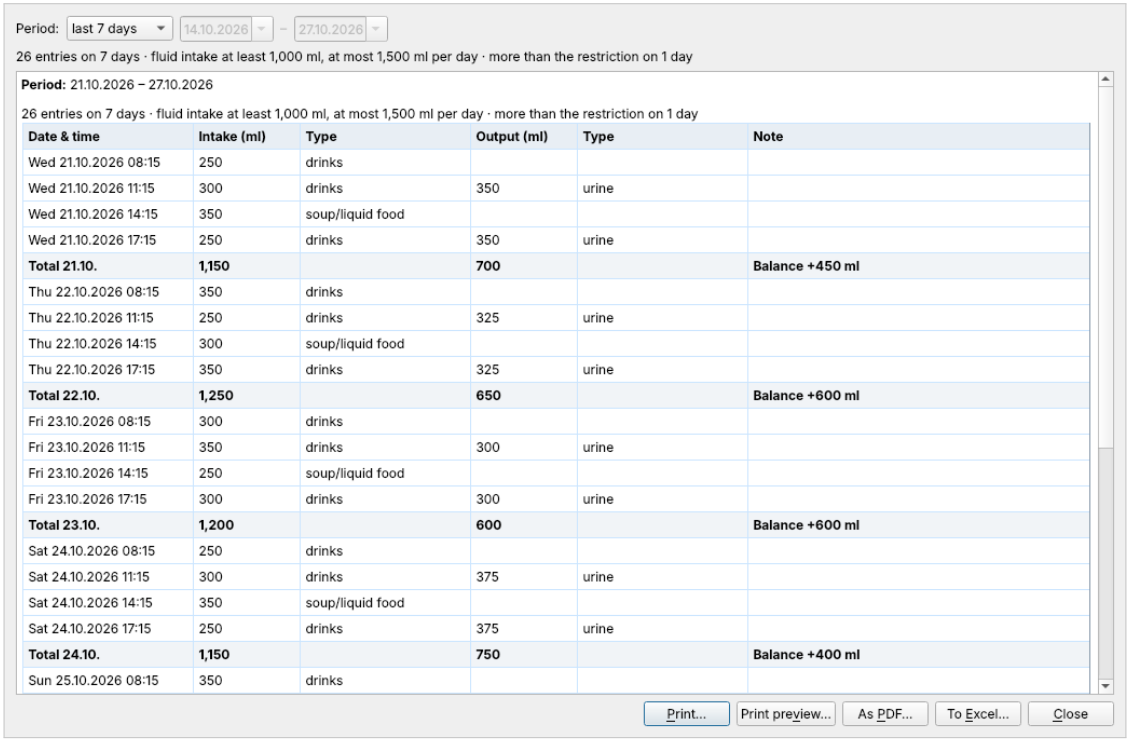
Date	fasting (07:00)	2 h after breakfast (09:30)	before lunch (12:00)	2 h after lunch (14:30)	before dinner (18:00)	2 h after dinner (20:30)	at bedtime (22:00)	other values	day Ø
Sat 24.10.2026	128 (07:05)	182 (09:40)	118 (12:10)	165 (14:35)				**62 (03:00) at night**	131
Mon 26.10.2026	131 (07:00)	171 (09:30)							151
Mean	**130**	**177**	**118**	**165**				**62**	**137**
Count	**2**	**2**	**1**	**1**				**1**	**7**

Assignment: by the timing of the measurement; values without timing go to the nearest profile time (up to 75 minutes away), otherwise under "other values". Time of day in brackets.

Highlighted: values below 70 or above 250 mg/dl.

Print... Print preview... As PDF... To Excel... Close

Fig. 7.11: Daily glucose profiles compiled from the readings



Period: 21.10.2026 – 27.10.2026

26 entries on 7 days · fluid intake at least 1,000 ml, at most 1,500 ml per day · more than the restriction on 1 day

Date & time	Intake (ml)	Type	Output (ml)	Type	Note
Wed 21.10.2026 08:15	250	drinks			
Wed 21.10.2026 11:15	300	drinks	350	urine	
Wed 21.10.2026 14:15	350	soup/liquid food			
Wed 21.10.2026 17:15	250	drinks	350	urine	
Total 21.10.	**1,150**		**700**		**Balance +450 ml**
Thu 22.10.2026 08:15	350	drinks			
Thu 22.10.2026 11:15	250	drinks	325	urine	
Thu 22.10.2026 14:15	300	soup/liquid food			
Thu 22.10.2026 17:15	350	drinks	325	urine	
Total 22.10.	**1,250**		**650**		**Balance +600 ml**
Fri 23.10.2026 08:15	300	drinks			
Fri 23.10.2026 11:15	350	drinks	300	urine	
Fri 23.10.2026 14:15	250	soup/liquid food			
Fri 23.10.2026 17:15	300	drinks	300	urine	
Total 23.10.	**1,200**		**600**		**Balance +600 ml**
Sat 24.10.2026 08:15	250	drinks			
Sat 24.10.2026 11:15	300	drinks	375	urine	
Sat 24.10.2026 14:15	350	soup/liquid food			
Sat 24.10.2026 17:15	250	drinks	375	urine	
Total 24.10.	**1,150**		**750**		**Balance +400 ml**
Sun 25.10.2026 08:15	350	drinks			

Fig. 7.12: Intake and output with daily totals and balance

Exports are not encrypted: PDF and Excel files leave the protected area of the database. Only store them where no unauthorised person has access, and delete them when they are no longer needed. To send data to a practice, the encrypted medical record is the better choice (Section 10.4 (p. 75)).

8 The medication plan

The medication plan is part of the master record. It records which medicines a person takes, when and how much – and it is the basis for the 'Prepare medication' assistant, the search for controlled narcotics, painkillers, blood thinners, blood pressure medicines and cortisone-containing products, and the medication appointments in the appointment planner.

8.1 The plan at a glance

Open the master record (Ctrl + E) and switch to the **Medication plan** tab (Fig. 8.1). The medicines are divided into four groups, each with a coloured header row:

Regular medication	Medicines taken according to a fixed plan – daily or on a fixed schedule.
As-needed medication	Only when needed, e.g. a painkiller for fever. The occasion and the maximum dose go into the note.
Emergency medication	For an emergency, e.g. a nitro spray for chest pain. Also shown in red in the header of the main window.
Other	Ointments, eye drops, food supplements, vaccinations, aids. The application is described in the note.

The *Dosage* column shows the dosage in the familiar short form morning – noon – evening – night, for example '1-0-1-0 pcs'. For medicines with a schedule, the intake days follow ('once a week (Wednesday)', 'monthly on day 1'). A small document symbol after the name shows that a package leaflet has been saved in the medical record.

Sorting: clicking a column heading sorts the medicines alphabetically in ascending order, a second click in descending order; a small arrow in the heading shows the direction. Sorting always takes place only *within* each type – the medicines stay in their group, and the order of regular, as-needed and emergency medication and other does not change. Start and end are sorted as dates, and empty fields always come last. VitaParms remembers the chosen sort order.

The buttons below the list:

Add… / **Edit…** open the medication dialog (double-clicking a row also works).

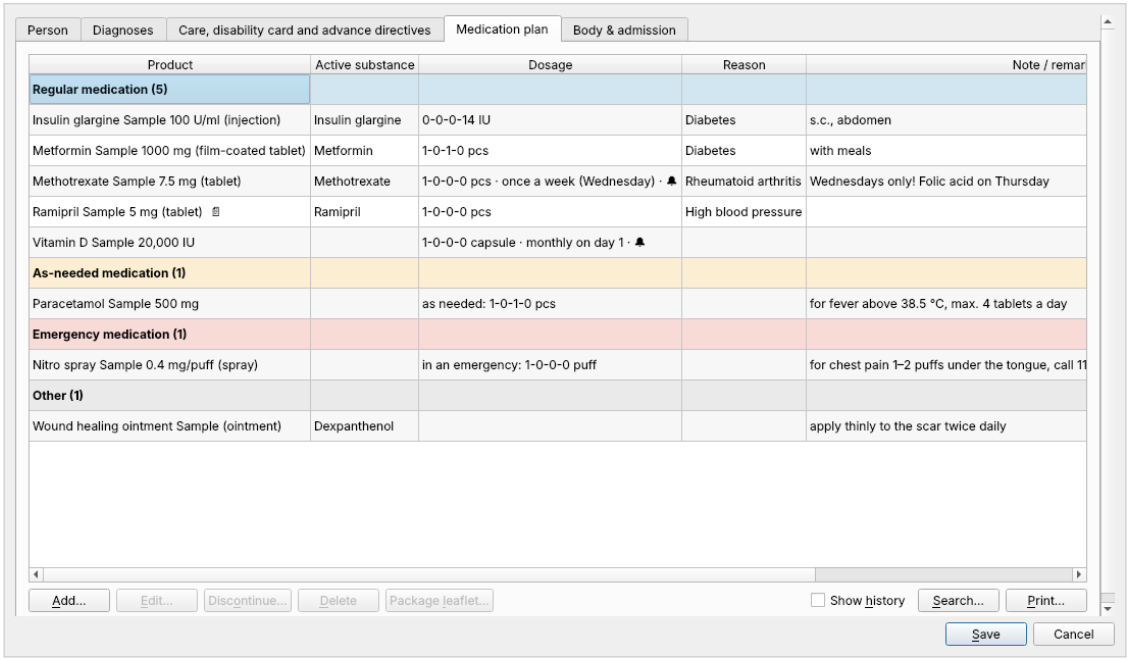


Fig. 8.1: The medication plan with all four groups

Discontinue… discontinues a medicine without deleting it: enter the date (default: today) and, if you like, the reason, e.g. ‘side effect’ or ‘replaced by …’. The medicine moves to the archive; the *Status* column shows ‘discontinued on …’, the reason appears in the tooltip. If the date is in the future – for a planned change, say – the medicine stays in the plan up to the day before (status ‘to be discontinued on …’) and moves to the history automatically on that date; Cancel discontinuation in the same dialog removes the reservation. Discontinued and ended medicines – the history – are shown with *Show history*; Resume brings them back.

Delete… removes a medicine permanently – only useful for incorrect entries.

Package leaflet… looks up the package leaflet (Section 8.4 (p. 56)).

Search… searches for controlled narcotics, painkillers, blood thinners, blood pressure medicines or cortisone-containing products (Section 8.5 (p. 57)).

Print… outputs the medication plan (Section 8.6 (p. 60)).

> **Tip:** Discontinue medicines instead of deleting them. This keeps a record of what was taken and when – often an important question at a doctor’s appointment or after a hospital stay.

Changes to the medication plan only take effect when you click Save in the master record.

8.2 Step by step: entering a medicine

1. On the **Medication plan** tab, click Add… . The medication dialog opens (Fig. 8.2).

2. **Product:** enter the *Brand name* as printed on the pack (required), plus *Active substance*, *Strength* (e.g. '5 mg') and *Dosage form* (tablet, capsule, drops, patch, injection …).

3. **PZN:** the eight-digit Pharmazentralnummer is the German pharmacy product number, printed on German packs next to the barcode. It is optional and only helps when looking up the package leaflet. VitaParms checks the check digit and warns about typing errors. For packs from other countries, simply leave the field empty.

4. **Dosage:** enter the amount for *morning*, *noon*, *evening* and *night* – whole numbers, decimals or fractions such as '½'. A '0' or an empty field means: nothing at this time. Choose the *Unit*: pcs, drops, puff, IU, ml and so on.

5. **Reason / indication** (e.g. 'high blood pressure') sits right next to the PZN.

6. **Type, period and intake:** choose the *Type* (regular, as-needed or emergency medication, or other). Under *Period* enter *from* when *until* when the medicine is taken. Without a tick at *until* it is permanent – this is the default. With an end date, e.g. for an antibiotic, VitaParms shows the duration in days (Fig. 8.3); reminders and the assistant end on that day. Once the end date has passed, VitaParms moves the medicine to the history automatically (status 'ended on …'): it no longer appears in the current medication plan or on the list for the pharmacy. A medicine without an end date stays in the plan until you discontinue it. For regular medication the *Schedule* follows (Section 8.3 (p. 51)); anything else to be observed goes into the *Note*.

7. **Instructions** for taking it (e.g. 'with meals').

8. **Discontinue:** if a medicine is no longer taken, tick *Discontinued* and enter the reason. Without an end date VitaParms enters today. The same works in the list with Discontinue… .

9. Confirm with OK and save the master record.

8.3 Intake schedule and note

Not every medicine is taken daily. For regular medication, VitaParms therefore offers the following schedules:

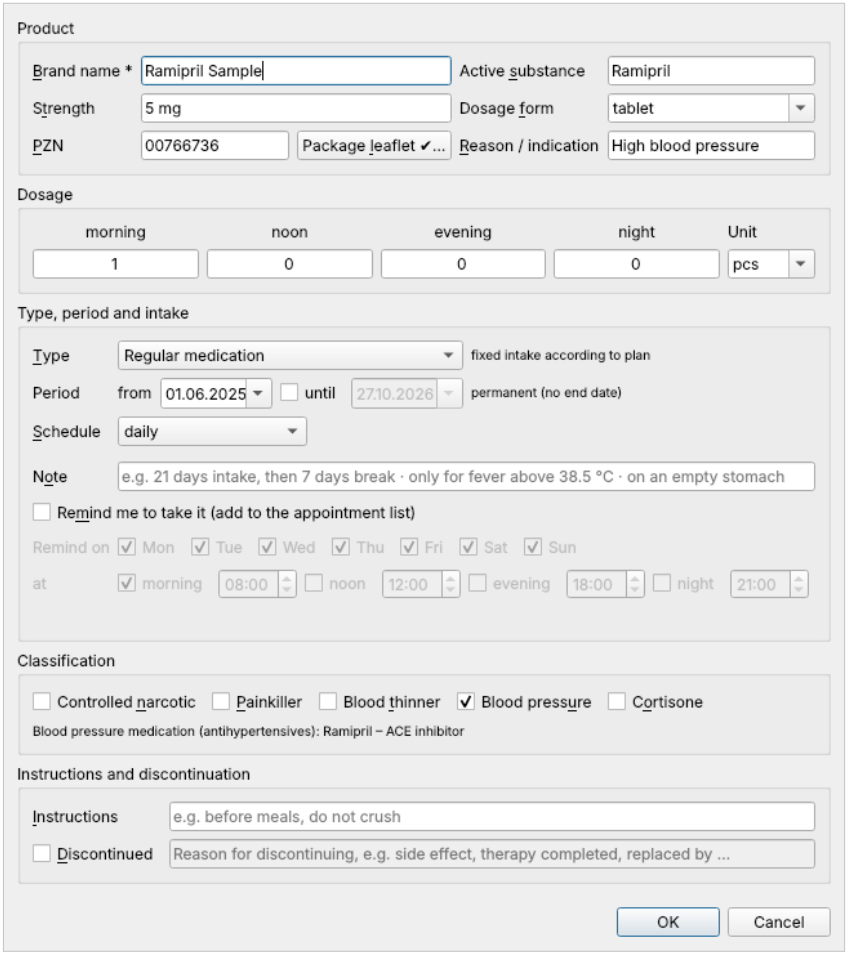


Fig. 8.2: The medication dialog. The tick on Package leaflet ✓... indicates a valid PZN.

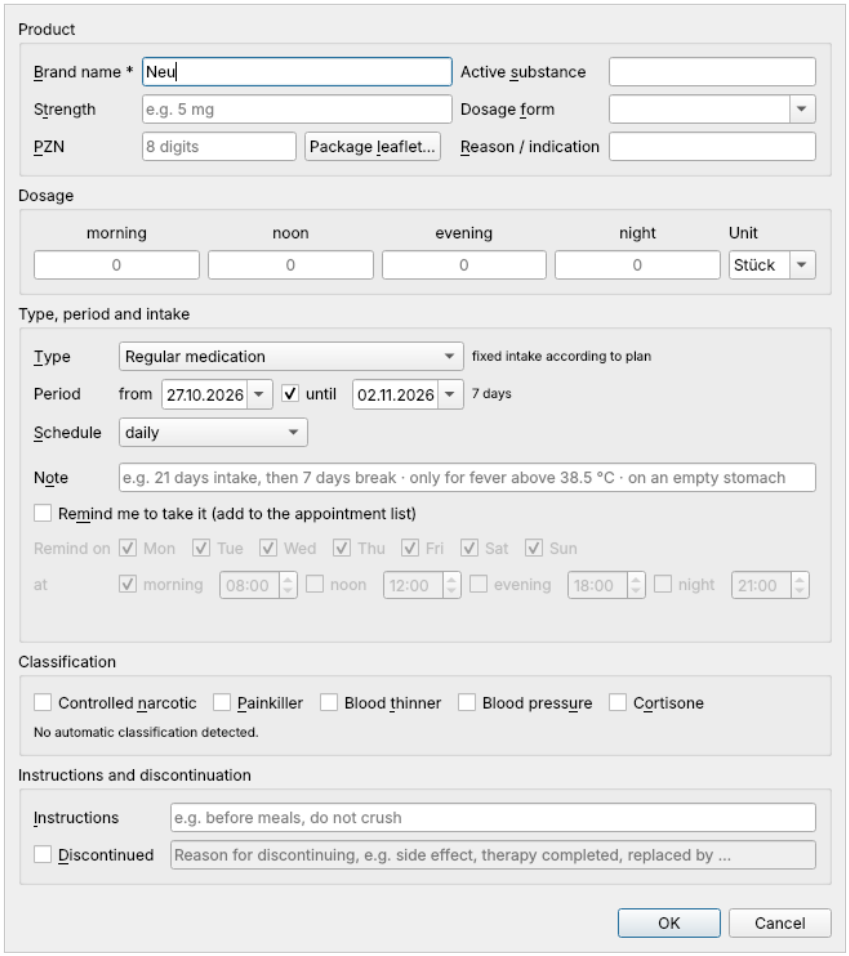


Fig. 8.3: Period from – until: one week, the duration is shown next to it

Schedule	Example
daily	most long-term medicines
every … days	change a pain patch every 3 days
on certain weekdays	methotrexate only on Wednesdays; also ‘every 2 weeks’
monthly	vitamin D on the 1st of each month; also ‘every 3 months’

Counting starts from the start day you enter next to the schedule (Fig. 8.4). Below the fields, VitaParms shows the next intake days – check there that the schedule is correct. The ‘Prepare medication’ assistant only lists such a medicine on the days it is due.

The *Note* adds what cannot be expressed as a schedule: ‘21 days intake, then 7 days break’, ‘Wednesdays only! Folic acid on Thursday’. For as-needed and emergency medication the field is called *Occasion / note*: it says *when* the medicine is given and the maximum amount – for example ‘for fever above 38.5 °C, max. 4 tablets a day’.

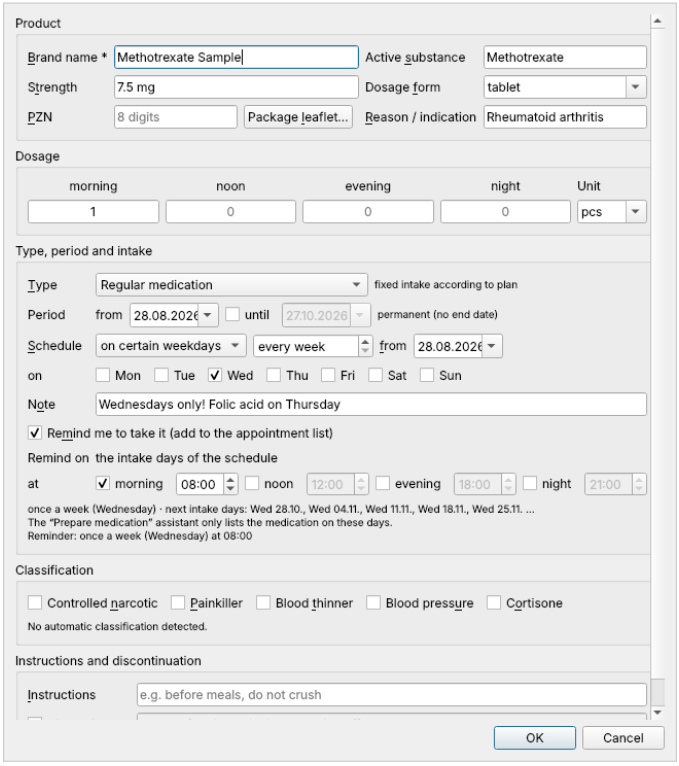


Fig. 8.4: Methotrexate: once a week on Wednesday, with a note and added to the appointment list

Caution: Especially with weekly medicines such as methotrexate, accidental daily intake can be dangerous. Enter the schedule carefully, check the intake days shown and also write important instructions into the note.

Intake reminders

With the tick *Remind me to take it (add to the appointment list)*, VitaParms creates appointments with a reminder in the appointment planner when you save the master record (Fig. 8.5, Fig. 11.5 on p. 83). Reminders are available for regular medication and for 'Other', not for as-needed and emergency medication.

Remind on For daily intake (and for 'Other') choose the weekdays – for example only Monday to Friday if someone else gives the medicines at the weekend. For any other schedule – every … days, on certain weekdays, monthly – VitaParms reminds you on the intake days of the schedule.

at Up to four times: morning, noon, evening and night. The times of day that have an amount in the dosage scheme are preselected with 8 am, 12 pm, 6 pm and 9 pm; each time can be changed. Each time gets its own appointment series 'name strength – amount' with a pill symbol.

Below the fields VitaParms summarises the reminder, e.g. 'Reminder: Mon–Fri at 08:00 and 18:00'; in the medication plan of the master record such medicines are marked with a bell symbol. If you change the medicine, the appointments are renewed from today; if you untick the box, archive or delete the medicine, they end. Past appointments remain. Medicines with a non-daily schedule that were already in the appointment list before version 0.10.2 keep their reminder.

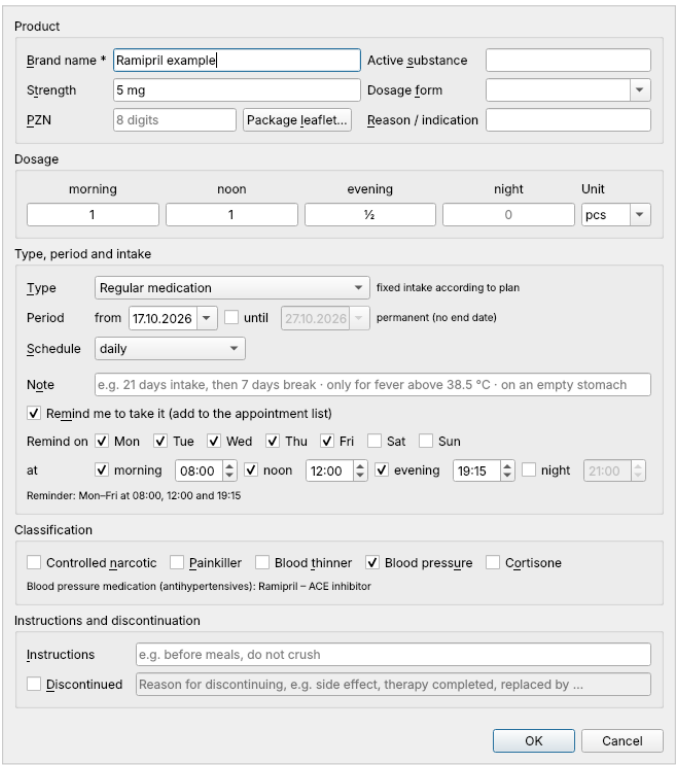


Fig. 8.5: Intake reminder: Monday to Friday at 8:00, 12:00 and 19:15

8.4 Looking up the package leaflet

Package leaflets are easily mislaid. VitaParms helps you find them on the internet and can store them as PDF in the medical record – encrypted and available offline.

1. Select the medicine in the medication plan and click Package leaflet… . (The button is also available in the medication dialog and in the assistant.) The window in Fig. 8.6 opens.

2. Under *Look up*, choose a source:

   - Apotheken Umschau (PZN) uses the PZN to find exactly your product on this German health portal and opens the page in the browser. It usually contains a link to the original package leaflet as PDF.
   - Gelbe Liste (a German drug database) and Web search search by name in the browser.
   - EMA product information loads the product information including the package leaflet directly into the window for medicines authorised centrally in the EU. The EMA publishes it in all EU languages, including English.
   - openFDA (USA, English) shows the US prescribing information – useful as a supplement, but *not* the package leaflet for your country.

3. If you have found the link to the original PDF in the browser, copy it and click Load address… . If you already have the package leaflet as a file, choose From file… . Under Tools » Settings… you can add further sources, for example the websites of your national medicines agency (Section 13.2 (p. 94)).

4. The PDF appears in the window. Save in the medical record stores it as a document of the type ‘Package leaflet’, linked to the medicine. The next time, Package leaflet… opens the stored copy directly.

Privacy: For the lookup, VitaParms only sends name, strength, active substance or PZN of the medicine to the chosen provider – never any details of the person. If you want to run the program completely offline, switch the lookup off in the settings (Section 13.2 (p. 94)).

Caution: The package leaflet enclosed with your medicine always takes precedence. Online versions may be out of date or belong to a different manufacturer.

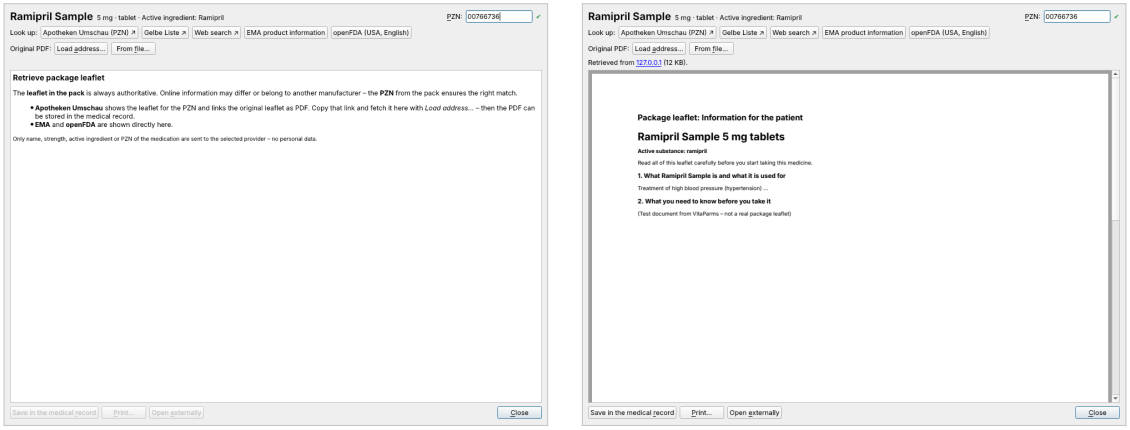


Fig. 8.6: Looking up the package leaflet (left) and the loaded original PDF (right)

8.5 Finding drug groups: narcotics, painkillers, blood thinners, blood pressure, cortisone

Before an operation, at the dentist, when travelling abroad or in the emergency department the question often comes up: 'Are you taking blood thinners?' 'Which painkillers do you take?' or 'Are you on cortisone?' The search answers it at a glance.

1. Choose Person » Search medications and then the group – for controlled narcotics also directly with Ctrl + Shift + B. Alternatively click Search… in the medication plan.

2. The search window lists all relevant medicines of the current plan (Fig. 8.7), arranged by regular, as-needed and emergency medication, each with

   - dosage and intake (schedule or note),
   - *Amount* per day, e.g. '2 pcs per day',
   - *Active amount* per day as strength times amount, e.g. '20 mg per day'; for interval medication 'per intake day', for as-needed medication 'per as-needed dose',
   - *Classification*: why the medicine belongs to the group.

3. Use *Search for* to switch the group or search with a free *Search term*. *include discontinued (archived)* also takes earlier medicines into account.

4. Print… and As PDF… output the list – handy for the anaesthetist or your travel documents.

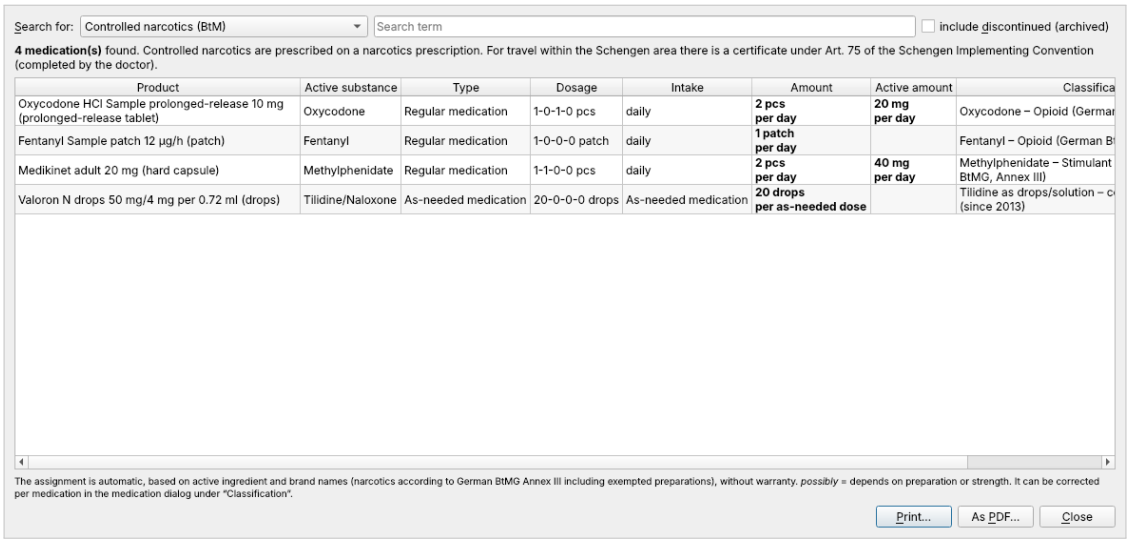


Fig. 8.7: Search for controlled narcotics with amount and active amount per day

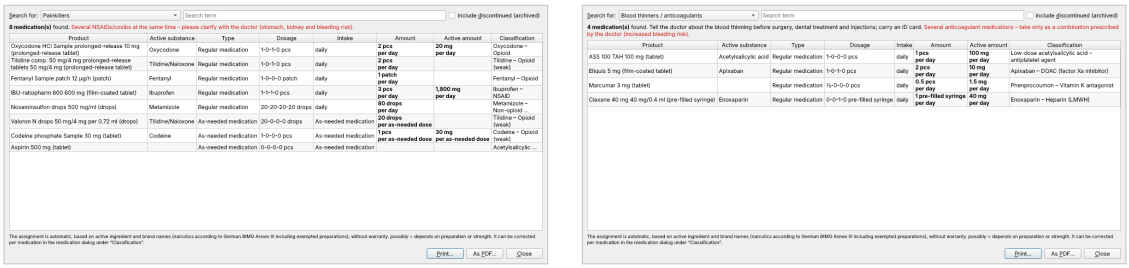


Fig. 8.8: Search for painkillers (left) and blood thinners (right) with warnings

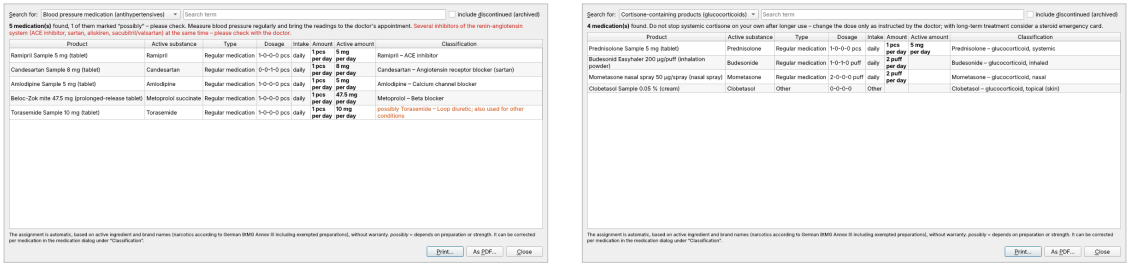


Fig. 8.9: Search for blood pressure medicines (left) and cortisone-containing products (right)

How VitaParms classifies

The classification is based on the active substance *and* the brand name – so VitaParms also recognises 'Eliquis' as apixaban or 'Marcumar' as phenprocoumon. Both German and international (INN) spellings are recognised, e.g. 'Oxycodon' and 'oxycodone', 'Paracetamol' and 'acetaminophen'.

Controlled narcotics (BtM) according to Schedule III of the German Narcotics Act (Betäubungsmittelgesetz, BtMG), including the *exempted preparations* listed there. For example, codeine up to 100 mg per unit is not a controlled narcotic, nor is tilidine as a prolonged-release tablet with naloxone, whereas tilidine drops are. Details in Section B.6 (p. 135).

Painkillers opioids, NSAIDs and coxibs (e.g. ibuprofen, diclofenac), metamizole, paracetamol, triptans and others; also medicines whose reason is 'pain', 'migraine' or 'neuralgia'.

Blood thinners vitamin K antagonists (phenprocoumon, warfarin), direct oral anticoagulants (apixaban, rivaroxaban, edoxaban, dabigatran), heparins, fondaparinux and antiplatelet agents (aspirin up to 160 mg, clopidogrel and others).

Blood pressure ACE inhibitors ('-pril'), sartans ('-sartan'), calcium channel blockers, beta blockers, thiazides, aliskiren and others; also unknown substances with these endings. Loop and potassium-sparing diuretics, alpha blockers, clonidine and minoxidil are also used for other conditions and are therefore marked 'possibly' – as are medicines whose reason is 'high blood pressure'.

Cortisone glucocorticoids such as prednisolone, dexamethasone, budesonide, fluticasone or mometasone. The classification also states the route – *systemic*, *inhaled*, *nasal*, *topical (skin)*, *eyes/ears* or *rectal* – derived from the dosage form (e.g. 'inhalation powder', 'cream', 'eye ointment').

If several NSAIDs or several anticoagulant medicines are taken at the same time, the search window points this out in red (Fig. 8.8); likewise if several inhibitors of the renin-angiotensin system (ACE inhibitor, sartan, aliskiren, sacubitril/valsartan) are entered at the same time (Fig. 8.9). For systemic cortisone VitaParms reminds you not to stop it on your own and to carry a steroid emergency card. 'possibly' marks cases that depend on the preparation or strength.

> **Outside Germany:** The classification as controlled narcotic follows German law. Other countries regulate controlled drugs differently – for example the UK Misuse of Drugs Regulations or the US Controlled Substances Act. Use the 'Classification' tick boxes described below to adapt the result to the rules that apply to you.

Correcting the classification

If VitaParms has classified a medicine wrongly or not at all, correct this in the medication dialog under *Classification* (Fig. 8.10): tick or untick *Controlled narcotic, Painkiller, Blood thinner, Blood pressure* or *Cortisone*. Only your deviation from the automatic detection is stored; below the boxes VitaParms shows how it would have classified the medicine itself.

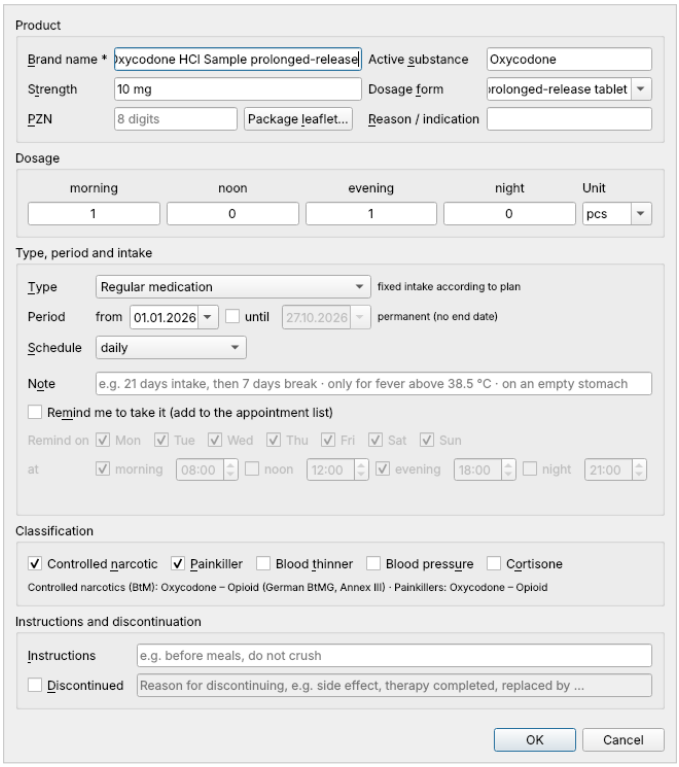


Fig. 8.10: Classifying a medicine as controlled narcotic and painkiller

Legal note: The automatic classification is an aid, not a binding statement. Whether a product falls under narcotics law depends on the active substance, the preparation and the amount per unit, and may change when the law changes. The prescription, the pack and the advice of your pharmacy are authoritative.

8.6 Printing the medication plan

Person » Print » Medication list… , Output » Print medication plan… or Print… in the master record produces a medication plan in landscape format (Fig. 8.12): arranged by regular, as-needed and emergency medication and other, with dosage, schedule, note, reason and instructions. It is a good copy for the fridge, the care folder or your wallet.

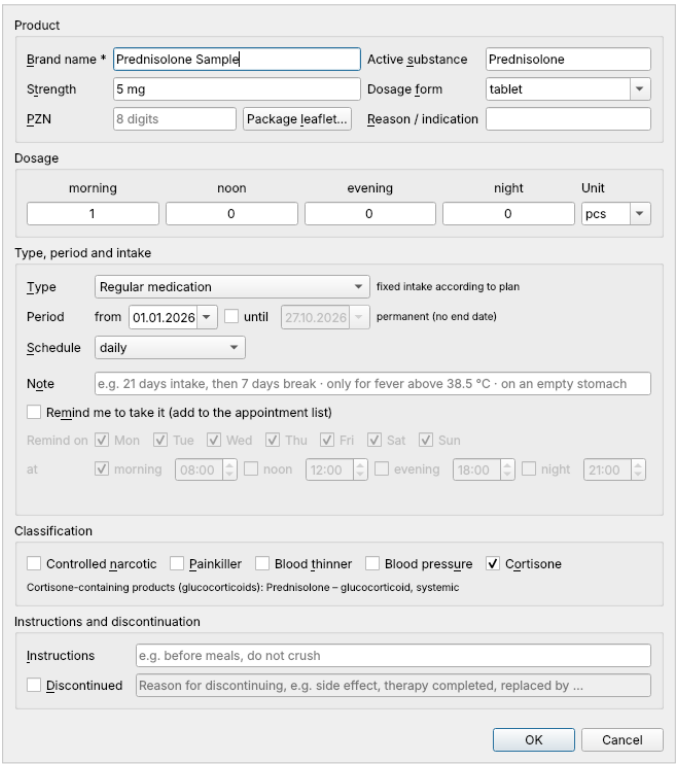
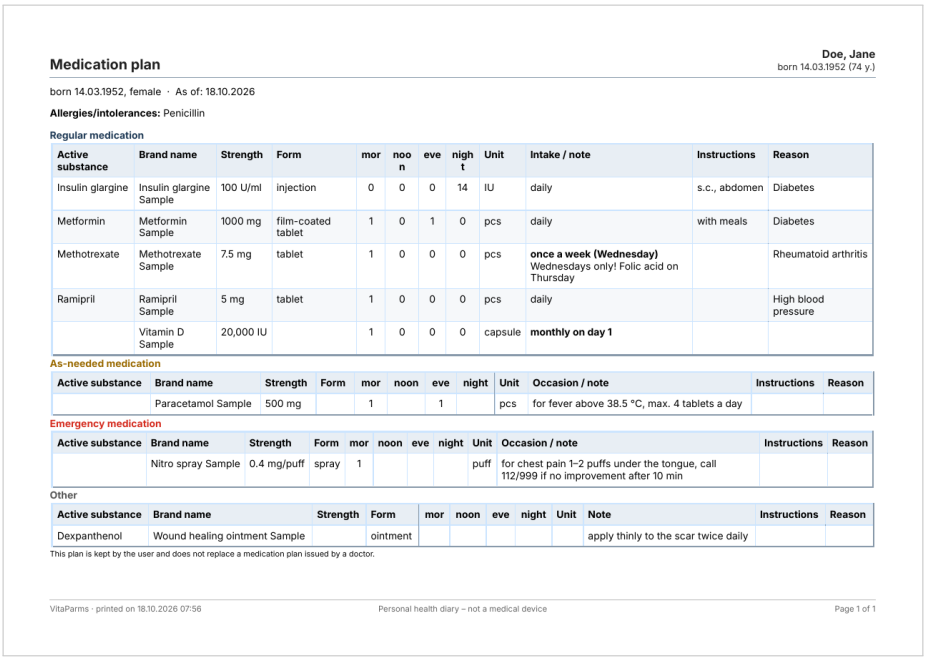


Fig. 8.11: Prednisolone: automatically classified as cortisone-containing (systemic)



Medication plan

Doe, Jane
born 14.03.1952 (74 y.)

born 14.03.1952, female · As of: 18.10.2026

Allergies/intolerances: Penicillin

Regular medication

Active substance	Brand name	Strength	Form	mor	noon	eve	night	Unit	Intake / note	Instructions	Reason
Insulin glargine	Insulin glargine Sample	100 U/ml	injection	0	0	0	14	IU	daily	s.c., abdomen	Diabetes
Metformin	Metformin Sample	1000 mg	film-coated tablet	1	0	1	0	pcs	daily	with meals	Diabetes
Methotrexate	Methotrexate Sample	7.5 mg	tablet	1	0	0	0	pcs	**once a week (Wednesday)** Wednesdays only! Folic acid on Thursday		Rheumatoid arthritis
Ramipril	Ramipril Sample	5 mg	tablet	1	0	0	0	pcs	daily		High blood pressure
	Vitamin D Sample	20,000 IU		1	0	0	0	capsule	**monthly on day 1**		

As-needed medication

Active substance	Brand name	Strength	Form	mor	noon	eve	night	Unit	Occasion / note	Instructions	Reason
	Paracetamol Sample	500 mg		1		1		pcs	for fever above 38.5 °C, max. 4 tablets a day		

Emergency medication

Active substance	Brand name	Strength	Form	mor	noon	eve	night	Unit	Occasion / note	Instructions	Reason
	Nitro spray Sample	0.4 mg/puff	spray	1				puff	for chest pain 1–2 puffs under the tongue, call 112/999 if no improvement after 10 min		

Other

Active substance	Brand name	Strength	Form	mor	noon	eve	night	Unit	Note	Instructions	Reason
Dexpanthenol	Wound healing ointment Sample		ointment						apply thinly to the scar twice daily		

This plan is kept by the user and does not replace a medication plan issued by a doctor.

VitaParms · printed on 18.10.2026 07:56 Personal health diary – not a medical device Page 1 of 1

Fig. 8.12: The printed medication plan

Note: The printout from VitaParms is a convenience copy, not an official medication record. In Germany, for example, patients with statutory health insurance who take at least three prescribed medicines long-term are entitled to a standardised medication plan with a barcode (§ 31a SGB V); other healthcare systems have similar documents. Ask your GP practice.

8.7 Having interactions checked: the list for the pharmacy

VitaParms deliberately does *not* check whether medicines influence each other: a reliable check needs a current, complete drug database and professional judgement – and software that supports treatment decisions would be a medical device. This check belongs in the pharmacy. VitaParms gives the pharmacy everything it needs.

Person » Print » Medication list for the pharmacy… or Output » Medication list for the pharmacy… creates a list in landscape format (Fig. 8.13) with

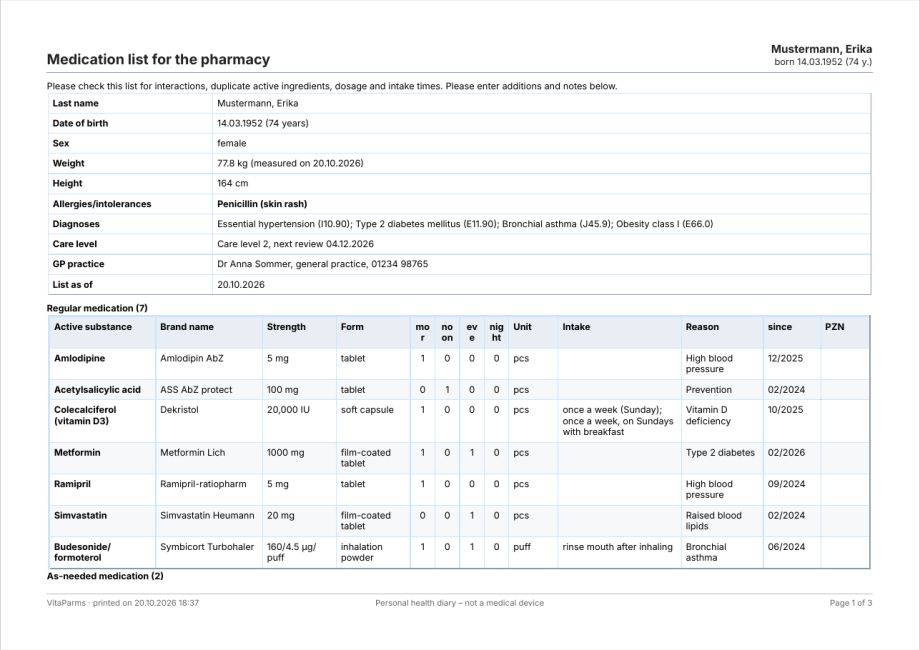
- the details that matter for the check: age, sex, latest weight with date, height, allergies and intolerances (bold), diagnoses, care level and GP practice,
- all current medicines – regular, as-needed and emergency medication and other items – with *active ingredient*, brand name, strength, form, dosage scheme, intake or occasion, reason, 'since' and PZN,
- a checklist of what is often missing (over-the-counter medicines, food supplements, herbal remedies, ointments and drops, medicines from other doctors), with blank rows to add them,
- questions for the pharmacy and space for its notes.

Medication review: In Germany, anyone taking five or more prescribed medicines long-term is entitled to an *extended medication review for polypharmacy* at the pharmacy every twelve months – a pharmaceutical service paid for by the health insurance. The list points this out when the regular medication has five or more entries. Take the packages along to the appointment as well.

The list is also available on the website (**Reports** tab) and in the tablet app (**Medication** → List for the pharmacy). If you enter the PZN of the medicines (Section 8.4 (p. 56)), the pharmacy can identify each product unambiguously.

8.8 Medication stock and orders

So that no medicine runs out, VitaParms can keep a stock: you receive each pack with quantity and expiry date, VitaParms deducts the prepared amount when you prepare the



Medication list for the pharmacy

Mustermann, Erika
born 14.03.1952 (74 y.)

Please check this list for interactions, duplicate active ingredients, dosage and intake times. Please enter additions and notes below.

Last name	Mustermann, Erika
Date of birth	14.03.1952 (74 years)
Sex	female
Weight	77.8 kg (measured on 20.10.2026)
Height	164 cm
Allergies/intolerances	**Penicillin (skin rash)**
Diagnoses	Essential hypertension (I10.90); Type 2 diabetes mellitus (E11.90); Bronchial asthma (J45.9); Obesity class I (E66.0)
Care level	Care level 2, next review 04.12.2026
GP practice	Dr Anna Sommer, general practice, 01234 98765
List as of	20.10.2026

Regular medication (7)

Active substance	Brand name	Strength	Form	mor	noon	eve	night	Unit	Intake	Reason	since	PZN
Amlodipine	Amlodipin AbZ	5 mg	tablet	1	0	0	0	pcs		High blood pressure	12/2025	
Acetylsalicylic acid	ASS AbZ protect	100 mg	tablet	0	1	0	0	pcs		Prevention	02/2024	
Colecalciferol (vitamin D3)	Dekristol	20,000 IU	soft capsule	1	0	0	0	pcs	once a week (Sunday); once a week, on Sundays with breakfast	Vitamin D deficiency	10/2025	
Metformin	Metformin Lich	1000 mg	film-coated tablet	1	0	1	0	pcs		Type 2 diabetes	02/2026	
Ramipril	Ramipril-ratiopharm	5 mg	tablet	1	0	0	0	pcs		High blood pressure	09/2024	
Simvastatin	Simvastatin Heumann	20 mg	film-coated tablet	0	0	1	0	pcs		Raised blood lipids	02/2024	
Budesonide/ formoterol	Symbicort Turbohaler	160/4.5 µg/ puff	inhalation powder	1	0	1	0	puff	rinse mouth after inhaling	Bronchial asthma	06/2024	

As-needed medication (2)

VitaParms · printed on 20.10.2026 18:37 — Personal health diary – not a medical device — Page 1 of 3

Fig. 8.13: The medication list for the pharmacy (first page)

medication, calculates how long the rest will last and reminds you to reorder in good time. Open the stock list with Person › Medication stock and orders… (Ctrl + Shift + L).

How VitaParms calculates

Stock A medicine is tracked as soon as a pack has been received (Receive pack…: quantity, expiry date, note). Packs and movements… shows every pack and every movement; there you can correct the stock after counting and book a pack out as disposed of.

Consumption After every preparation with the assistant (Chapter 9) VitaParms deducts the prepared amount – only once per medicine, day and intake time, even if the same day is prepared again. The pack that expires first is used first. As-needed and emergency medication is not deducted automatically.

Expiry Expired packs do not count as stock. If this leaves too little usable stock of a medicine that is still prescribed, it has to be reordered.

Coverage VitaParms calculates day by day according to the dosage scheme and intake schedule. *lasts until* is the last day for which the stock is sufficient.

Preparing If the medication is prepared weekly (weekly pill box, Chapter 9), the stock must suffice for the whole week on the preparation day – VitaParms then calculates in weeks. The preparation rhythm is saved when preparing and can also be chosen by hand at the bottom of the stock list under *Prepared* (*daily* or *weekly from* a weekday).

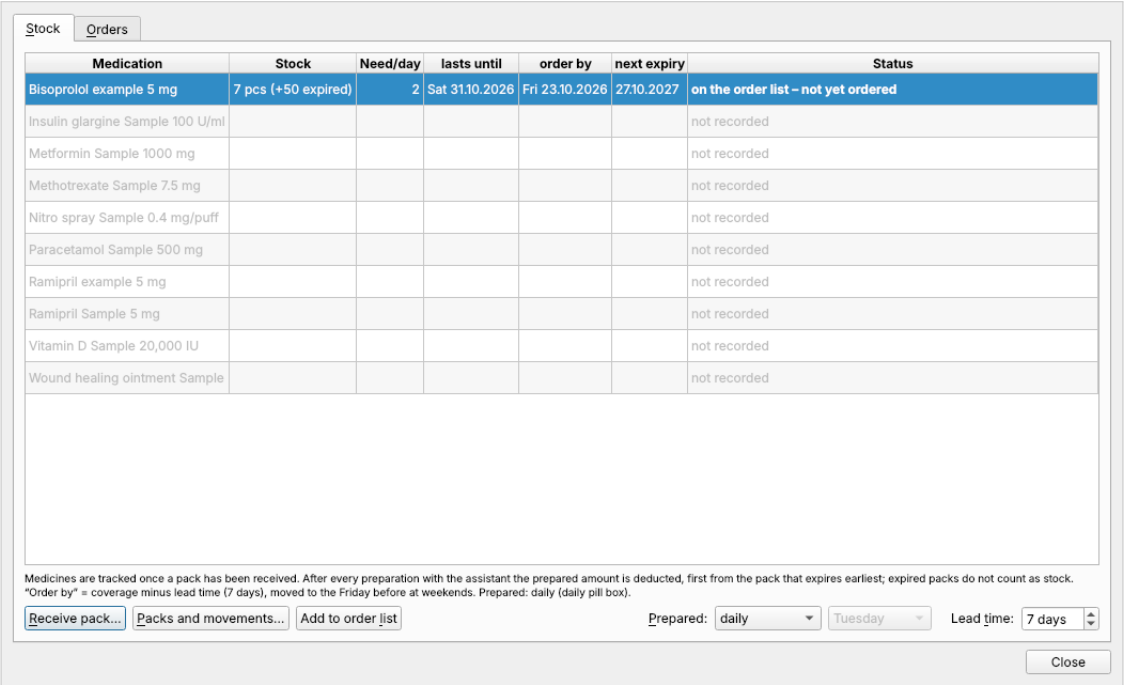


Fig. 8.14: The stock list: stock, need per day, coverage and latest order day

Order by Coverage minus *Lead time* (default 7 days, adjustable at the bottom right) – the time for prescription, order and delivery. If that day falls on a weekend, the Friday before applies; no orders can be placed at weekends, so the stock must last over the weekend.

Order list and reminder

Once the latest order day has been reached or there is no usable stock left, VitaParms puts the medicine on the open order list and reports at start-up and after every preparation, for example: 'Medicine Bisoprolol 5 mg: is almost used up – please get a prescription and order!' (Fig. 8.15). The header of the main window shows it, too.

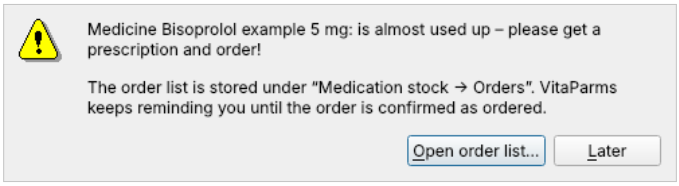


Fig. 8.15: Reminder to reorder

The **Orders** tab (Fig. 8.16) lists all order lists, including older ones – to print again or save as PDF:

1. Print the order list or save it as PDF and request the prescription from the practice.

Change quantity… adjusts the quantity (e.g. ‘1 pack of 100’); VitaParms remembers it for the next order.

2. After ordering, Confirm as ordered… – with date and note. Only then does the reminder stop.

3. When the goods arrive, Book delivery… for each item with quantity and expiry date. The pack is added to the stock; once all items have been delivered, the order is closed.

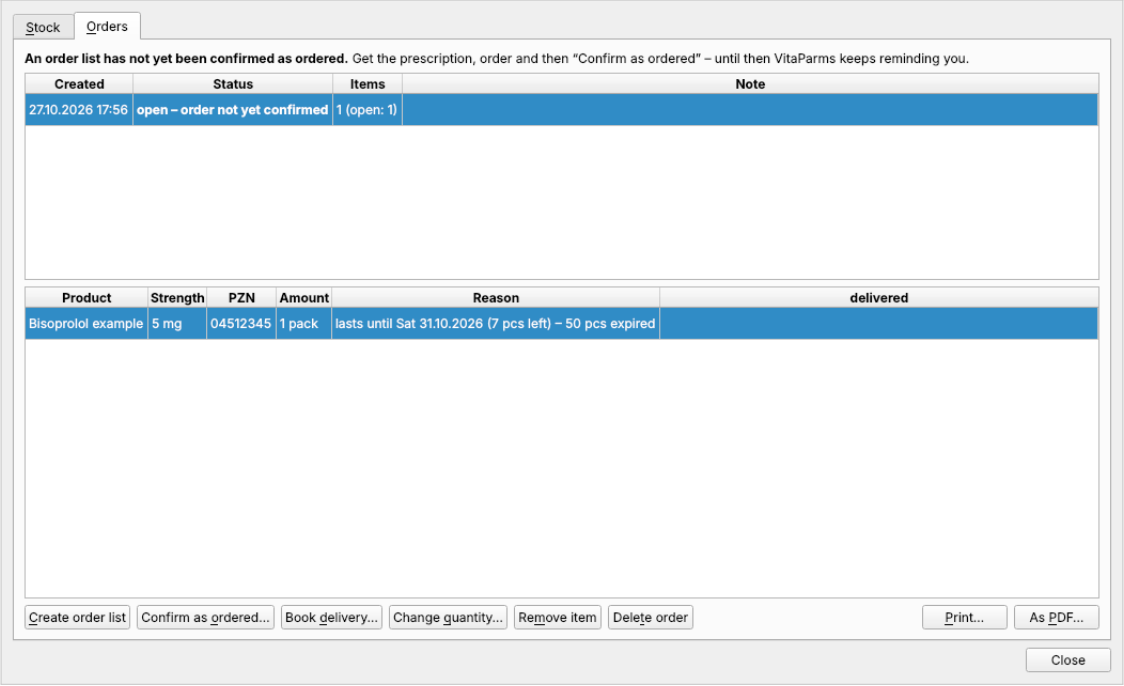


Fig. 8.16: Orders: open list with reasons, confirmation and delivery

Note: The quantities on the order list are suggestions. The prescription is what counts. The stock can be kept on the computer and in the tablet app (section 15.8); synchronisation keeps both up to date.

9 Preparing medication with the assistant

Anyone who fills a weekly pill box for a relative needs both hands and their eyes on the tablets – not on the screen. The ‘Prepare medication’ assistant therefore goes through the medication plan one medicine at a time, shows everything in large type and reads it aloud if you wish. It only moves on when you confirm.

9.1 Step by step: preparing medication

1. Choose the person and start the assistant with Person » Prepare medication (assistant)…, Ctrl + Shift + M or via the toolbar.

2. Check the settings in the top line (Fig. 9.1):

   - *Prepare*: *daily* for a daily pill box or *weekly (7 days)* for a weekly pill box (Fig. 9.5). In weekly mode the assistant shows a table with all seven days for each medicine, reads identical days together (‘All 7 days: morning 1 pcs’) and states the total for the week. Medicines with an intake schedule only appear on their intake days. VitaParms remembers the choice per person and suggests the usual weekday next time.
   - *Time of day*: *all times* or only *in the morning*, *at noon*, *in the evening*, *at night*.
   - *for* or *from*: the day for which you are preparing – in weekly mode the first of the seven days. The assistant takes the intake schedule of each medicine into account.
   - *Voice* and *read aloud automatically* (Section 9.3 (p. 69)).

3. The assistant shows the first medicine: name, strength, dosage form, active ingredient and the dose for each time of day. The fields with a dose are highlighted in blue. Below are the note and the reason.

4. Prepare the medicine and click Next ▶ – or press the space bar, the Enter key or the right arrow key.

5. If you did not catch something, ⟳ Repeat (key R) repeats the same medicine. ◀ Back (left arrow) goes back one medicine.

6 After the last medicine, ‘Medication is prepared’ appears in large letters, accompanied by a signal tone and an announcement (Fig. 9.2). If you like, enter a *Note*, for example ‘Weekly pill box, week 43’.

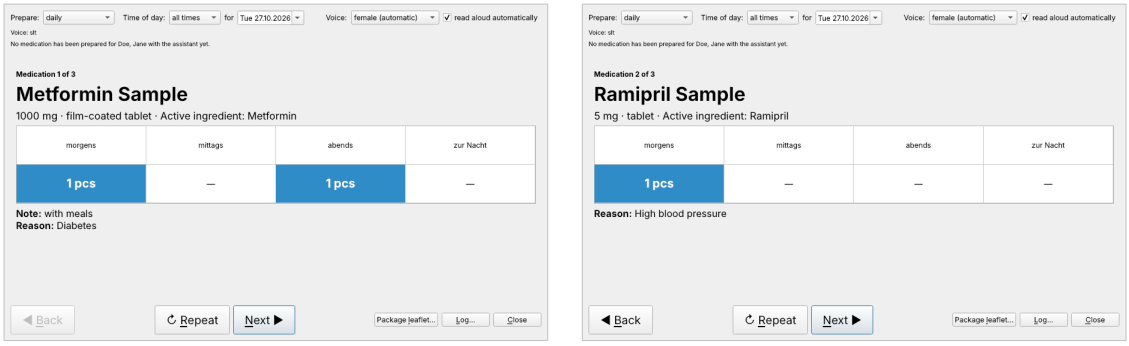


Fig. 9.1: The assistant shows one medicine after another – here the first and the second of three

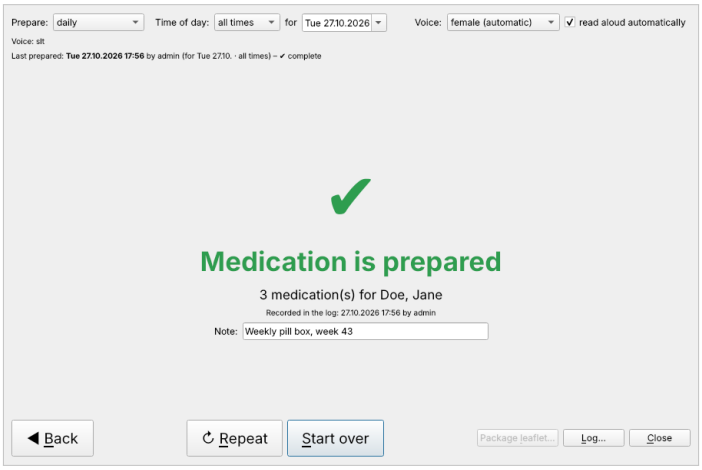


Fig. 9.2: Done: the medication is prepared and recorded in the log

9.2 What the assistant shows – and what it does not

The assistant reads out the **regular medication** that is **due** on the selected day: not archived, between start and end, and – for interval medication – only on its intake days. The order follows the day: what is taken in the morning comes first.

Everything else is shown as information above the medicine (Fig. 9.3):

- **Not due on this day:** interval medication with its next intake day.
- **As-needed medication** and **Other** with their note.
- **Emergency medication** in red.

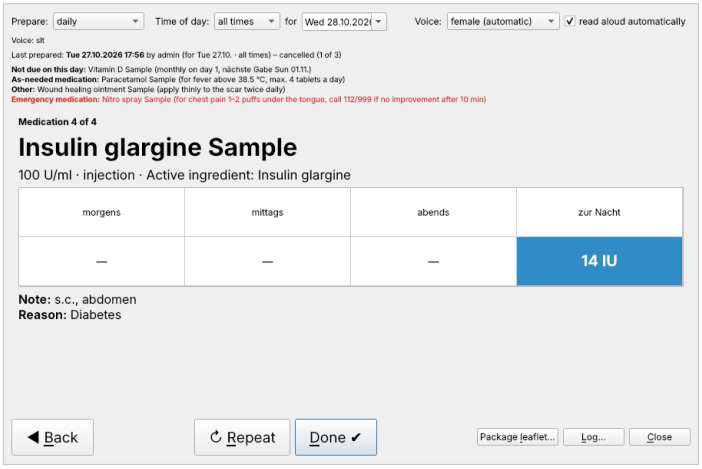


Fig. 9.3: Notes on medication that is not due, as-needed and emergency medication above the current medicine

9.3 Reading aloud and voice

If *read aloud automatically* is ticked, the assistant speaks each medicine: name, strength, dosage form, the dose for each time of day as well as schedule and note. Abbreviations are spelled out – 'mg' becomes 'milligrams', 'IU' 'international units', '½' 'half a', 's.c.' 'under the skin'.

Under *Voice*, choose *female (automatic)* or *male (automatic)* – VitaParms then looks for a suitable voice in the language of the user interface – or one of the voices installed on the computer. The speech rate is set in the settings.

> **Note:** VitaParms speaks with the voices of the operating system. If no voice for the language of the user interface is installed, the pronunciation sounds foreign; if there is no voice of the desired gender, VitaParms changes the pitch of an available voice. In both cases the assistant points this out. You can install additional voices on Windows in the settings under *Time & language* (language pack with text-to-speech), on macOS in the system settings under *Accessibility* → *Spoken Content*, and on Linux via `speech-dispatcher` with `espeak-ng` or `RHVoice`.

9.4 The preparation log

Every run of the assistant is logged: when the medication was prepared, for which day and time of day, by which login account, which medicines with which dose, and your note. If you end the assistant early, the entry appears in the log as 'cancelled (2 of 3)' – medicines that were not confirmed are marked.

The assistant shows at the top when medication was last prepared. Open the complete log with Log… in the assistant or via Person » Medication preparation log… (Fig. 9.4). Choose

the *Period* at the top; clicking an entry shows its details below. Print… and As PDF… output the list. Only administrators may delete entries.

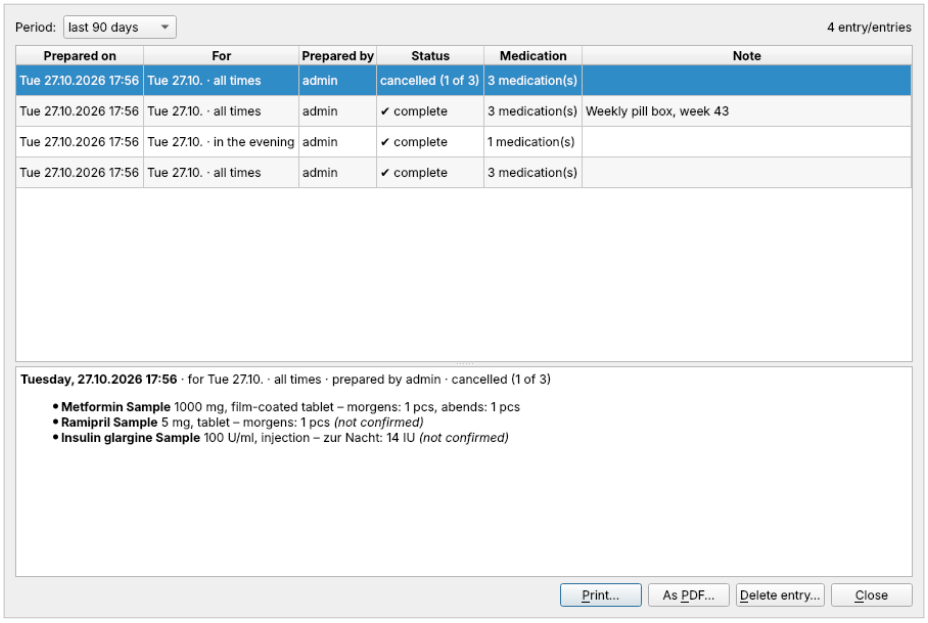


Fig. 9.4: The log with a cancelled run

Tip: When several people share the care, the log creates clarity: who filled this week's pill box? Was the evening dose prepared? Make sure everyone logs in with their own account – otherwise the log shows the wrong name.

Caution: The assistant does not replace the four-eyes principle or checking the pill box before the medication is taken. For every medicine, check that pack, strength and display match.

Stock: If you keep a stock for a medicine (Section 8.8 (p. 62)), VitaParms deducts the prepared amount automatically after preparing – for all seven days with the weekly pill box – and reminds you to reorder as soon as it runs low.

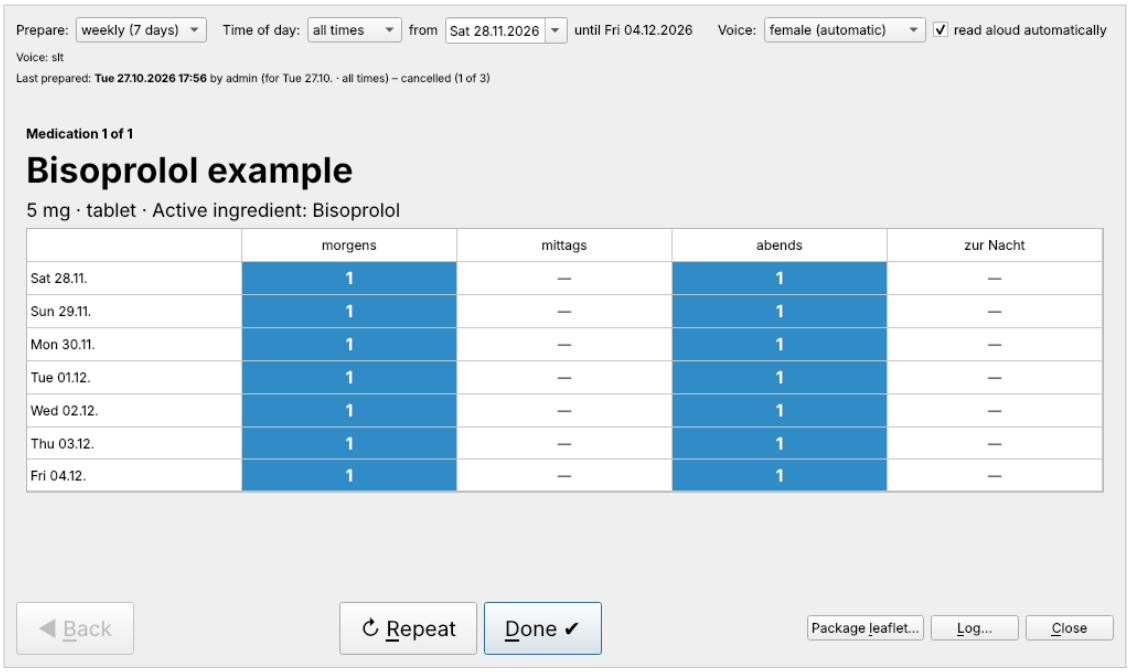


Fig. 9.5: Weekly pill box: seven days at a glance

10 The digital medical record

Doctors' letters, lab results, reports, discharge letters, X-ray reports – a lot of paper accumulates over time. On the **Documents (medical record)** tab you file these documents as PDF, per person and sorted by date. The documents are stored encrypted in the database and are therefore included in every backup.

10.1 Adding documents

1. Open the **Documents (medical record)** tab (Ctrl + 3).
2. Click Add… or choose Medical record › Add documents… (Ctrl + D) and select one or more files. It is even quicker to drag the files from the file manager onto the window with the mouse.
3. For each file the dialog in Fig. 10.1 opens. VitaParms suggests details:
   - *Date* from the file name (`2026-09-30_Lab.pdf`), from the letterhead or from the PDF properties – dates such as '30.09.2026', '30/09/2026', '2026-09-30', '30 September 2026' and 'September 30, 2026' are recognised,
   - *Type* based on keywords in the text (lab results, discharge letter, findings …), in German and English,
   - *Title* from the document or the file name.

   On the right you see a preview of the first page.
4. Check and complete the details, especially *From* (practice, hospital, laboratory) and – if needed – a *Note* with keywords for searching.
5. Click Import. Skip leaves out a file, Cancel import ends the process.

Photos and scans VitaParms converts images in JPG, PNG, TIFF, BMP or WebP format into a PDF when importing them. So you can simply photograph a letter with your smartphone.

Duplicate files are recognised by their content, even if the file name differs, and VitaParms asks what to do.

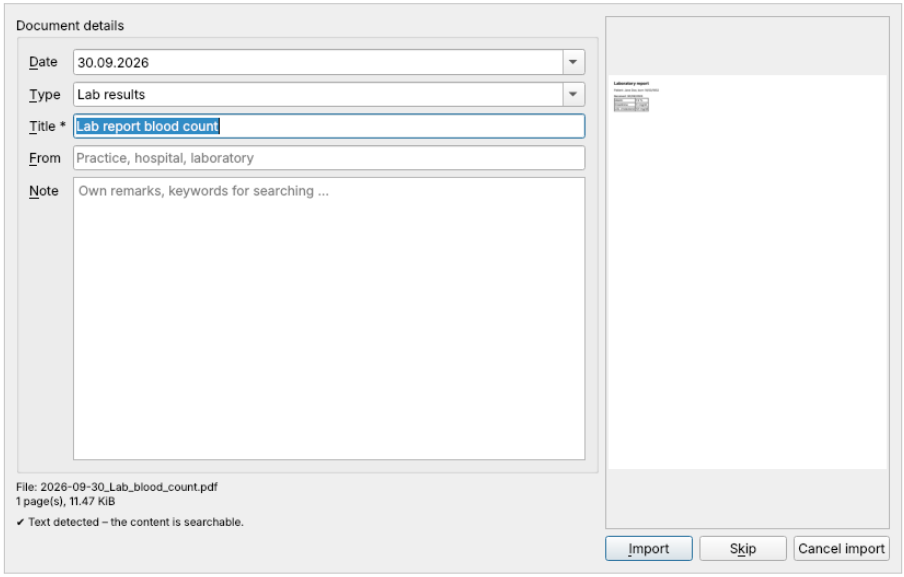


Fig. 10.1: Details of a new document – VitaParms has recognised date, type and title by itself

Document type Predefined are Doctor's letter, Findings, Lab results, Discharge letter, Imaging, Surgery report, Referral, Prescription, Vaccination record, Certificate, Other and – for stored package leaflets – Package leaflet. You can type in your own types at any time.

Change the details later with Edit… or by double-clicking.

10.2 Finding documents

The list shows all documents of the person, newest first, with a preview of the selected document next to it (Fig. 10.2). Use *All types* and *All years* to filter by type and year.

The *Search* field searches title, type, sender, note, file name and date – and, if the PDF contains text, the entire content. If you enter several terms, all must occur. The *Match* column shows the hit in context. In Fig. 10.2, searching for 'LVEF' finds the cardiology letter because the term occurs in the text of the letter.

> **Scanned documents:** A scan or photo is initially just an image; its text cannot be searched (the *Text* column is then empty). VitaParms has no built-in text recognition. Use a scanning app that creates 'searchable PDFs', or write the most important keywords into the note.

10.3 Viewing, saving and printing documents

- Open shows the document in the default PDF viewer.

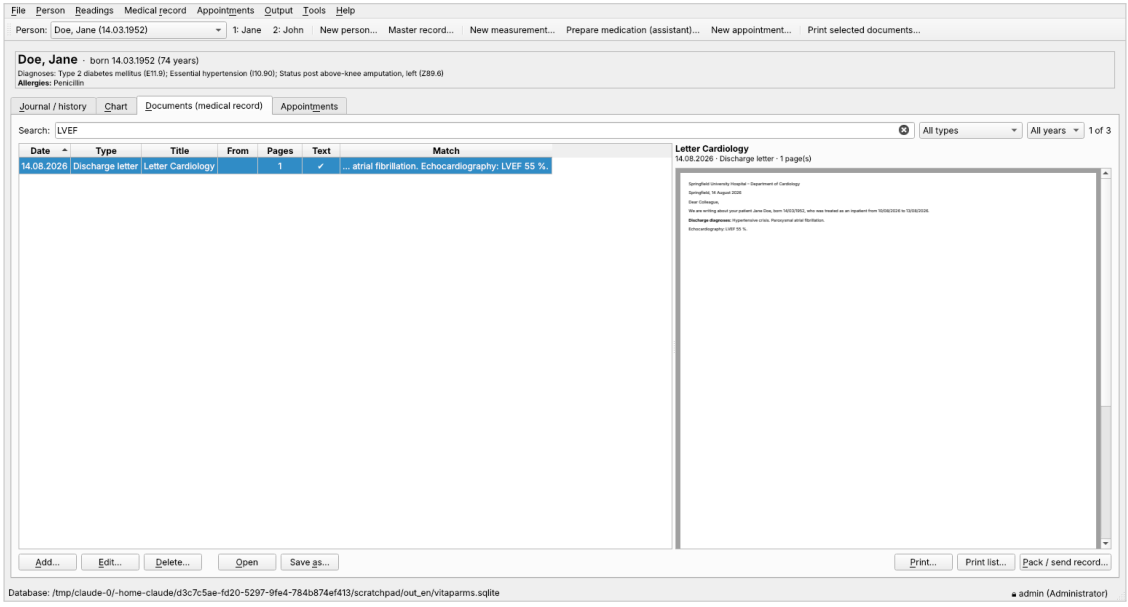


Fig. 10.2: Full-text search in the medical record

- Save as… stores a copy as a file – this copy is *not* encrypted.
- Print… prints all selected documents in the order of the list. Select several documents while holding down Ctrl or Shift.
- Print list… prints a table of contents of the documents shown.
- Delete… removes the document from the database.

> **Documents from the tablet:** Documents photographed or added on the tablet arrive in the record with synchronisation (section 15.7). Their contents are transferred separately from the database; until the content has arrived, the title is followed by '⧖' and the preview says it is still missing. Opening, saving and printing skip such documents.

10.4 Packing and sending the record

If a specialist appointment or a hospital stay is coming up, VitaParms compiles the documents into a ZIP archive – if you wish with master record, medication plan, history and chart, with a table of contents and encrypted with a password.

1. Select or filter the documents you want and click Pack / send record… (Ctrl + Shift + S).

2. Under *Contents* (Fig. 10.3), choose which documents are included: the selected ones, those currently displayed, all or none. Tick which reports VitaParms should create in addition: master record, medication plan, and history and chart of the vital signs for the last days. A *Note for the table of contents* explains to the recipient what the documents are for.

3. Leave *Encrypt archive with a password (AES-256) – recommended* ticked and choose a password – or let Generate create a secure password.

4. Save the archive with Save ZIP… or send it with Send by e-mail…. For this, enter recipient, subject and text; VitaParms opens a new message in your e-mail program with the archive attached. It is only sent when you click ‘Send’ there.

5. Tell the recipient the password by a *different* channel – by phone or text message, never in the same e-mail.

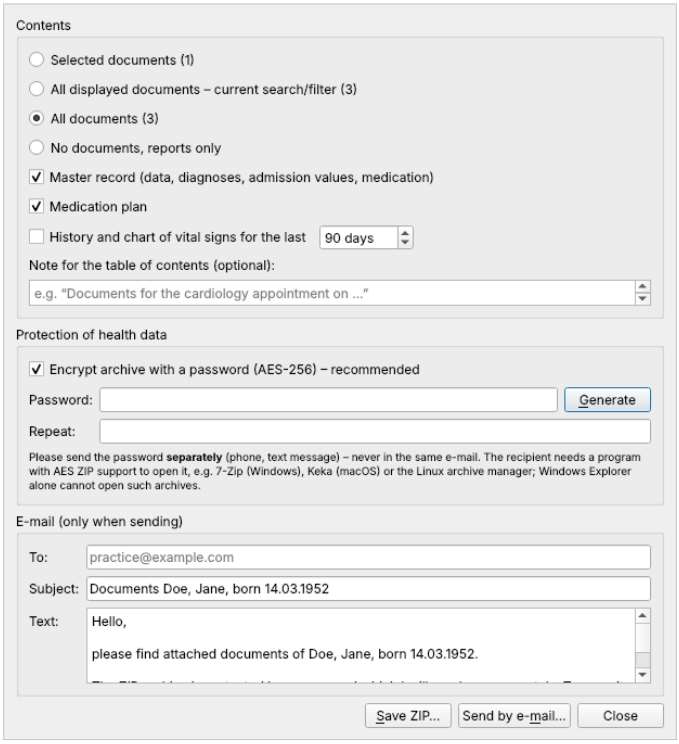


Fig. 10.3: Packing the record: contents, encryption and e-mail

The archive is clearly structured:

```
MedicalRecord_Doe_Jane/
  00_TableOfContents.pdf
  01_MasterRecord.pdf
  02_MedicationPlan.pdf
  03_VitalSigns_History.pdf
```

```
04_TrendChart.pdf
Documents/001_2026-10-18_Imaging_Chest_X-ray_p.a.pdf
          002_2026-09-30_Lab_results_Lab_report_blood_count.pdf
          ...
```

Opening encrypted ZIP archives: VitaParms encrypts with the widely used AES-256 method (WinZip AES). Such archives can be opened with 7-Zip, WinRAR, WinZip, PeaZip, Keka, The Unarchiver and the usual Linux archive tools – *not*, however, with Windows Explorer or the macOS Archive Utility alone. Point this out to the recipient.

If handing over to the e-mail program does not work, VitaParms opens a new message without an attachment and shows the ZIP file in the file manager so that you can attach it by hand. The copy for sending is kept in the folder `Documents/VitaParms-Versand` and can be deleted after sending.

11 Appointments and reminders

Doctor's appointments, physiotherapy, picking up prescriptions, dressing changes: the appointment planner on the **Appointments** tab records everything and reminds you in good time – with a notification window, a gong or by e-mail.

11.1 The appointment planner

Open the **Appointments** tab or choose Appointments » Show appointment planner (Ctrl + 4). On the left you see a calendar, on the right the list of appointments (Fig. 11.1).

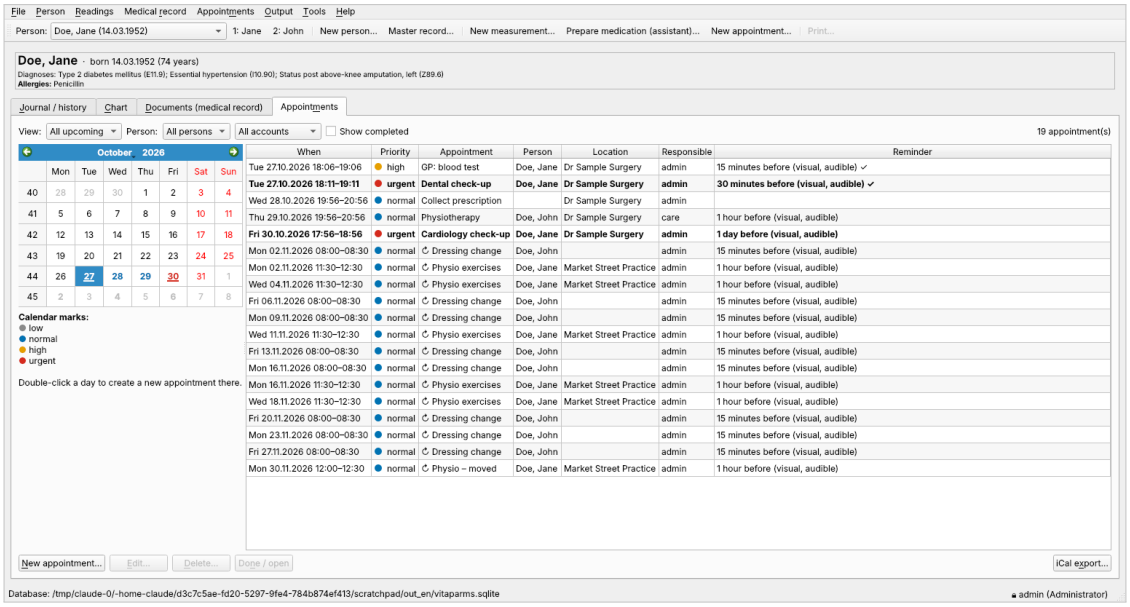


Fig. 11.1: The appointment planner with calendar and list of appointments

View *All upcoming*, *Day*, *Week* or *Month* – relative to the day selected in the calendar – or *All*.

Person all persons or just one. Appointments without a person (such as a team meeting) are possible.

Accounts Administrators see the appointments of all login accounts and can filter by account; users only see their own.

Show completed also shows appointments that have been ticked off.

Calendar Days with appointments carry a coloured mark in the colour of the highest priority: grey for *low*, blue for *normal*, orange for *high*, red for *urgent*. Double-clicking a day creates a new appointment on that day.

The list shows time, priority, appointment, person, location, responsible account and reminder. Appointments that belong to a series are marked with '↻', medication appointments with a pill symbol (Section 8.3 (p. 51)). Urgent appointments are shown in bold.

11.2 Step by step: creating an appointment

1. Choose Appointments › New appointment… (Ctrl + T) or click New appointment…. The appointment dialog opens (Fig. 11.2).
2. Enter a *Title* (required), *Start* and *End*. For appointments without a time, tick *all day*.
3. Add *Location* and *Person*. *Responsible* is the login account that is reminded of the appointment; administrators can choose a different account here.
4. Choose the *Priority* and note under *Note* what to bring – referral, insurance card, 'come fasting'.
5. Under *Reminder*, set the *Advance notice* – from *at the time of the appointment* to *1 week before* – and the type: *visual*, *audible* and/or *by e-mail*.
6. Click Save.

Edit an appointment by double-clicking it or with Edit…. Done / open ticks it off; completed appointments disappear from the list and no longer trigger a reminder.

11.3 Recurring appointments

Recurring appointments are created once as a series. In the appointment dialog, choose under *Recurrence*:

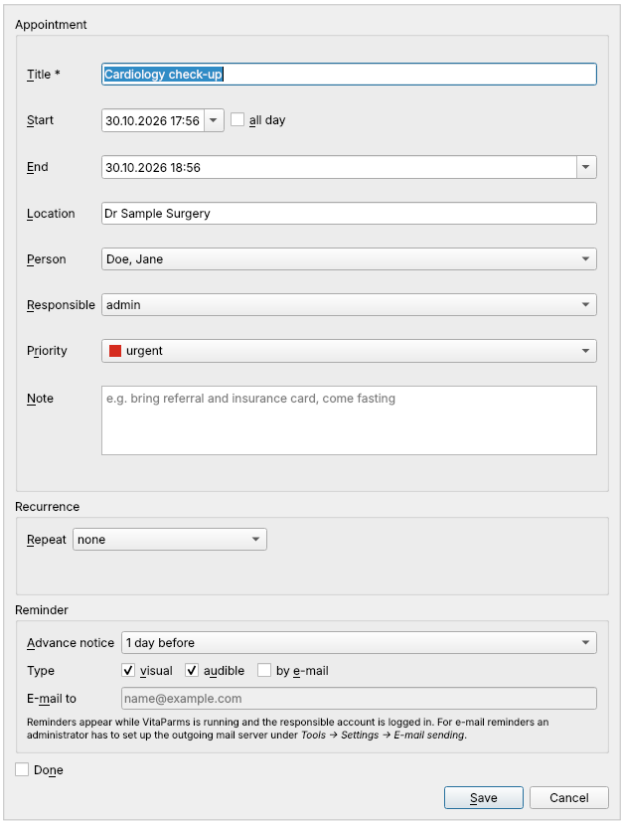


Fig. 11.2: The appointment dialog

Repeat	Options
daily	every day or every *n* days
every … days	every *n* days, e.g. ‘every 2 days’ for a dressing change; 2 days are suggested
on weekdays (Mon–Fri)	every working day
weekly	on any weekdays, every week or every *n* weeks
every … weeks (on weekday)	every *n* weeks on the selected weekdays, e.g. ‘every 3 weeks on Wednesday’; 2 weeks are suggested
monthly	on the same day of the month, on the *n*th weekday (‘on the 2nd Tuesday’) or on the last weekday
yearly	every year or every *n* years

Under *End of series* choose *no end*, *until (inclusive)* a date, or *after a number of appointments*. Below, VitaParms describes in words what you have set and lists the next dates – a quick check that everything is right (Fig. 11.3).

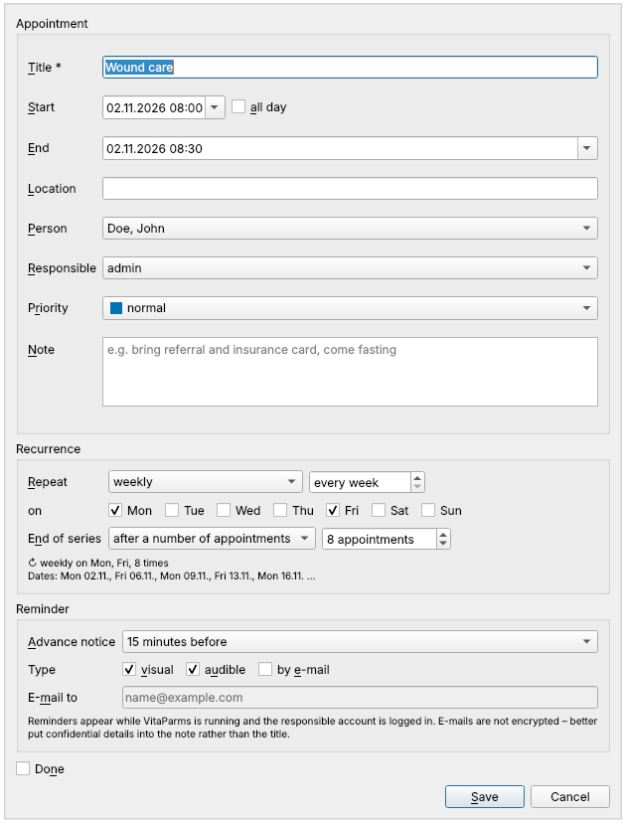


Fig. 11.3: A series: wound care on Mondays and Fridays, eight appointments

Each occurrence of a series is an appointment of its own with its own reminder and its own ‘Done’. Series without an end are created continuously about 13 months in advance. If you change or delete an appointment that belongs to a series, VitaParms asks about the scope (Fig. 11.4):

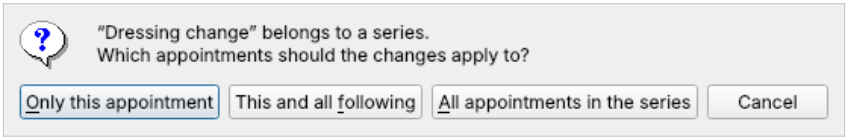


Fig. 11.4: Which appointments should the change apply to?

Only this appointment – for example if physiotherapy is postponed once.

This and all following – for example if the time changes from now on. Past appointments stay unchanged.

All appointments in the series – for example to correct a typing error in the title.

11.4 Medication appointments

Medicines with the tick *Remind me to take it* are added to the appointment planner as series, one per intake time (Section 8.3 (p. 55)). You can recognise them by the pill symbol in front of the name (Fig. 11.5). These appointments are maintained via the medication plan: if you change the schedule, days or times there, the series adapt when the master record is saved.

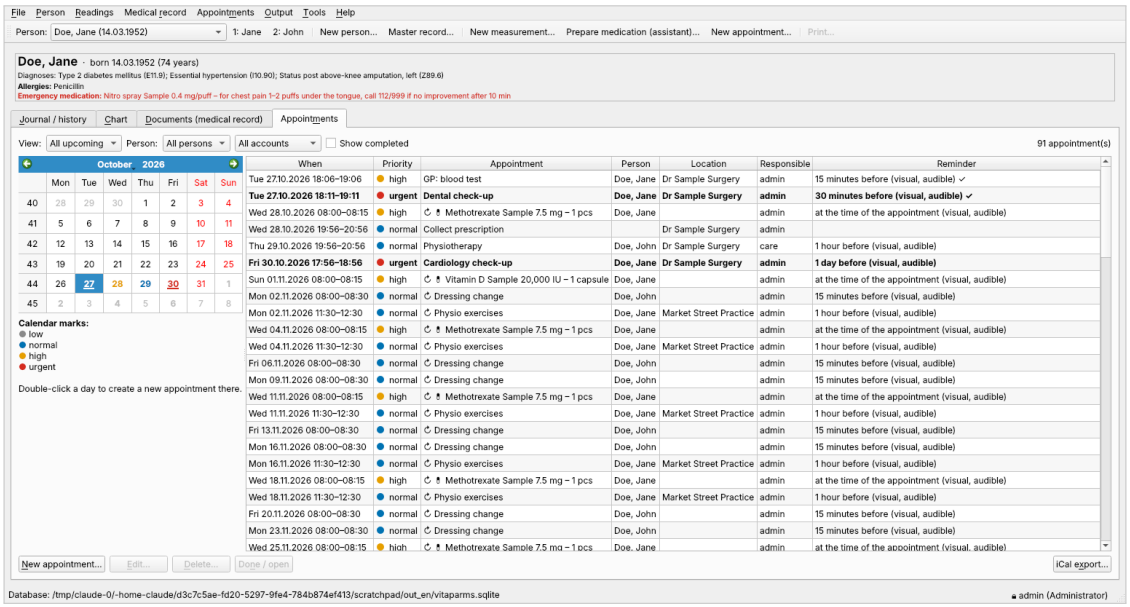


Fig. 11.5: Medication appointments for methotrexate (Wednesdays) and vitamin D (monthly) among the other appointments

11.5 Reminders

At the set time, VitaParms reminds you of the appointment:

visual A notification window appears at the bottom right of the screen, always on top, with a coloured bar in the colour of the priority (Fig. 11.6). Remind me later reminds you again after 5 to 60 minutes, Open… shows the appointment, Done ticks it off.

audible A gong sounds.

by e-mail VitaParms sends an e-mail to the address entered. For this, an administrator must set up e-mail sending once (Section 13.4 (p. 95)).

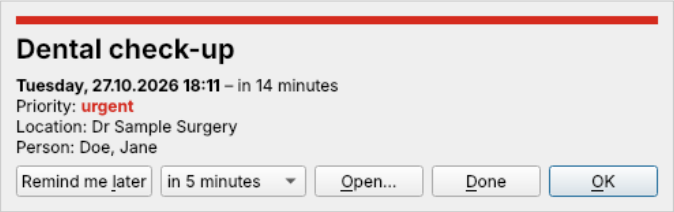


Fig. 11.6: Reminder of an urgent appointment

Note: Reminders appear as long as VitaParms is running and the responsible account is logged in. If the program was closed, VitaParms catches up on missed reminders up to six hours after the start of the appointment the next time you log in. Appointments » Check reminders now triggers the check immediately. If you want reliable reminders even when the computer is switched off, also transfer the appointments to your smartphone calendar (next section).

11.6 Change of quarter: having the insurance card read

From two days before until two days after each change of quarter (1 January, 1 April, 1 July, 1 October), VitaParms reminds you by itself to have your health insurance card read at the practices you use (Fig. 11.7). During this time the same message is shown at the top of the appointment planner.

Done – stop reminding confirms the reminder for this quarter.

Remind me tomorrow shows it again the next day (at most until two days after the change).

Close shows it again at the next program start.

The reminder can be switched off under Tools » Settings… (*Remind me of the change of quarter*).

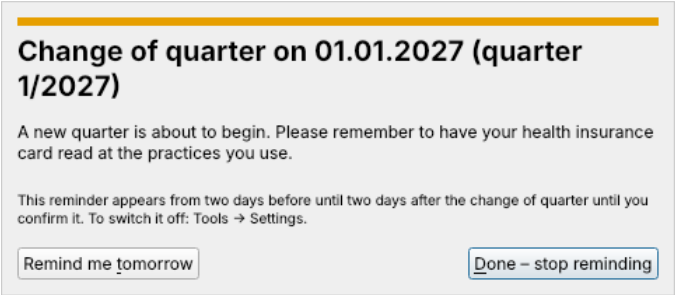


Fig. 11.7: Reminder of the change of quarter

11.7 Appointments at start-up

So that no appointment is overlooked, VitaParms shows the open appointments of the selected person for **today and tomorrow** straight after logging in – including medication appointments (Fig. 11.8). Appointments earlier today that have already passed are shown in grey. If there are no appointments, the window does not appear.

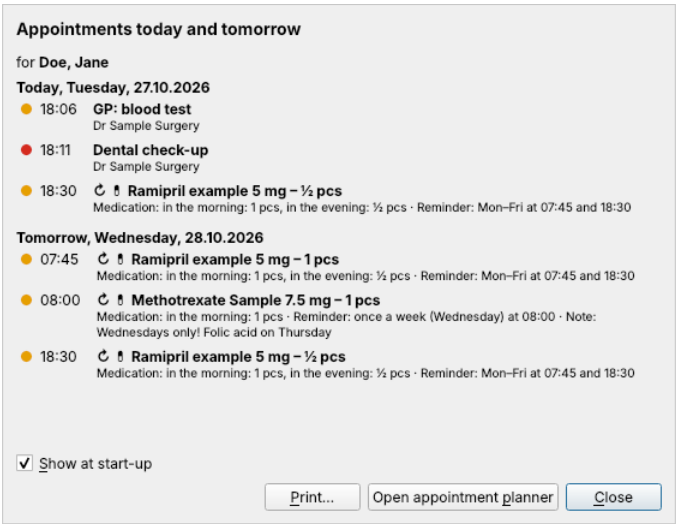


Fig. 11.8: Today's and tomorrow's appointments at start-up

- Print… opens the reminder printout (next section) for today and tomorrow.
- Open appointment planner switches to the **Appointments** tab.
- If you untick *Show at start-up*, the window no longer appears; you can switch it on again under Tools » Settings….
- Show it again at any time: Appointments » Appointments today and tomorrow….

The person shown is the one selected at start-up – the person edited last, or main person 1. Users only see appointments for which their account is responsible.

11.8 Printing from the ‘Person’ menu

Person › Print brings together the three printouts most often needed for a person (Fig. 11.9):

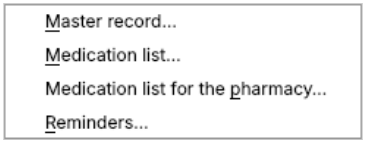


Fig. 11.9: The ‘Print’ submenu in the ‘Person’ menu

Master record... personal details, diagnoses, admission values and medication (Chapter 5).

Medication list... the medication plan, arranged by regular, as-needed and emergency medication (Section 8.6 (p. 60)).

Reminders... the upcoming, still open appointments of the person.

Step by step: printing reminders

1. Choose the person and then Person › Print › Reminders... (Fig. 11.10).
2. Choose the *Period*: today and tomorrow, the next 7, 14, 30 or 90 days, or all upcoming appointments. Below, VitaParms shows how many appointments that is.
3. Leave *Include medication appointments (intake reminders)* ticked if the medication intake reminders should also be listed.
4. Click Preview / print... or As PDF....

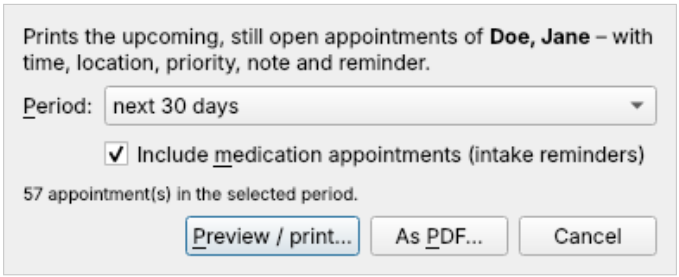


Fig. 11.10: Printing reminders: period and medication appointments

The printout is arranged by day and shows for each appointment the time, title with note, priority, location, the responsible account and the reminder set (Fig. 11.11) – handy for the notice board, the care folder or relatives without access to VitaParms.



Upcoming appointments

Doe, Jane
born 14.03.1952 (74 y.)

Period: 18.10.2026 to 16.11.2026 · 14 open appointment(s)

Time	Appointment	Priority	Location	Responsible	Reminder
Today, Sunday, 18.10.2026					
08:06–09:06	GP: blood test	high	Dr Sample Surgery	admin	15 minutes before (visual, audible)
08:11–09:11	**Dental check-up**	urgent	Dr Sample Surgery	admin	30 minutes before (visual, audible)
Tomorrow, Monday, 19.10.2026					
11:30–12:30	Physio exercises	normal	Market Street Practice	admin	1 hour before (visual, audible)
Wednesday, 21.10.2026					
07:56–08:56	**Cardiology check-up**	urgent	Dr Sample Surgery	admin	1 day before (visual, audible)
08:00–08:15	Methotrexate Sample 7.5 mg Medication: in the morning: 1 pcs Schedule: once a week (Wednesday) Note: Wednesdays only! Folic acid on Thursday	high		admin	at the time of the appointment (visual, audible)
11:30–12:30	Physio exercises	normal	Market Street Practice	admin	1 hour before (visual, audible)
Wednesday, 28.10.2026					
08:00–08:15	Methotrexate Sample 7.5 mg Medication: in the morning: 1 pcs Schedule: once a week (Wednesday) Note: Wednesdays only! Folic acid on Thursday	high		admin	at the time of the appointment (visual, audible)
11:30–12:30	Physio exercises	normal	Market Street Practice	admin	1 hour before (visual, audible)
Sunday, 01.11.2026					
08:00–08:15	Vitamin D Sample 20,000 IU Medication: in the morning: 1 capsule Schedule: monthly on day 1	high		admin	at the time of the appointment (visual, audible)
Monday, 02.11.2026					
11:30–12:30	Physio exercises	normal	Market Street Practice	admin	1 hour before (visual, audible)
Wednesday, 04.11.2026					
08:00–08:15	Methotrexate Sample 7.5 mg Medication: in the morning: 1 pcs Schedule: once a week (Wednesday) Note: Wednesdays only! Folic acid on Thursday	high		admin	at the time of the appointment (visual, audible)
11:30–12:30	Physio exercises	normal	Market Street Practice	admin	1 hour before (visual, audible)
Wednesday, 11.11.2026					
08:00–08:15	Methotrexate Sample 7.5 mg Medication: in the morning: 1 pcs Schedule: once a week (Wednesday) Note: Wednesdays only! Folic acid on Thursday	high		admin	at the time of the appointment (visual, audible)
Monday, 16.11.2026					
12:00–12:30	Physio – moved	normal	Market Street Practice	admin	1 hour before (visual, audible)

VitaParms · printed on 18.10.2026 07:56 Personal health diary – not a medical device Page 1 of 1

Fig. 11.11: Printout of the upcoming appointments

11.9 Transferring appointments to other calendars (iCalendar)

With Appointments » Export appointments (iCalendar)... or iCal export... you save appointments as an `.ics` file. Outlook, Thunderbird, Apple Calendar, Google Calendar and the calendar apps of smartphones understand this standard format.

1. Choose the *Scope*: the selected appointment or its series, or all appointments in the current view (Fig. 11.12).
2. Set the *Options*: include completed appointments, export advance notices as calendar alarms, include location, person and note.
3. Click Save... or Send by e-mail... and import the file into your calendar program.

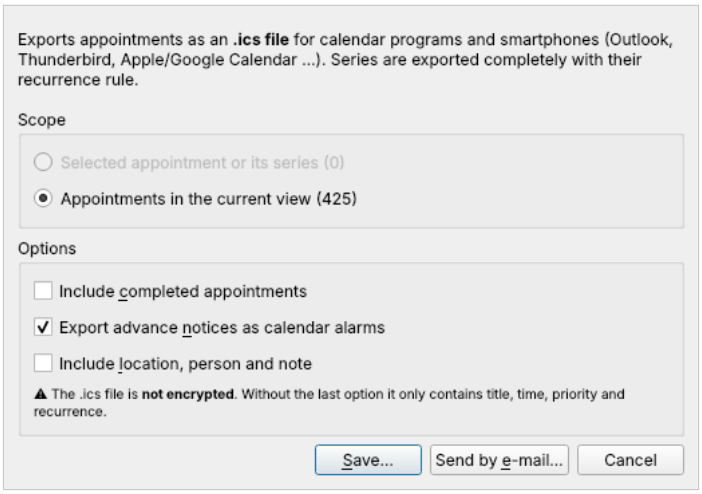


Fig. 11.12: Exporting appointments as an iCalendar file

Series are transferred as a real recurrence rule, individually changed or deleted appointments as exceptions, and daylight saving time is handled correctly. If you import the same appointments again later, the calendar program updates them instead of duplicating them.

> **Caution:** The `.ics` file is *not* encrypted. Without the last option it only contains title, time, priority and recurrence. Choose titles that reveal nothing confidential to third parties if you transfer appointments to a cloud calendar.

12 Accounts, security and backups

Health data is among the most sensitive data of all. VitaParms therefore protects it on several levels: the database is fully encrypted, access is only possible through login accounts, and the program can lock itself automatically. This chapter explains how to manage accounts and make backups – and where the limits of the protection lie.

12.1 Login accounts and roles

Several people can work with the same database, each with their own account and password. Each account has one of two roles:

Role	may
User	create and edit persons; enter readings, medication, documents and appointments; print, export, send the record; change their own password; change settings
Administrator	in addition: manage login accounts, show the recovery key, delete persons, create and open databases, create and restore backups, customise forms, delete entries in the preparation log, see the appointments of all accounts

Step by step: creating another account

1. Log in as an administrator and choose Tools » Login accounts… . The account manager shows all accounts with role and creation date (Fig. 12.1).
2. Click New account… .
3. Enter *Account name* and *Role* and set a provisional password (Fig. 12.1).
4. Give the password to the person in person. After the first login they can change it under Tools » Login accounts… with Change own password… .

Further buttons in the account manager: Set password… gives another account a new password, Switch role turns a user into an administrator and vice versa, Delete… removes an account, and Show recovery key… shows the key again – for example if the printout has been lost. There must always be at least one administrator account.

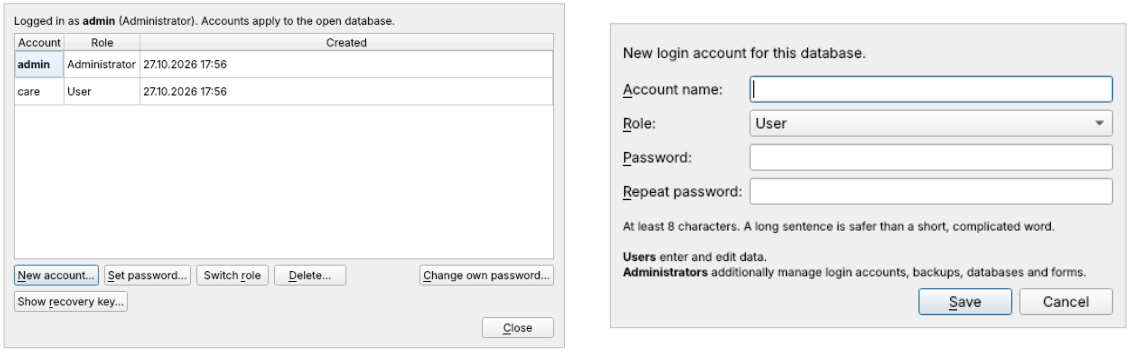


Fig. 12.1: Account manager (left) and new account (right)

> **Tip:** A long sentence makes a better password than a short, complicated word: ‘Granny drinks two cups of coffee every morning’ is easier to remember and harder to guess than ‘X7#kq!’.

12.2 Backups

A hard disk can fail, a laptop can be stolen. Make backups regularly and keep them separate from the computer, for example on a USB stick.

Creating a backup

1. Choose File › Create backup…
2. Choose folder and file name, for example on a USB stick the file `VitaParms-Backup-2026-10-18.sqlite`.
3. VitaParms creates two files: the encrypted database (`….sqlite`) and the key file with the accounts (`….sqlite.keys`). Both belong together.

The backup is encrypted and compact: space freed by deleted documents is no longer contained in it.

Restoring a backup or importing a database

1. Choose File › Restore backup / import… and the `.sqlite` file of the backup or a transfer file `.vpdb`. The same is possible right *at login* with Import database… and at the first start (no database yet) with Import database… instead of a new database.

2. Confirm the prompt. As a precaution, VitaParms keeps the previous state as a copy with the extension `.vor-import-...` ('before import') and then replaces the database.

3. Log in with an account *of the backup*. If the key file of the backup is missing, you need the recovery key.

File » Export database (for another device)... saves the database including the login accounts in *one* file (`.vpdb`) – handy for taking it to another computer or the tablet without a website, via USB cable or your own cloud storage (Section 15.3 (p. 110)). The content remains encrypted.

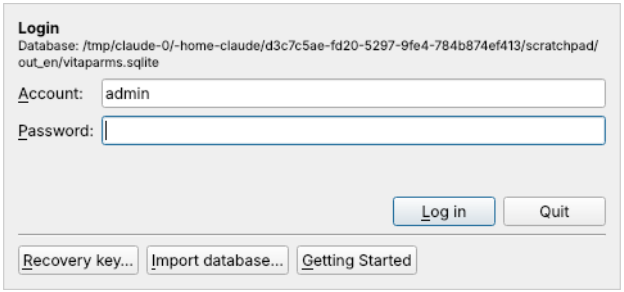


Fig. 12.2: The login window with Import database...

> **Caution:** A backup that has never been tested is not a backup. From time to time, check on a second computer or with File » Open database... that your backup can be opened.

12.3 Several databases

Normally one database is enough for all persons. If you want to keep data strictly separate – for example your own readings and those of the parents you look after – create another one with File » New database.... Each database has its own accounts and its own recovery key. Use File » Open database... to switch between them; VitaParms remembers the database opened last.

12.4 How VitaParms protects your data

Encrypted database The entire database including all documents is encrypted with ChaCha20-Poly1305 (SQLite3 Multiple Ciphers). Deleted content is overwritten, temporary data is only kept in memory.

Key file The random 256-bit database key is stored for each account in the `.keys` file, protected by the account's password (PBKDF2 with 200,000 iterations, AES-256,

HMAC-SHA256). If the list of accounts is manipulated from outside, VitaParms notices and only grants user rights.

Recovery key It is the database key itself, in readable form with a checksum. It also works without the key file.

Automatic lock After 10, 20, 30 or 60 minutes without use, VitaParms logs out (Section 13.2 (p. 94)).

Limits of the protection – stated honestly

- Encryption protects against people *without* an account – for example if a laptop is lost or someone copies the file. The roles, by contrast, are a rule within VitaParms: anyone who can unlock the database with a user account could read it completely with their own tools. So only give accounts to people you would entrust with the data.
- Exports – PDF, Excel, `.ics`, unencrypted ZIP archives, documents stored with Save as… – are not encrypted. The same applies to files that VitaParms passes to other programs for viewing.
- E-mails are usually not end-to-end encrypted.
- A computer with malware can record what you type. Keep your operating system and virus protection up to date and lock the screen when you leave the computer.

13 Settings and customisation

Under Tools » Settings… you adapt VitaParms to the way you work (Fig. 13.1). Most changes take effect immediately; only changing the language requires a restart, which VitaParms offers to do.

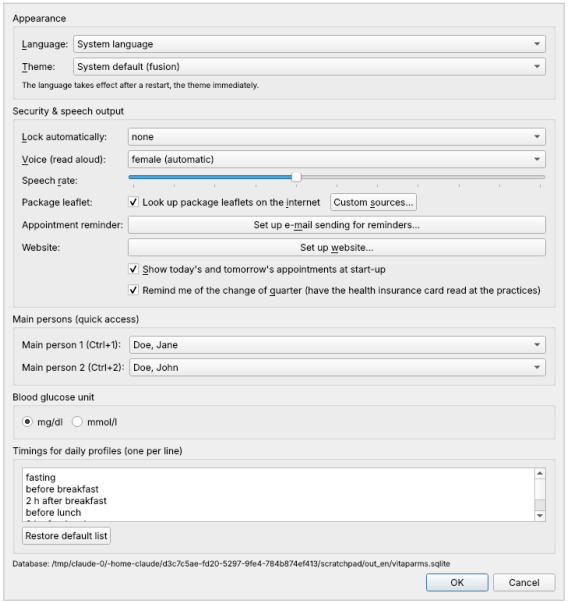


Fig. 13.1: The settings

13.1 Language and appearance

Language *System language*, *Deutsch* or *English*. The user interface is translated; your own entries – diagnoses, timings, document types – stay as you wrote them. The sample fields of the standard forms (such as 'Respiratory rate') are shown in the selected language.

Theme *System default*, the styles of your operating system (e.g. Fusion, Windows, macOS) and two dark themes: *Dark (AmigaED)* and *Visual Studio Code Dark (AmigaED)*. The change takes effect immediately. On screen the charts adapt to the dark background; printouts remain on white paper (Fig. 13.2).



Fig. 13.2: The dark theme: weekly chart and assistant

13.2 Security and speech output

Lock automatically *none* or after 10, 20, 30 or 60 minutes without use. The login window then appears; unsaved input in open dialogs is lost.

Voice and speech rate Default for the 'Prepare medication' assistant (Section 9.3 (p. 69)).

Package leaflet *Look up package leaflets on the internet* switches the online lookup on or off. Under Custom sources... you add further websites, one per line in the form `Name | Address`. VitaParms replaces placeholders such as `{name}`, `{substance}`, `{strength}` and `{pzn}` with the details of the medicine. Example: `NHS | https://www.nhs.uk/search/results?q={substance}`

Appointments at start-up *Show today's and tomorrow's appointments at start-up* (Section 11.7 (p. 85)).

Appointment reminder Set up e-mail sending for reminders... (Section 13.4 (p. 95)).

13.3 Main persons, blood glucose and timings

Main persons 1 and 2 the persons for Ctrl + 1 and Ctrl + 2 (Section 4.3 (p. 17)).

Blood glucose unit *mg/dl* or *mmol/l*. Values are stored internally in one unit; switching converts all displays, charts and printouts (1 mmol/l ≈ 18 mg/dl).

Timings for daily profiles the selection list in the measurement form, one entry per line. Adapt it to your daily routine; Restore default list resets it.

13.4 Setting up e-mail sending

For VitaParms to send reminders by e-mail, it needs the access data of the outgoing mail server (SMTP) of your e-mail provider. Only an administrator may change this setting.

1. In the settings, click Set up e-mail sending for reminders… (Fig. 13.3).
2. Enter *Server*, *Encryption* (STARTTLS or SSL/TLS) and *Port* – usually 587 for STARTTLS and 465 for SSL/TLS. You can find these details in the help pages of your e-mail provider.
3. Enter *User name*, *Password* and the *Sender* address.
4. Test the settings with Send test e-mail… and click Save.

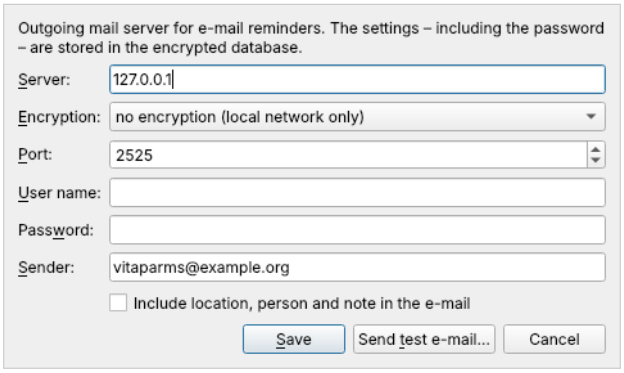


Fig. 13.3: Outgoing mail server for e-mail reminders (here a test server on the local network)

The access data is stored encrypted in the database. With a password, VitaParms only logs in over an encrypted connection. Because e-mails are usually not end-to-end encrypted on their way, reminders only contain title, time and priority by default. Location, person and note are only added if you tick *Include location, person and note in the e-mail*.

> **Tip:** Many providers require a separate ‘app password’ for programs such as VitaParms, which you create in the security settings of your e-mail account. Your normal password will then not work.

13.5 Customising forms

The input forms of VitaParms – master record, measurement, medication, document and appointment – are Qt Designer files that the program loads when a dialog opens. They can be changed without rebuilding the program: rearrange, rename or remove fields, or add your own fields. This feature is aimed at technically experienced users and is reserved for administrators.

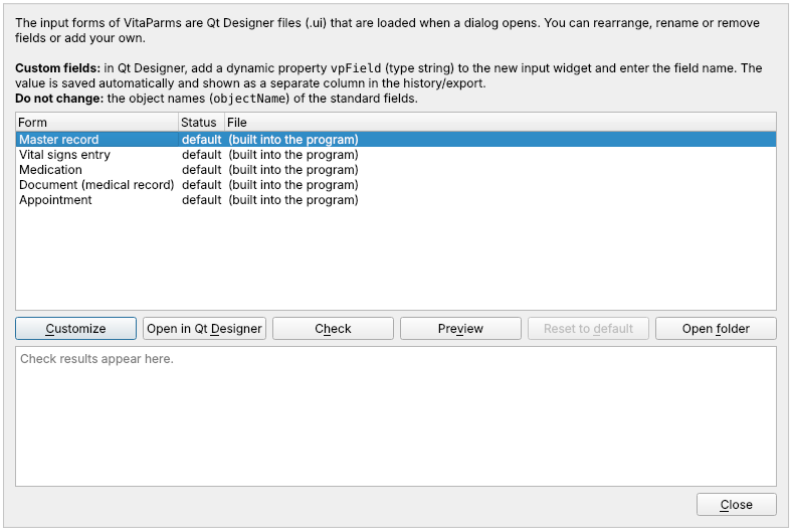


Fig. 13.4: Customising forms

1. Choose Tools › Customize forms…, select a form and click Customize. VitaParms creates an editable copy in the user forms folder.

2. Open in Qt Designer starts the Designer with this copy (it must be installed; it is part of the Qt tools).

3. Change and save the form. Check verifies that all required fields are present; Preview shows the result.

4. The next time the dialog opens, the new form is used. Reset to default discards the customisation.

Custom fields

In the Designer, create an input element – for example a number field for the waist circumference – click '+' in the property editor, choose *String* and create the dynamic property `vpField`. As its value, enter the field name, for example `Waist circumference`. VitaParms then stores the value automatically; custom fields of a measurement appear as a column of their own in the journal, in the printout and in the Excel export – like the 'Waist circumference' column in Fig. 4.1.

> **Peak flow:** Since version 0.7.4, peak flow is a built-in reading. If you previously created it as a custom field `Peak-Flow`, VitaParms transfers the stored values to the new reading the first time the database is opened. Then remove the custom field from your form.

Note: The field name is stored with the data exactly as you enter it and is not translated. Choose a name in the language in which you and your colleagues work.

Rules for customised forms:

- The object names (`objectName`) of the standard fields must be kept, otherwise these fields are no longer read. If a field is missing, its value in the database stays unchanged.
- Required fields are `lastNameEdit` (master record), `dateTimeEdit` (measurement), `nameEdit` (medication) and `titleEdit` (document). If they are missing, VitaParms uses the standard form.
- Do not remove the `formVersion` property on the top-level element. VitaParms uses it to recognise when a customisation is based on an older template, and points this out.

For a whole team, customised forms can also be placed in the `forms` subfolder of the program folder. VitaParms looks for a form first in the user forms folder, then in the program folder and finally uses the built-in standard version.

14 Your own website

Would relatives, the doctor's practice or you yourself like to view the readings while away, without VitaParms being installed on that device? VitaParms can publish the data of *all persons* on your own website for this – for reading, not for editing.

The website shows:

- the personal details from the master record and the emergency medication,
- the *complete* history of readings as a table and as trend charts (day, week, month, year or a custom period),
- the medication plan (optionally including discontinued medications) and appointments,
- reports to download: master record, medication plan and history as PDF, and the history as Excel and CSV file.

The website does *not* show the *medical record* with its documents. For synchronising with the tablet the documents are stored there encrypted in the `sync` folder, but the website can neither decrypt nor display them (section 15.5).

> **How the data is protected:** VitaParms encrypts everything *on your computer* with the website's master password (AES-256) before uploading. The server only stores scrambled data; not even your web host can read it. Only the browser decrypts the data after the password has been entered. After 30 minutes without use, the website logs out automatically. Anyone who knows the master password sees all persons – so only share it with people who may see all of the data.

14.1 What you need

- web space with *PHP 7.4 or later* and an *HTTPS certificate* (free with most hosts via Let's Encrypt),
- your own (sub)domain, for example `https://vitaparms.example.com`,
- an FTP or SFTP program such as FileZilla.

A database on the server is *not* required.

14.2 Step by step: setting up the website

1. Create the subdomain with your host, point it to an empty folder (such as `vitaparms`) and switch on HTTPS.

2. Upload the *contents* of the `website` folder from the VitaParms package into this folder – that is `index.html`, `.htaccess`, `robots.txt` and the folders `assets`, `api`, `data` and `sync` (for synchronising with the tablet, Chapter 15). Make sure the hidden `.htaccess` files are transferred too.

3. The `data` and `sync` folders must be writable for PHP (permissions 755, with some hosts 775).

4. As an administrator, choose Tools » Website… in VitaParms or click Set up website… in the settings (Fig. 14.1).

5. Tick *Use website* and enter the *Address*.

6. Choose a *Master password* (at least 10 characters) and repeat it. A long sentence such as 'Our family password 2026' is safer and easier to remember than a short, complicated word.

7. Click Test connection. VitaParms reports the PHP version and whether the website has already been set up.

8. Click Upload now. After a few seconds 'Website updated' appears. Open website shows the result in your browser.

The upload key. In addition to the master password, VitaParms creates an *Upload key*. It authorises uploading and is separate from the master password: someone who only views the website cannot upload anything. On the first upload the website remembers this key and afterwards only accepts data with the same key. To use a new key – for example after reinstalling VitaParms – click New and then delete the file `data/.uploadkey.php` on the server. The new key is accepted on the next upload.

14.3 Updating the website

If *Update automatically* is ticked (the default), VitaParms uploads the data in the background about one minute after each change – you can keep working meanwhile. On exit, VitaParms waits until pending changes have been transferred. To update manually at any time, choose File » Update website (Ctrl + Shift + U). The status bar shows when the last upload took place.

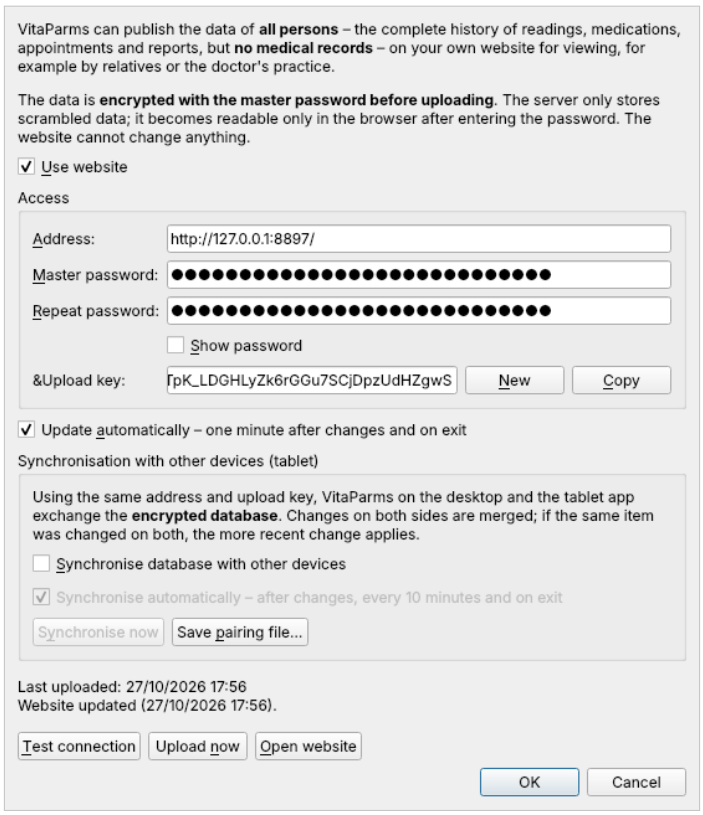


Fig. 14.1: The website settings

If you change the master password, the new password applies after the next upload; the old one then stops working.

14.4 Using the website

After signing in with the master password, choose the person on the left (at the top on a smartphone). The tabs correspond to those in VitaParms (Fig. 14.2):

Overview the latest readings with a small trace of the last 30 days, upcoming appointments, regular medication and the latest observations. Clicking a reading opens the matching trend chart.

Trend charts choose the period and show or hide curves with the coloured buttons (Fig. 14.3). Moving the mouse over a chart shows the values at that time. For longer periods the website shows daily values: the mean, and as a vertical line the minimum to maximum.

Readings all readings, newest first, with search in observations and measurement times, and download as a CSV file.

Medication and **Appointments** as in VitaParms.

Reports the reports to download; they are decrypted in the browser only when downloaded.

Print prints charts or tables on A4 paper. The website follows the light/dark setting of the device.

14.5 Troubleshooting

HTTP 404 / upload.php missing The `website` folder was uploaded incompletely or to the wrong folder. The file `api/upload.php` must be reachable at `https://.../api/upload.php`.

The folder data/ is not writable In your FTP program, set the permissions of the `data` folder to 755 or 775.

The upload key does not match The website was set up with a different key. Delete `data/.uploadkey.php` on the server.

The package exceeds post_max_size The server only accepts small uploads. Increase `upload_max_filesize` and `post_max_size` in your host's PHP configuration (to 64 MB, for example).

Wrong password Mind upper and lower case. If the master password was changed in VitaParms, it only applies after the next upload.

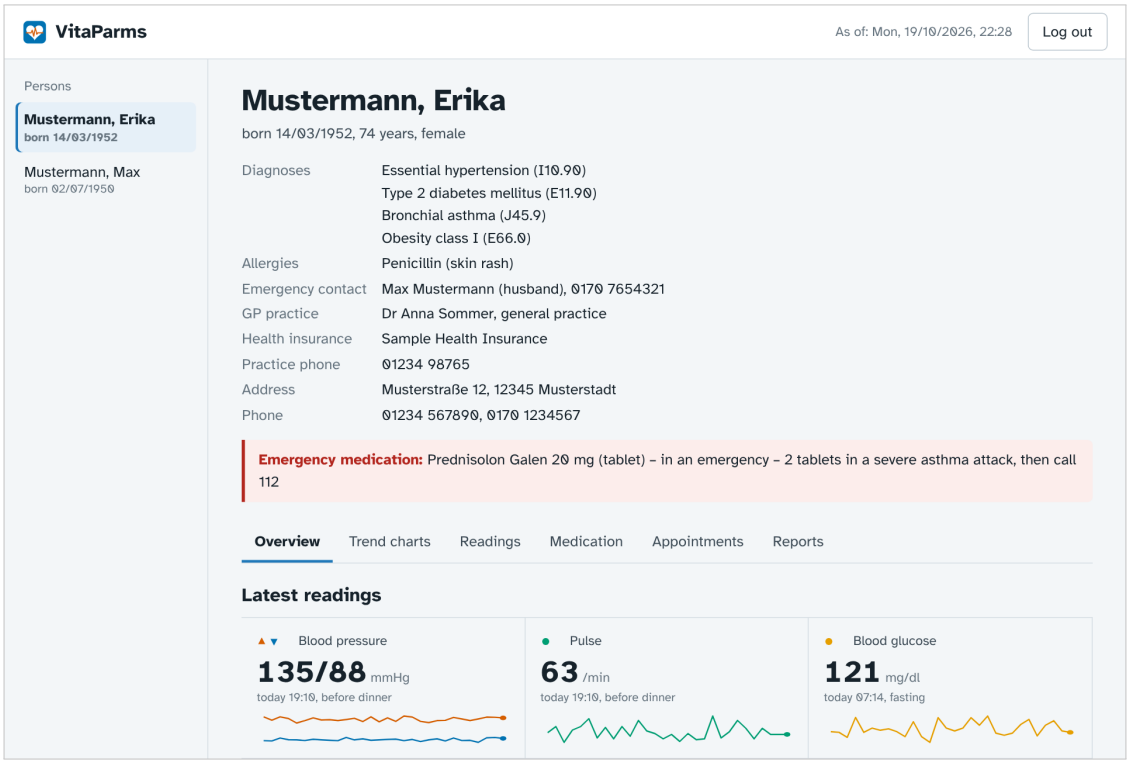


Fig. 14.2: A person's overview on the website

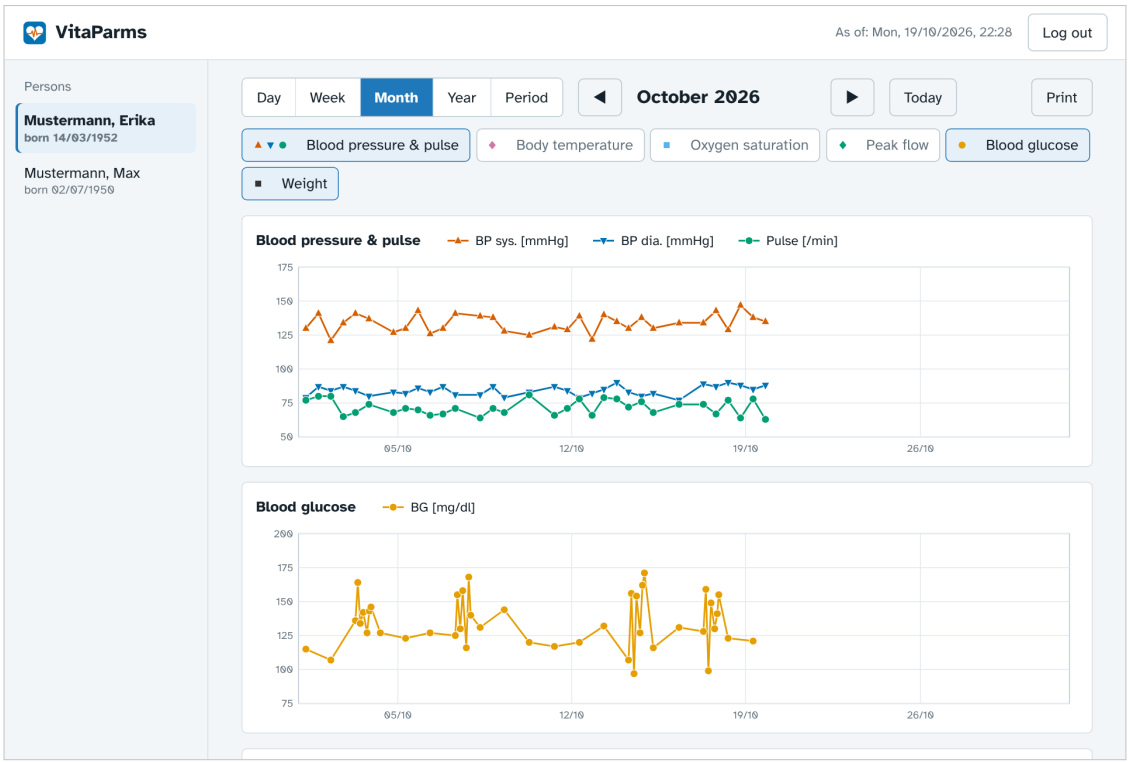


Fig. 14.3: Trend charts on the website

15 VitaParms on a tablet

At the bedside, on a home visit or on the sofa, a tablet is often handier than a computer. The VitaParms tablet app is designed for use with a finger – large buttons, number pads, highly legible type – and works with the *same* encrypted database as the program on the computer. Both synchronise via your own website (Chapter 14).

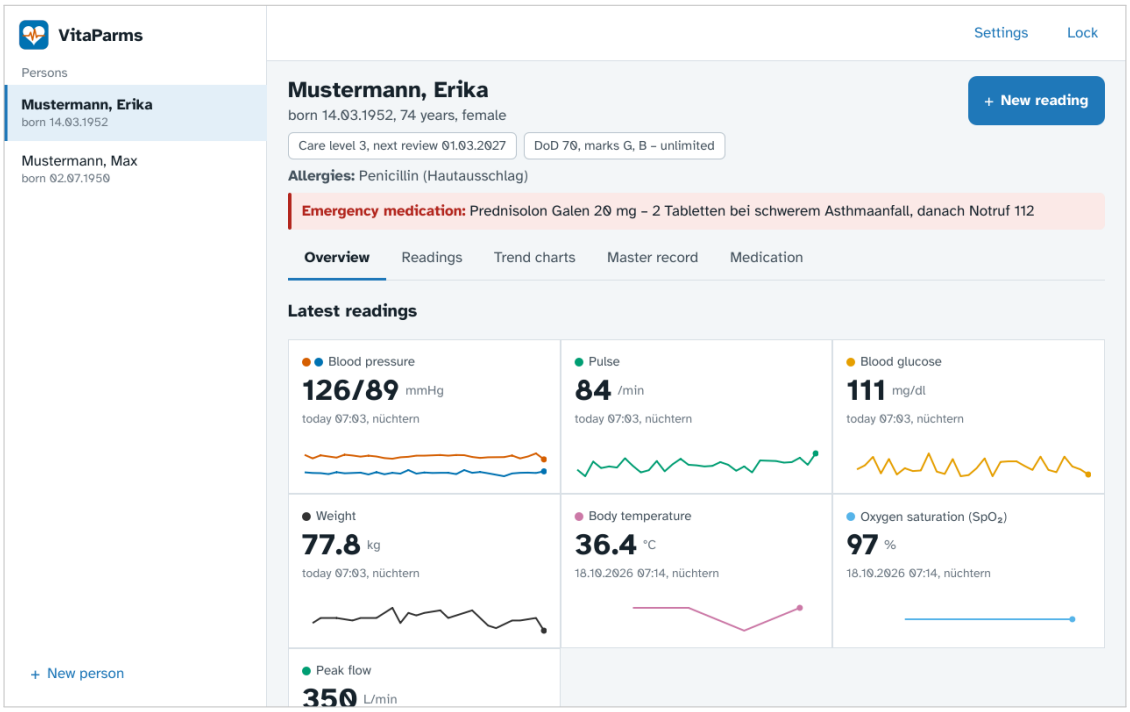


Fig. 15.1: The tablet app in landscape: persons on the left, overview on the right

15.1 What the tablet app can do

- select and create persons, view and edit the master record – including care level and severe disability card (Section 5.5 (p. 22)), contacts and advance directives
- enter and edit readings (Fig. 15.2), with measurement times for daily profiles, the minute timer for counting breaths (Section 6.2 (p. 29)) and a pain scale from 0 to 10
- all readings with search, trend charts by day, week, month and year (Fig. 15.12)
- view and maintain the medication plan as on the computer: intake schedule (every *n* days, on weekdays, monthly), as-needed and emergency medication, PZN,

markings (classification: narcotics, painkillers, anticoagulants …) and *Remind me to take it* with days and times (Fig. 15.3, Fig. 15.7)

- appointment planner with series and reminders, BG day profile, reminder at the change of quarter
- the medical record: view and search documents, photograph doctor's letters and reports with the camera (several pages become one PDF) or add them as a file, edit details and share documents (section 15.7)
- prepare medication with the read-aloud assistant (daily or weekly pill organiser), stock with packs and expiry dates, order lists (section 15.8)
- synchronisation with the computer via the website or the USB cable
- INR with target range, intake and output with daily balance, oedema with site and side; notes on the overview (Fig. 15.5)
- bowel movements on the Bristol scale with the details for the log; a note after three days without (Fig. 15.6)
- create the master record, medication plan, history, charts, bowel movement log, fluid balance and daily glucose profiles as PDF or Excel files; they open in the tablet's viewer app, from where they can be printed or shared
- light and dark, portrait and landscape, automatic lock

Since version 0.22 the tablet app can also prepare medication (with reading aloud), keep the medication stock and manage orders (section 15.8); an advance directive from the medical record can be linked in the master record and opened.

15.2 Installation

The app is not distributed via Google Play but installed directly on the tablet as an installation file (APK). It is built with Qt Creator from the `mobile` folder of the source code; the instructions are in `mobile/README.md`. In short:

1. Enable the developer options on the tablet: Settings » About tablet and tap *Build number* seven times. Then switch on *USB debugging* under Settings » System » Developer options.
2. Connect the tablet to the computer via USB and choose Run in Qt Creator – or copy the finished APK file to the tablet and open it there (allow installation from unknown sources first).

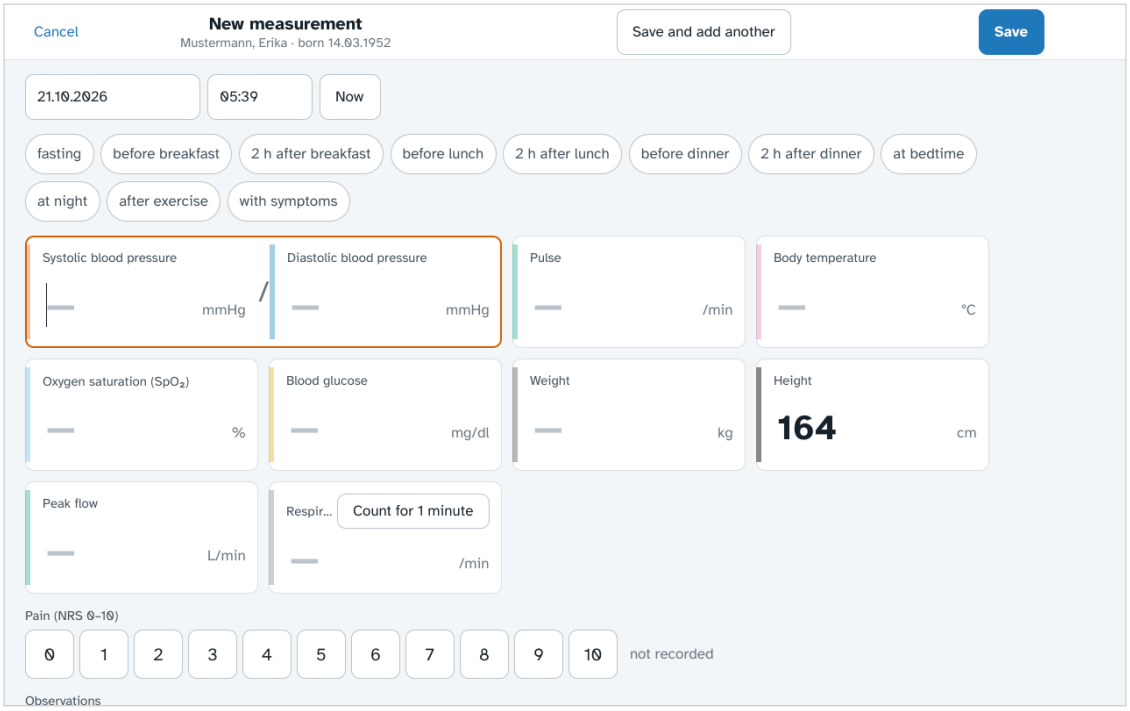


Fig. 15.2: Entering a reading on the tablet

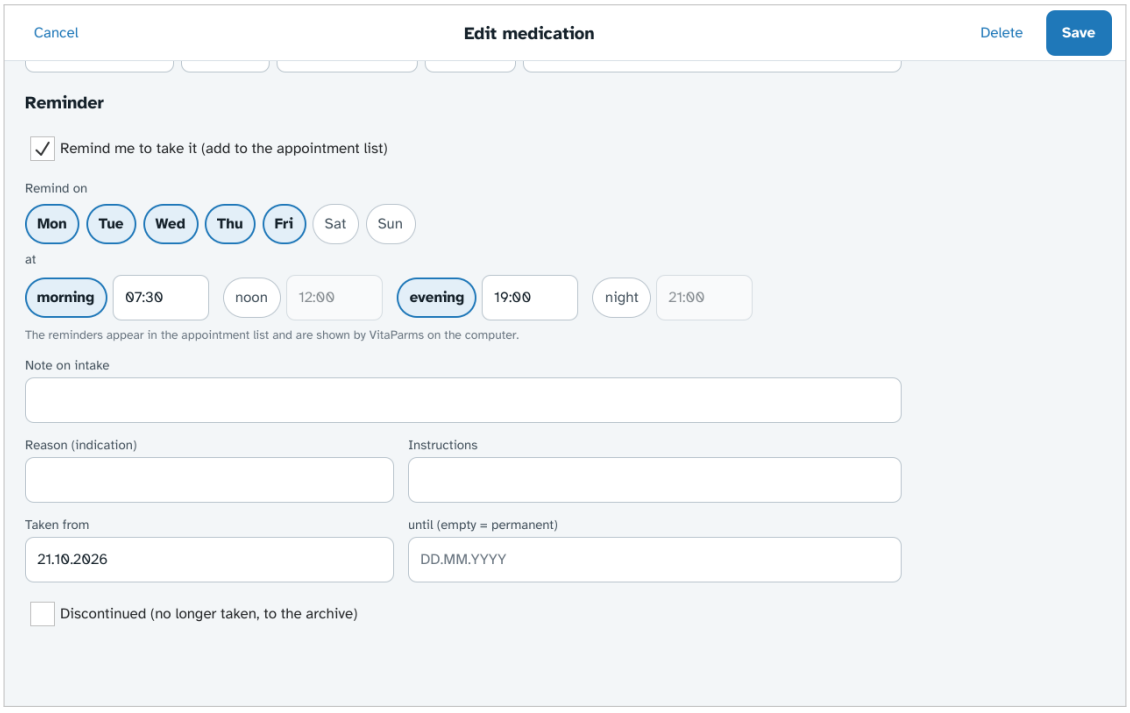


Fig. 15.3: Intake reminder on the tablet

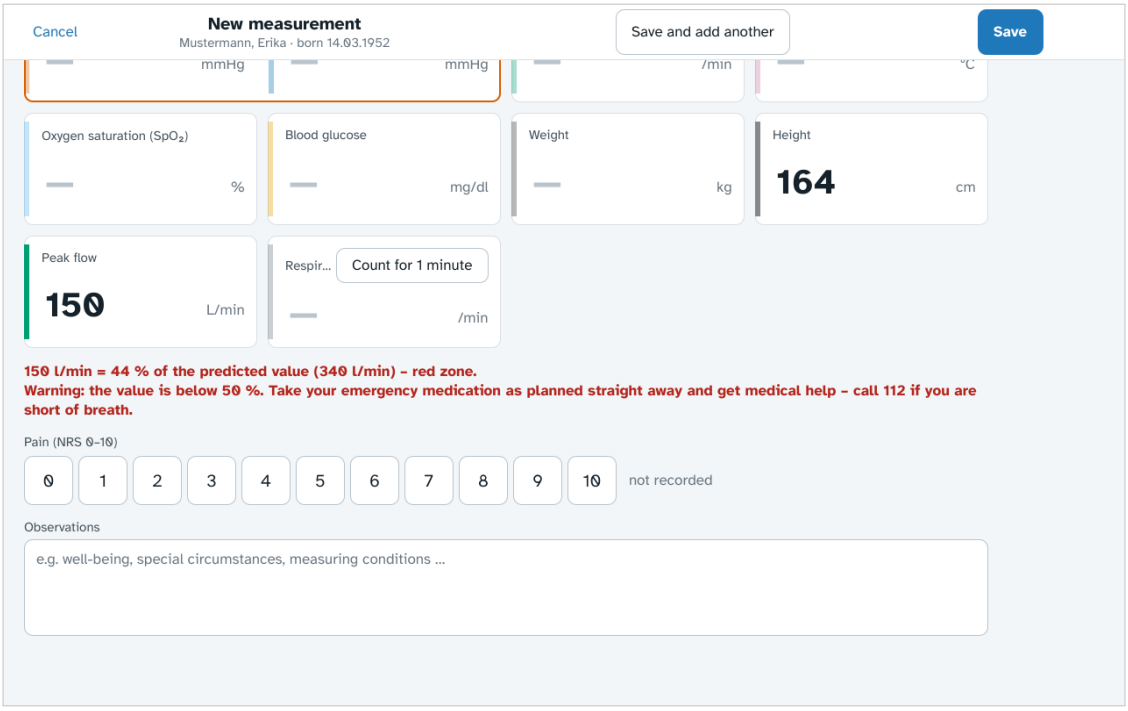


Fig. 15.4: Peak flow in the red zone on the tablet

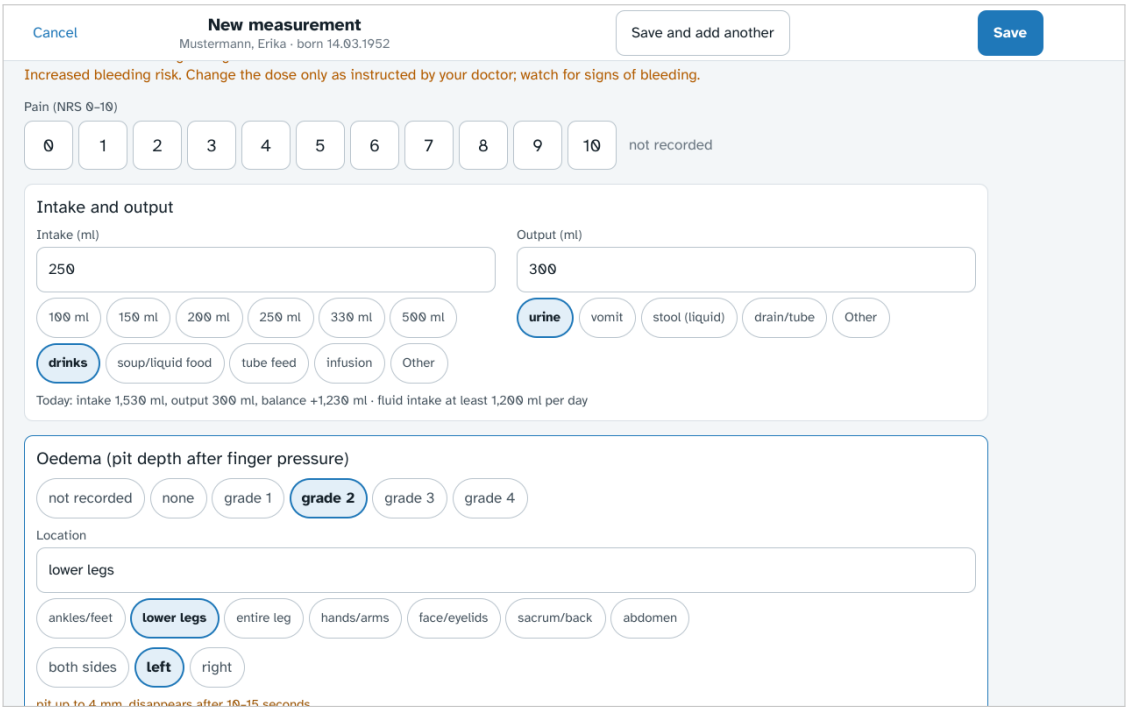


Fig. 15.5: Intake, output and oedema on the tablet: tap amounts to add them, notes below the fields

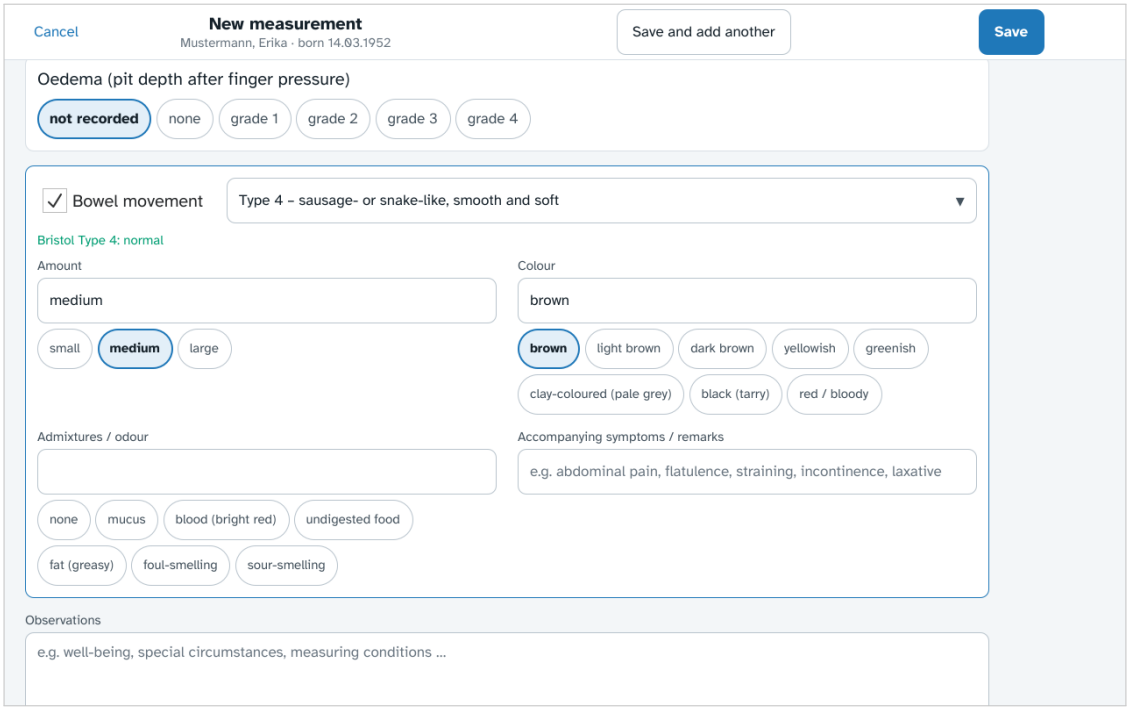


Fig. 15.6: Recording a bowel movement on the tablet: tick, Bristol type, amount and colour

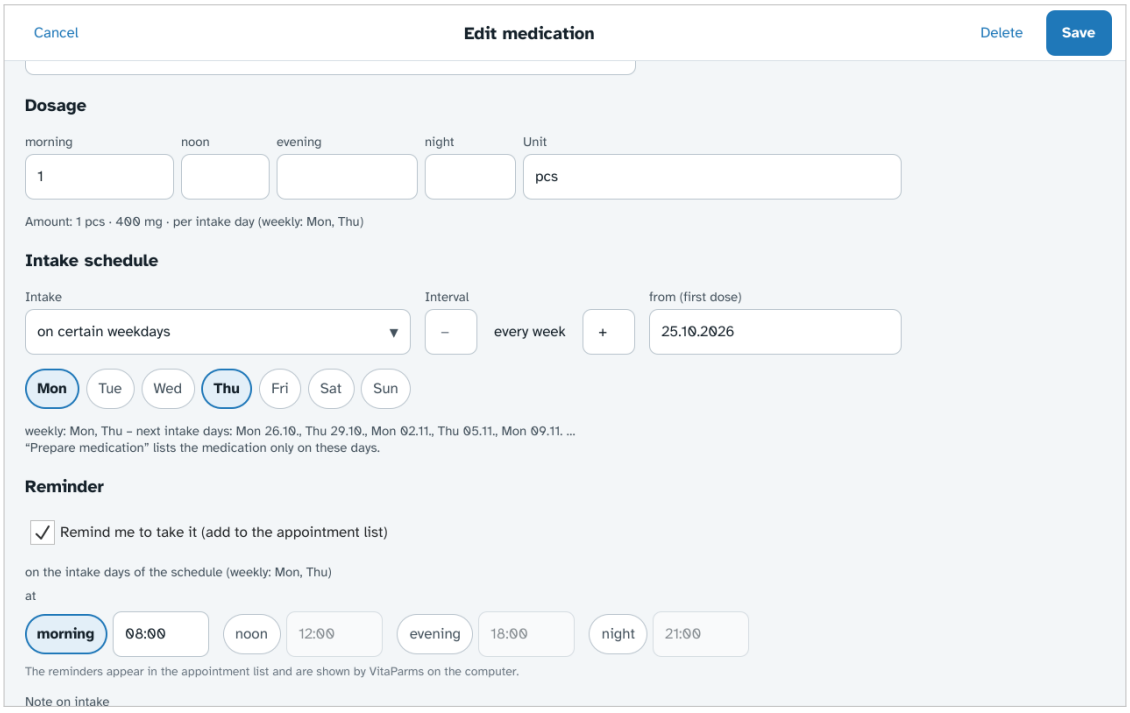


Fig. 15.7: Intake schedule 'Mon, Thu' on the tablet

15.3 Step by step: connecting tablet and computer

1. On the computer open Tools » Website… . Enter the website address (if not done yet) and tick *Synchronise database with other devices*.
2. Synchronise now – the encrypted database is now on your website.
3. Save pairing file… creates a small file with the address and upload key. Transfer it to the tablet (USB cable, your own cloud storage) – or type both values on the tablet.
4. On the tablet's first start choose Take over data from the computer (Fig. 15.9), Open pairing file… or enter address and key, then Load database.
5. Log in with a login account of the computer. Delete the pairing file afterwards.

Without a website On the computer choose File » Export database (for another device)… and take the `.vpdb` file to the tablet via USB cable or your own cloud storage. On the tablet choose Take over database file at the first start – or Import another database… in the login window –, select the file and Take over database. Instead of the `.vpdb` file, a backup (`.sqlite`) together with its key file (`.keys`) also works. There is no synchronisation then; it can be set up later via the website or the USB cable (section 15.4).

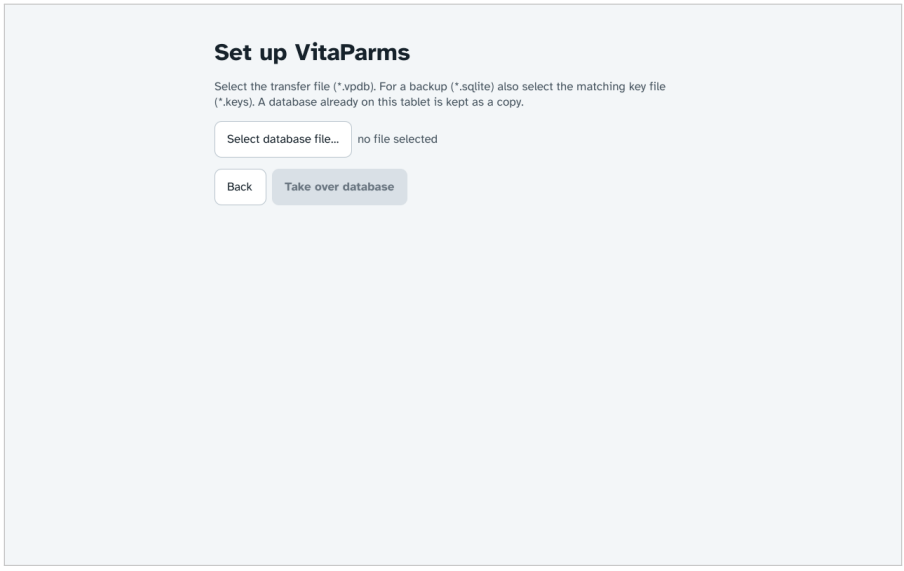


Fig. 15.8: Taking over a database file on the tablet

> **Security:** Only the encrypted database is stored on the website. Anyone who knows the upload key can download it, but cannot open it without a login account (name and password). Still, treat the upload key like a password.

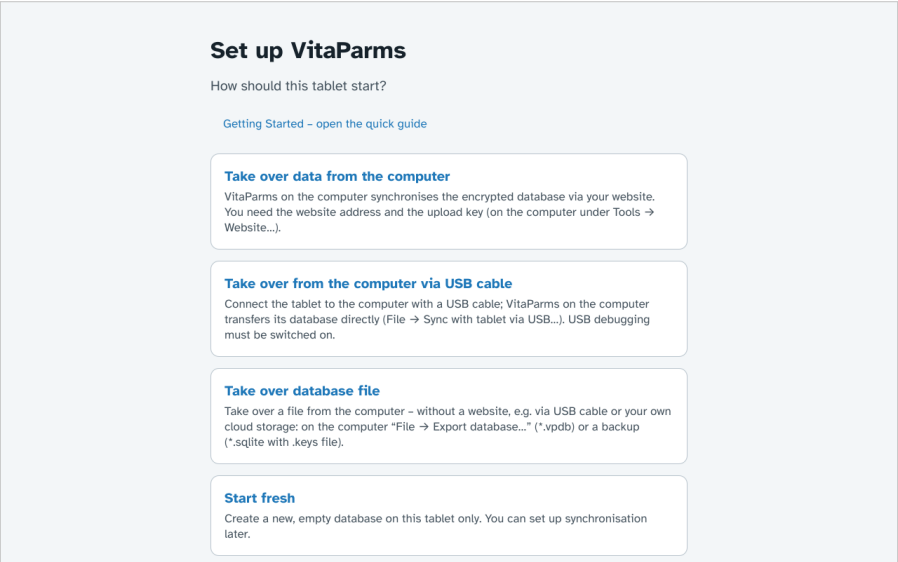


Fig. 15.9: First start of the tablet app

15.4 Synchronising over the USB cable

Without a website, the computer and the tablet (or smartphone) can also be synchronised directly over the USB cable. The same rules apply as for synchronising via the website (section 15.5): new and changed entries of both devices are merged; if the same item was changed on both, the more recent change wins. Afterwards both devices have the same, most recent data.

One-time preparation

- Switch on *USB debugging* on the tablet: Settings » About tablet, tap *Build number* seven times, then Settings » System » Developer options → *USB debugging*.
- The computer needs Google's free Android tool `adb` (part of the 'SDK Platform-Tools'). VitaParms looks for it in the Android SDK and in the search path; otherwise download the Platform-Tools, unpack them and select `adb.exe` with Choose… in the dialog.

1. Connect the tablet to the computer with the USB cable. Confirm the prompt 'Allow USB debugging?' – preferably with *Always allow from this computer*.
2. Open VitaParms on the tablet and log in.
3. On the computer choose File » Sync with tablet via USB…. The tablet appears under *Device* and VitaParms connects by itself: if both devices use the same database, they prove it to each other with their key – no code is needed then.

4. Only for the first setup (or different databases): on the tablet open Settings → USB sync with the computer, enter the six-digit code (Fig. 15.11) on the computer and choose Connect. The code stays valid and unchanged until VitaParms on the tablet is closed.

5. Synchronise. The dialog reports how many entries were added, changed or deleted on each device (Fig. 15.10).

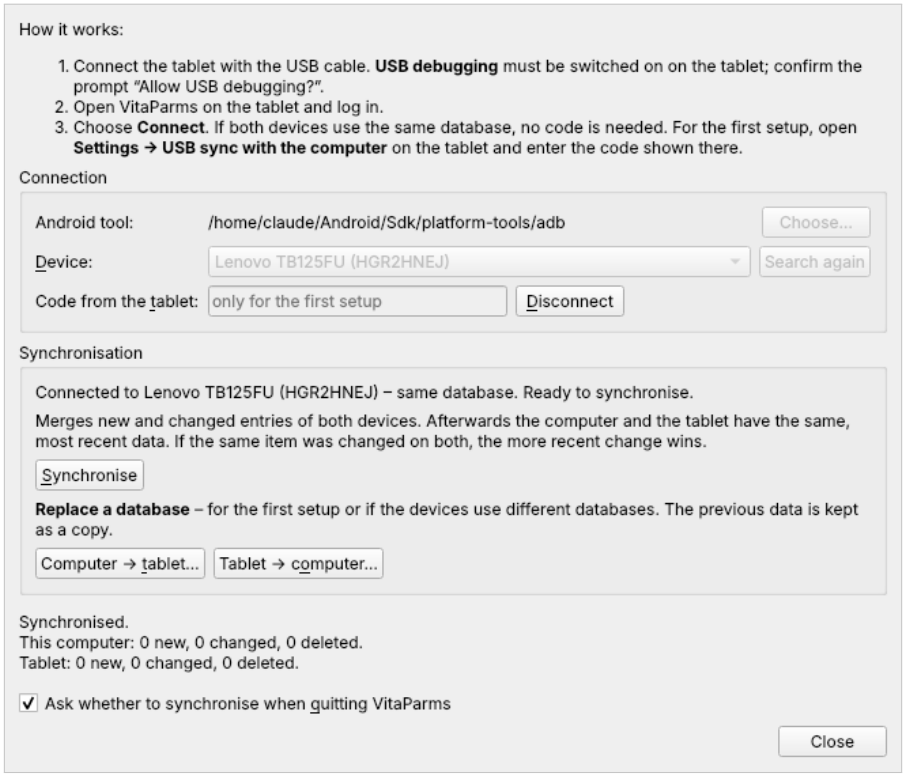


Fig. 15.10: Synchronising over the USB cable on the computer

Don't forget: syncing when quitting On the tablet, VitaParms asks when quitting or locking, and on the computer after the first USB connection when quitting, whether to synchronise – Synchronise... on the computer connects and synchronises right away (the tablet must be connected, with VitaParms open and logged in); Open USB sync on the tablet shows the page with the code. The question can be switched off in the computer's USB dialog or on the tablet's 'USB sync' page. The connection is always made by the computer over the cable. The tablet app can be closed with Quit at the top right (and in the login window).

Replacing a database For the first setup of a tablet – or if the devices use different databases – the same dialog has two more buttons (administrator accounts only):

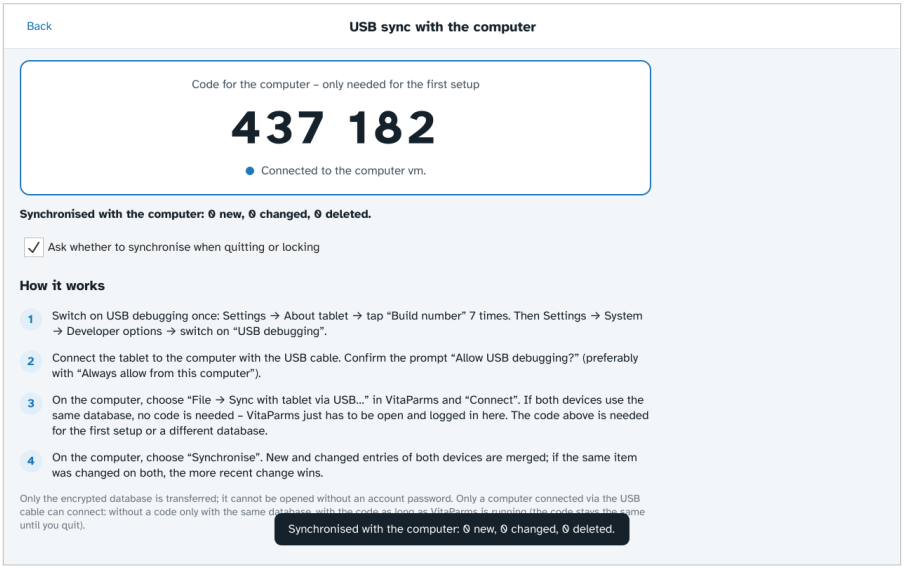


Fig. 15.11: The tablet waiting for the computer

Computer → tablet transfers the computer's database including the login accounts to the tablet. This also works when nobody is logged in on the tablet: at the first start choose `Take over from the computer via USB cable`, or `USB sync with the computer` in the login window.

Tablet → computer replaces the computer's database with the tablet's; afterwards you log in with an account of the tablet.

The previous data of the replaced device is kept as a copy (`*.vor-import-...`).

> **Security:** Only the encrypted database goes over the cable; the key never leaves a device. VitaParms checks whether both devices use the same database with a proof, without transferring the key. The tablet accepts connections only over the USB cable: without a code only from a computer with the same database (mutual proof via the key), otherwise only with the code, which is valid from the first time it is shown until VitaParms is closed. After five wrong entries a new code applies.

15.5 How synchronisation works

VitaParms does not exchange individual values but the whole encrypted database – and merges it on the device row by row:

- New entries from both sides are kept – even if readings were taken on the computer and the tablet at the same time.
- If the same item was changed on both devices (the master record, say), the *more recent* change applies.

- Deleted entries are also deleted on the other device.
- New login accounts and changed passwords are adopted.
- Settings and the note that an appointment reminder was already shown remain per device.
- The *contents* of the medical record (the PDF files) travel separately from the database and one by one – only those the other device lacks. This keeps every synchronisation small, even as the record grows. Until its content has arrived, a document is shown with '⧖' in the list.

Documents via the website: For the separate transfer the website needs the file `api/files.php` (from VitaParms 0.19). Upload the `api` folder from the package again. The contents are then stored in the `sync/files` folder – encrypted on the device (AES-256) like the database, under names that reveal nothing about the document. VitaParms deletes contents that are no longer needed after one day. All devices should use the same version; older versions still transfer the complete database.

Synchronisation is automatic: about one minute after each change, at start-up, every ten minutes and on exit or lock. Manually on the computer with File › Synchronise database (Ctrl + Shift + A), on the tablet with the Synchronise button in the header. A green dot means 'synchronised', an orange one 'changes not yet transferred'.

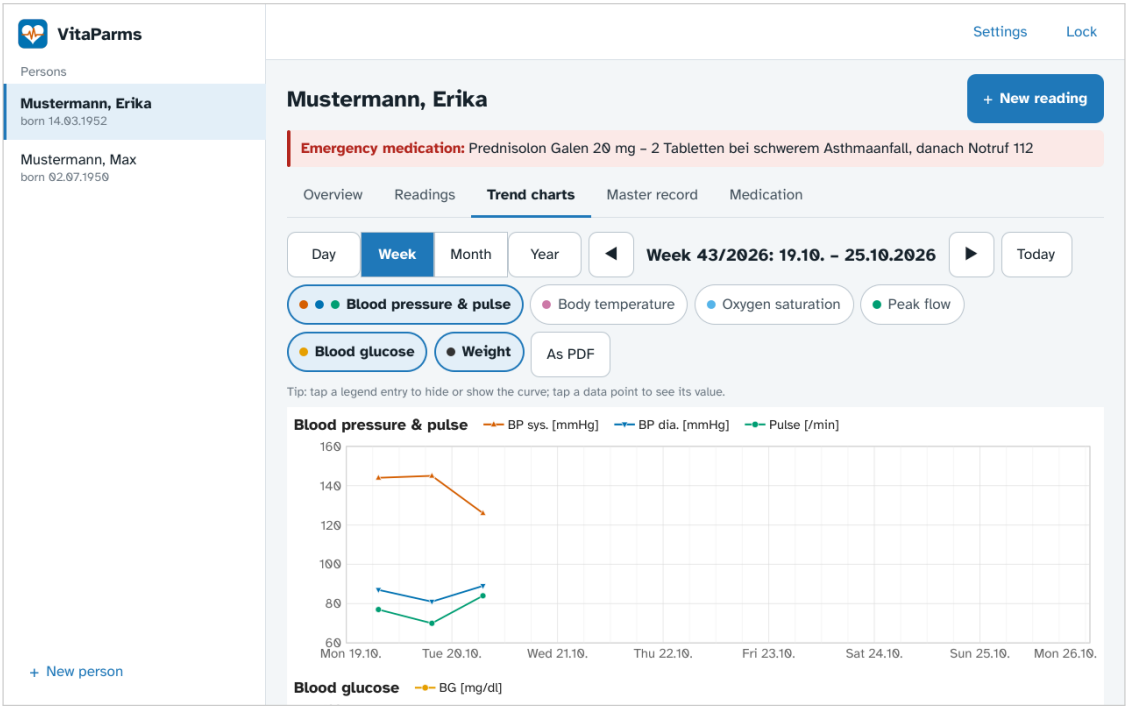


Fig. 15.12: Trend charts on the tablet

15.6 Using the app

Persons on the left in landscape, in the drop-down list at the top in portrait.

New reading Tap a value field – the number pad appears. After three digits of the systolic blood pressure, input jumps to the diastolic value. The height is preset with the known height. Save and add another keeps the form open for the next reading of a daily profile. Unusual values are outlined in red; swapped blood pressure values and times in the future are queried before saving.

Readings Tapping opens the reading for editing or deleting.

Trend charts Tapping a legend entry hides or shows the curve, tapping a data point shows its value.

Peak flow Below the values, VitaParms shows the reference value and – once a value is entered – the percentage and the zone; in the yellow or red zone with a note on what to do, and after saving also as a message (Fig. 15.4). On the **Readings** tab such values are coloured. The personal best and the biological sex are in the master record under *Peak flow* and *Biological sex* (section 6.5).

Bowel movement At the bottom of the reading form: tick *Bowel movement*, then choose the type on the Bristol scale. Amount, colour and admixtures can be tapped as suggestions or typed freely, plus accompanying symptoms (section 6.6). After three days without an entry a note appears on the **Overview**; the *Bowel movement* tile shows when the last one was.

INR, intake/output, oedema In the reading form: INR as a value tile with a note on the target range; for intake the amount buttons (100, 150 … 500 ml) add up, the type is tapped; below is the day's balance. For oedema, tap grade, site and side. Target range and fluid limits are in the master record (sections 6.7–6.9).

Logs On the **Readings** tab, Logs … creates the bowel movement log, the fluid balance or the daily glucose profiles for the period chosen above as a PDF or Excel file (section 7.4).

Time point Tap a predefined time point ('fasting', 'after exercise' …) – or enter free text under *Or your own time point*, e.g. 'after the walk'.

Edit medication **Medication** tab, tap the medication. Under *Type* choose regular, as-needed or emergency medication or other; for as-needed medication enter the single or maximum amount as the dose, and the occasion and minimum interval under *Occasion / use*. For regular medication, *Intake schedule* sets when it is taken: daily, every *n* days, on certain weekdays (chips 'Mon' to 'Sun', interval with − and +) or monthly, each counted from the first dose. Below, the next intake days are shown (Fig. 15.7); the reminder follows these days. VitaParms asks before saving an invalid PZN. Under *Classification*, VitaParms suggests the groups from the active ingredient and name; ticking or unticking a box sets them (section 8.5).

Appointments see section 15.9.

Editing the master record as on the computer – plus *Contacts* (health insurer with insurance number, GP, specialists, hospitals …: tap to edit, Add contact) and *Advance directives* (living will, lasting and care power of attorney with person, address, telephone and validity). First add the signed document to the medical record (take photos or import a PDF, type 'Advance directives'); then Choose from the medical record links it to the power of attorney or living will (Fig. 15.17). On the **Master record** tab, e.g. Open Lasting power of attorney then opens the document.

BG day profile **Readings** tab → BG day profile: choose the day and times, Start day profile (Fig. 15.26). A notification reminds you at each time; in the open app VitaParms asks 'Record measurement?' and opens the entry form with subject, time and time point, the cursor in the blood glucose field. After saving, the reminder is done (see also section 6.4).

Change of quarter reminds you as on the computer (section 11.6); can be switched off under Settings → 'Reminders'.

Language and help At the first start you choose the language; later under Settings → 'Language and help' – the app switches immediately, without a restart. There and on the set-up and login pages, Getting Started opens the quick guide as a PDF (an app for viewing PDF files must be installed).

Lock closes the database; after the set time without use, or after a longer time in the background, the app locks itself.

15.7 The medical record on the tablet

The **Medical record** tab shows the documents of the selected person by year, with a small image of the first page (Fig. 15.13). The search field searches titles, types, practices, notes and the text of the documents; the buttons below restrict the list to one type ('Doctor's letter', 'Lab results' …).

Viewing Tapping opens the document; the pages are shown one below the other, + and − zoom (Fig. 15.14). Open in PDF app shows it in the tablet's viewer app (where it can also be printed).

Photographing Take photo opens the tablet's camera app. After the photo the page appears in 'New document' (Fig. 15.15); Photograph another page adds the next page, '↻' rotates and '×' removes a page. All pages become one PDF. Tip: lay the sheet flat on a dark surface and photograph it from above without flash.

Adding a file Choose file adds a PDF file or images from the tablet's storage (downloads, e-mail attachments, cloud).

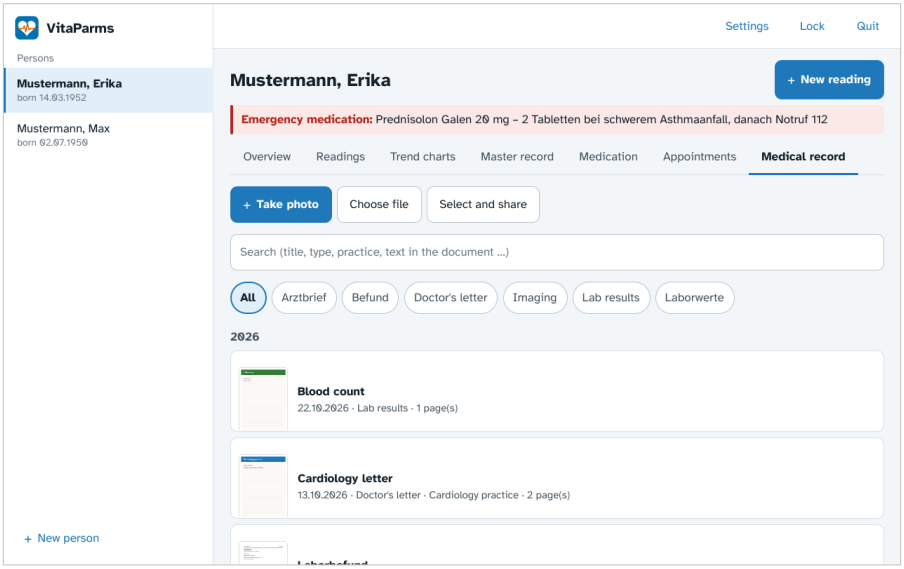


Fig. 15.13: The medical record on the tablet

Details Title, date, type and issuer as on the computer; the types and the practices from the contacts in the master record are offered as suggestions to tap. Add to record saves. VitaParms recognises a document that is already in the record.

Sharing Select and share (or a long tap on a document) selects documents; Share ... offers them as PDF files or as a ZIP archive with table of contents, master record and medication plan – AES-256 encrypted with a password (Fig. 15.16, as in section 10.4). Then choose e-mail, messenger or cloud in the Android share menu. Share the password by another channel.

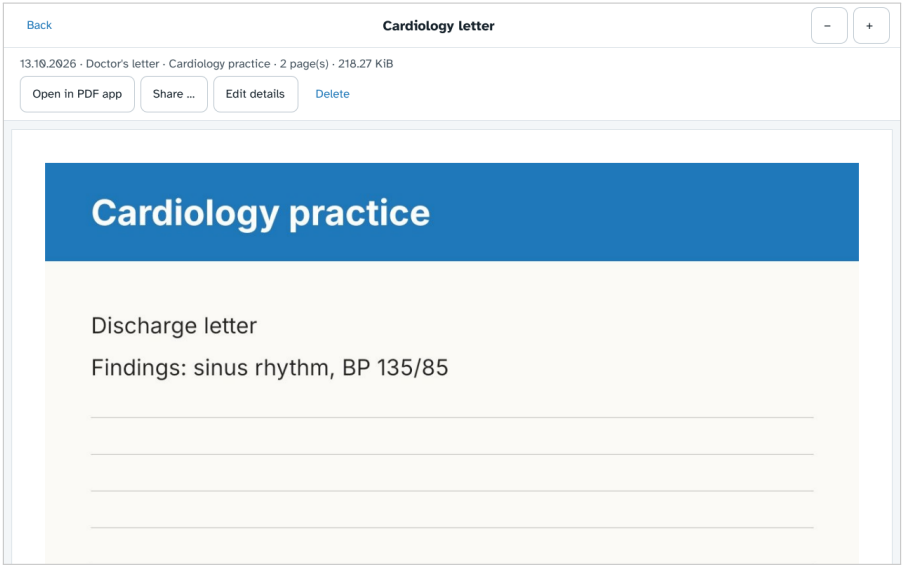


Fig. 15.14: Viewing a document

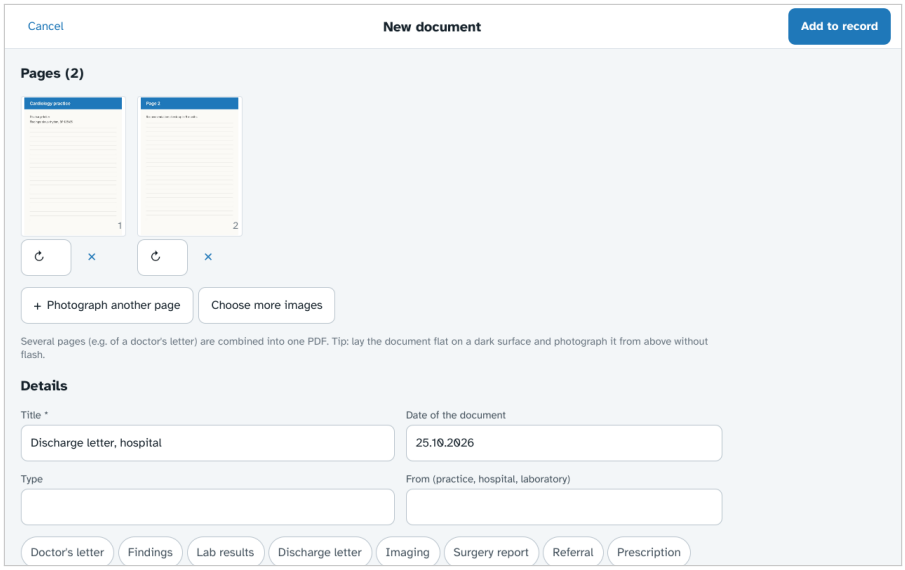


Fig. 15.15: New document from two photos

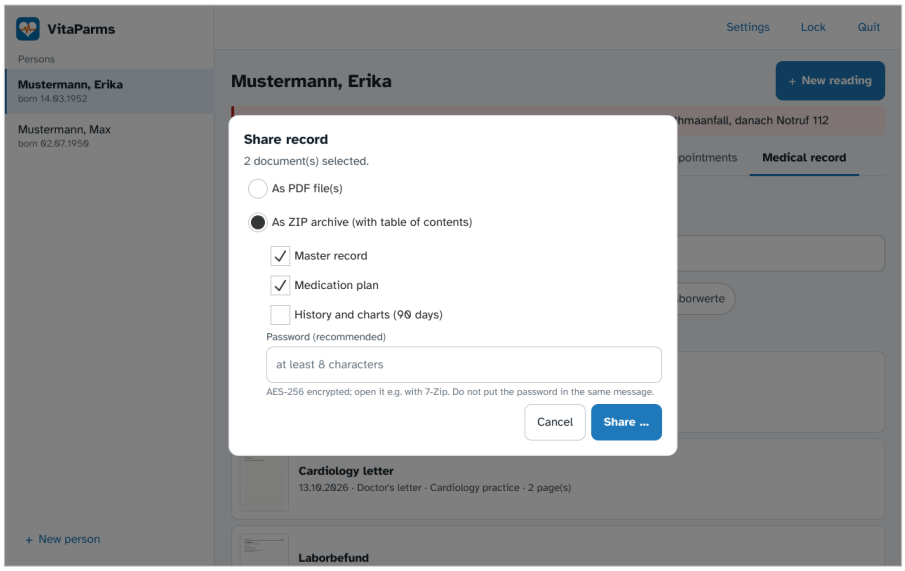


Fig. 15.16: Sharing documents as an encrypted ZIP archive

VitaParms needs no camera permission of its own – the photo is taken by the device's camera app. Photos are reduced to about 200 dpi (A4); a page takes about 100 KB.

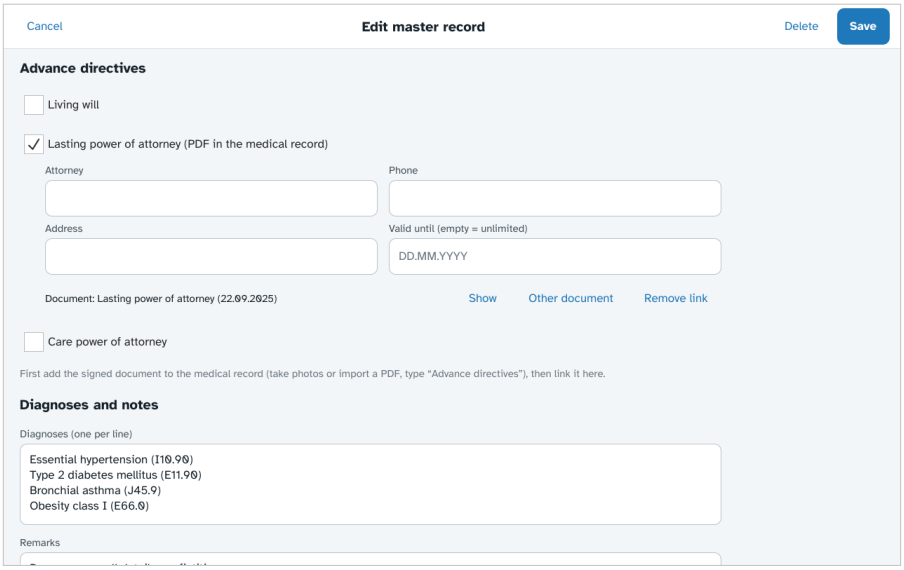


Fig. 15.17: Linking a lasting power of attorney in the master record to a document from the medical record

15.8 Preparing medication, stock and orders on the tablet

At the top of the **Medication** tab are Prepare medication and Stock and orders (Fig. 15.18). Below each medicine whose stock is kept, VitaParms shows the stock, how long it lasts and its state ('OK', 'order soon', 'order now' …). If an order list has not yet been confirmed as ordered, a yellow note reminds you – here and on the **Overview**; tapping it opens the stock.

Preparing medication The assistant works as on the computer (Chapter 9):

- At the top choose daily (daily pill organiser) or weekly (7 days) (weekly pill organiser), the *Time* (all, morning, noon, evening, night), the day and the voice. *read aloud automatically* reads each medicine aloud when you move on.
- Below are the medicines not due on that day as well as as-needed and emergency medication and other items – these are not prepared.
- The card shows the medicine in large type with strength, dosage form and active substance, the dose per time (Fig. 15.19) or, for the weekly organiser, one row per day with the weekly quantity (Fig. 15.20), followed by intake schedule, remark and note.
- The buttons at the bottom always stay visible: ◀ Back, ↻ Repeat (read it again) and Next ▶; for the last medicine Done ✓.

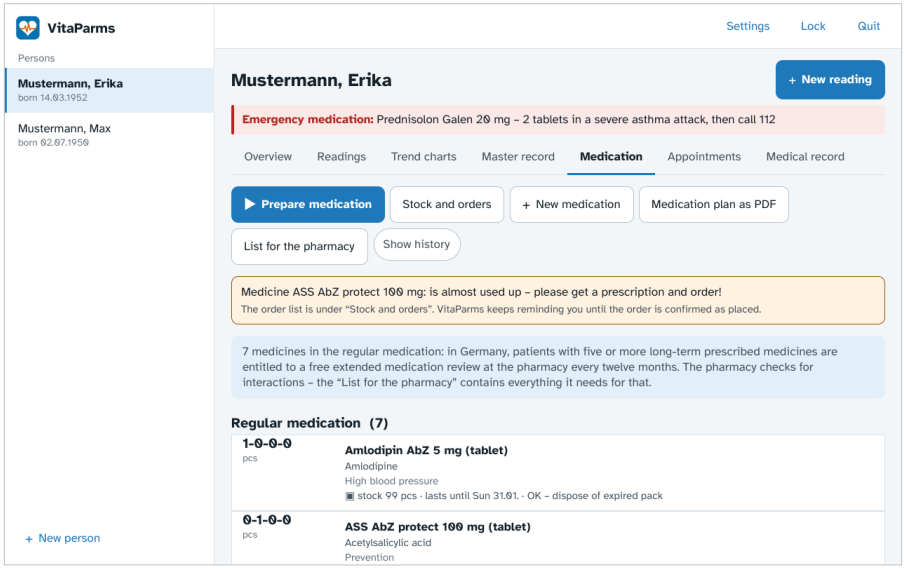


Fig. 15.18: Medication tab with stock and a reminder to reorder

- After the last medicine VitaParms reports 'Medication is prepared' (Fig. 15.21), records the round in the log and books the prepared quantities out of the stock. You can add a *Note*. If a stock was not sufficient or something has to be reordered, the note appears below; Stock and orders leads straight to the order list.
- Close in the middle of a round records it as cancelled; only medicines confirmed with Next are booked out. Log shows the rounds of the last 90 days and outputs them as a PDF.

Speech uses the Android text-to-speech output (Settings » Accessibility » Text-to-speech output). If no voice for the selected language is installed, VitaParms points this out.

Stock Stock and orders shows for each medicine the stock, need per day, 'lasts until', 'order by' and the next expiry date (Fig. 15.22); the rules are the same as on the computer (section 8.8).

Preparation daily or weekly from a weekday – with the weekly organiser the stock must last for the whole week on the preparation day.

Lead time the days needed for prescription, order and delivery (−/+).

Receive pack quantity, expiry date and note; this starts stock keeping for the medicine.

Packs and bookings lists the packs (the top one is used first) and all bookings. Count/correct takes over a counted quantity, Dispose of books an (expired) pack out.

Add to order list adds the medicine to the open order list by hand.

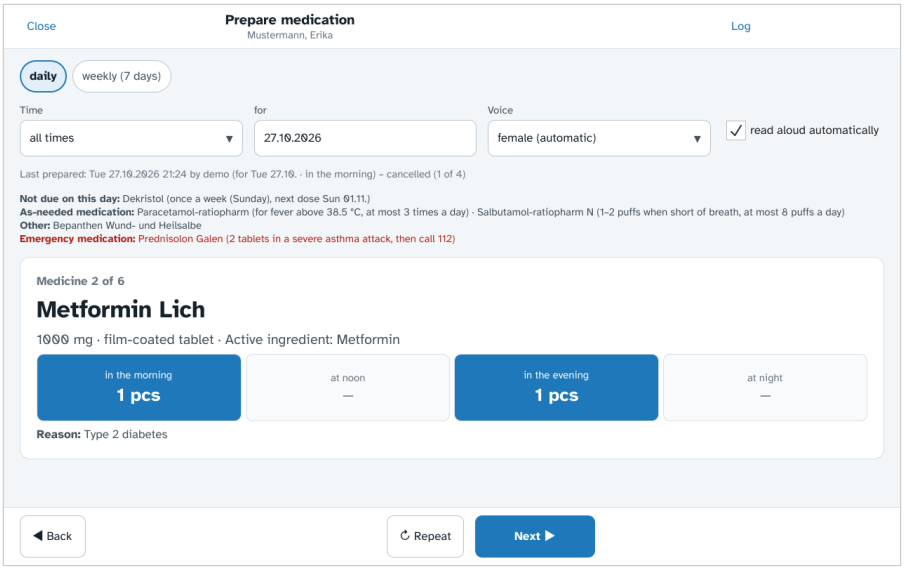


Fig. 15.19: Preparing medication on the tablet: daily pill organiser

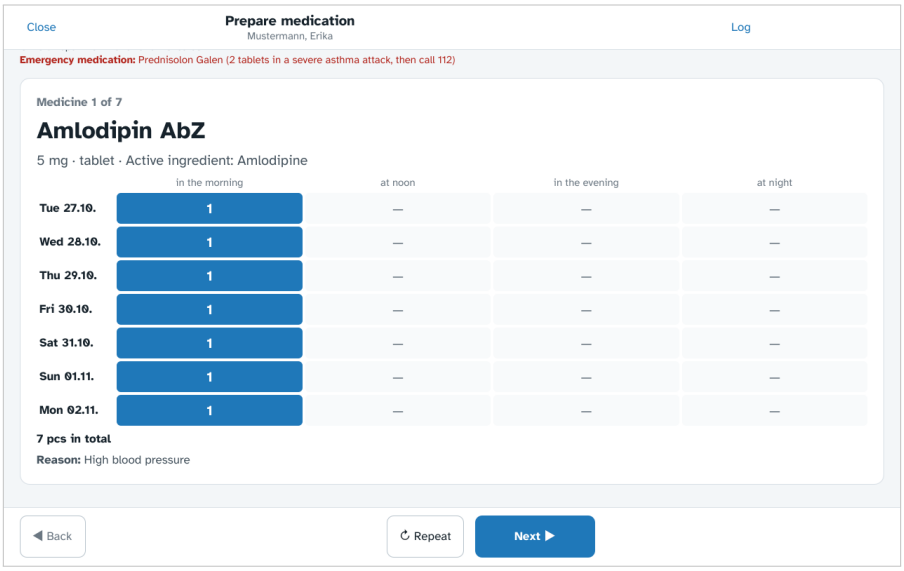


Fig. 15.20: Weekly pill organiser: one row per day

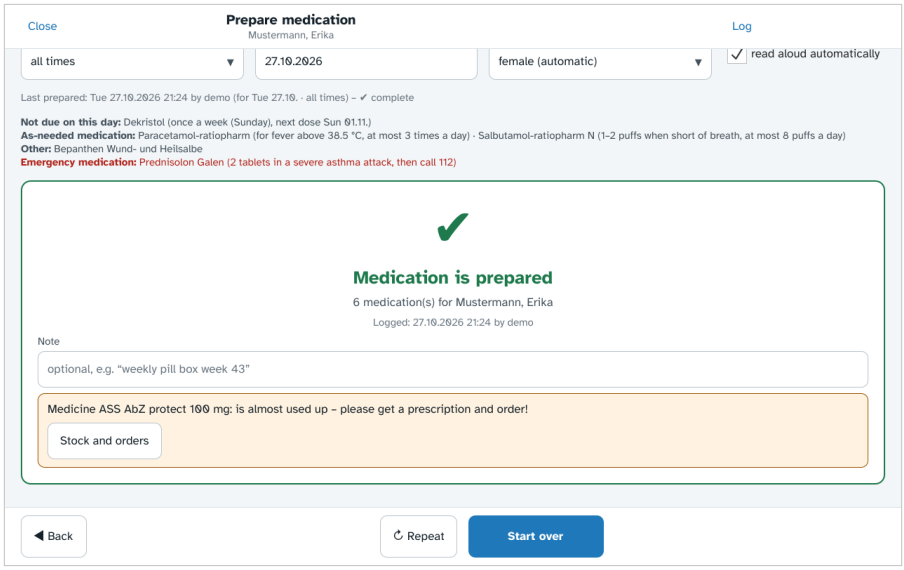


Fig. 15.21: 'Medication is prepared' – with a note to reorder

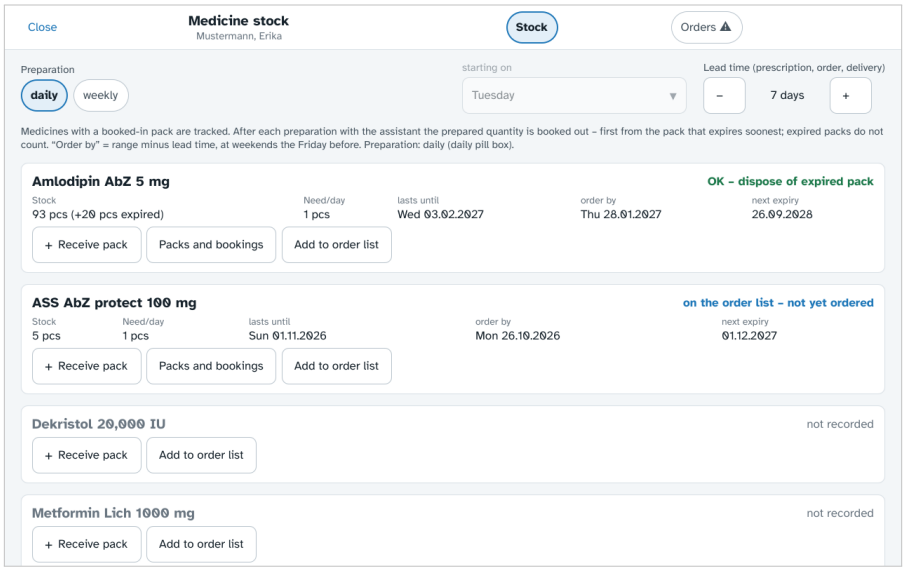


Fig. 15.22: Medication stock on the tablet

Orders Orders lists all order lists (Fig. 15.23). Create order list adds the medicines that must be ordered now. For each item you can Change quantity (e.g. '1 pack N3'; VitaParms remembers it for next time) or remove the item. As PDF outputs the list for the practice or pharmacy. Once ordered, tap Confirm as ordered (date and note) – the reminder then ends. When the delivery arrives, Book delivery books the new pack in with quantity and expiry date.

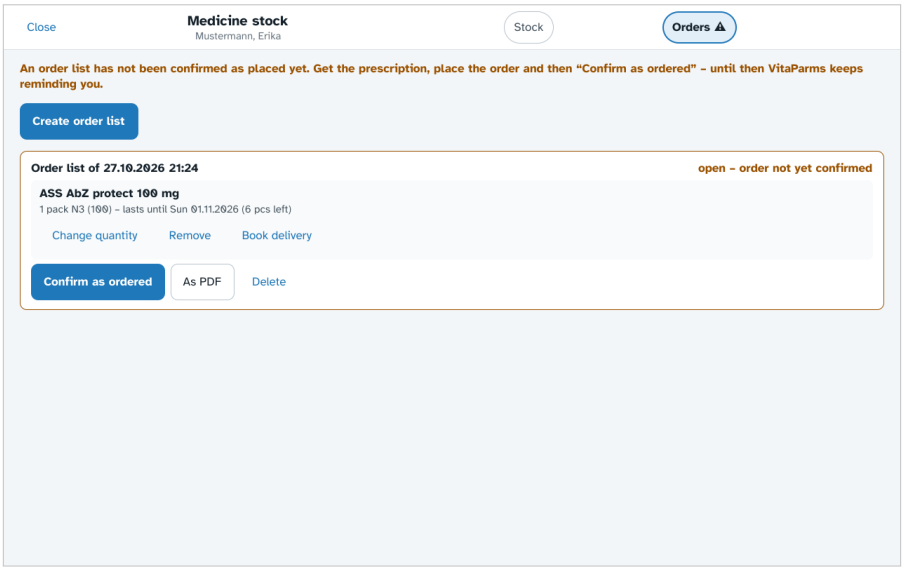


Fig. 15.23: Open order list on the tablet

15.9 Appointments on the tablet

The **Appointments** tab shows the selected person's appointments grouped by day – either Upcoming, Today, This week, This month or Past (90 days). All persons shows everybody's appointments, Show completed also the ticked ones. The coloured bar on the left stands for the priority, '↻' for an appointment from a series. The round tick on the right marks an appointment as done (tap again: open).

New appointment or tapping an appointment opens the form (Fig. 15.25): title, start and end (or *all day*), location, person, priority, note and reminder. *Recurrence* offers the same rules as on the computer – daily, every … days, weekdays, weekly (with weekdays), every … weeks, monthly, yearly – plus the end of the series (no end, until a date, after a number of appointments). Below, VitaParms lists the next dates of the series. If the appointment belongs to a series, VitaParms asks when saving or deleting whether the change applies to Only this appointment, This and all following or All appointments of the series.

Reminders The tablet reminds you with a *notification* – on time, even when VitaParms is in the background, locked or closed, and also after the tablet restarts. This applies to appointments of the logged-in account that have a reminder, including medication with

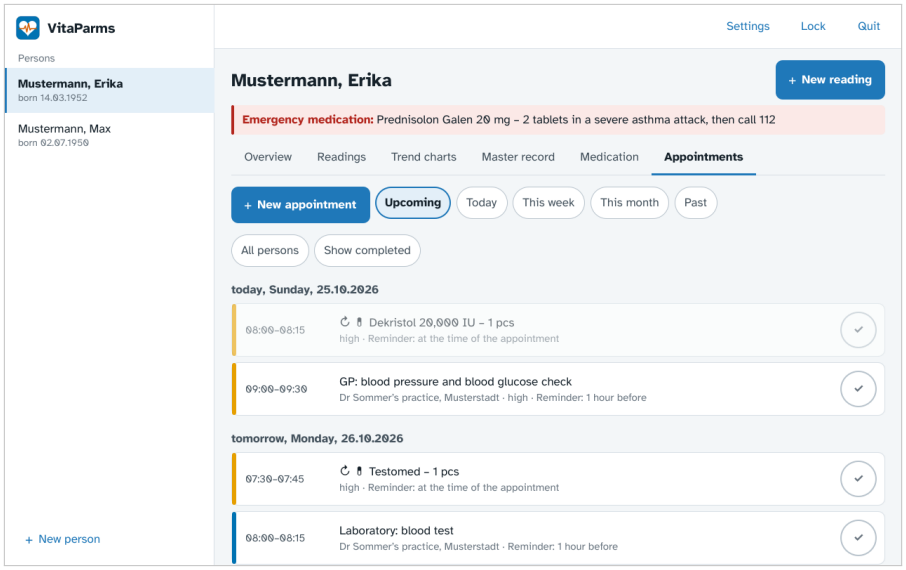


Fig. 15.24: Appointments on the tablet

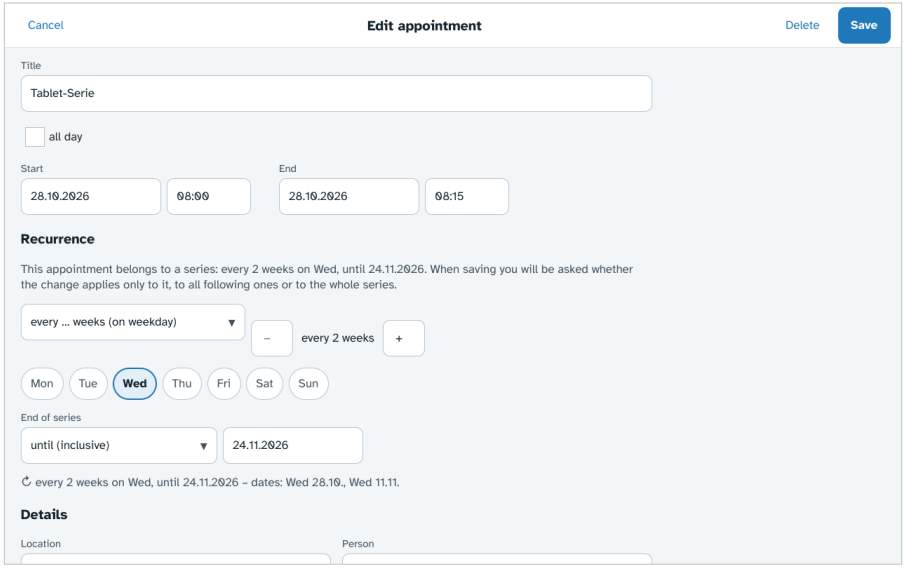


Fig. 15.25: Editing an appointment series on the tablet

a reminder. At the first login Android asks whether VitaParms may send notifications – please choose *Allow* (later under Settings › Apps › VitaParms › Notifications). *Sound* decides whether the notification plays a sound. Reminders for the next 14 days are scheduled; every time the app is opened it plans further ahead. The lock screen only shows ‘VitaParms – Appointment reminder’, no details. E-mail reminders are still sent by VitaParms on the computer.

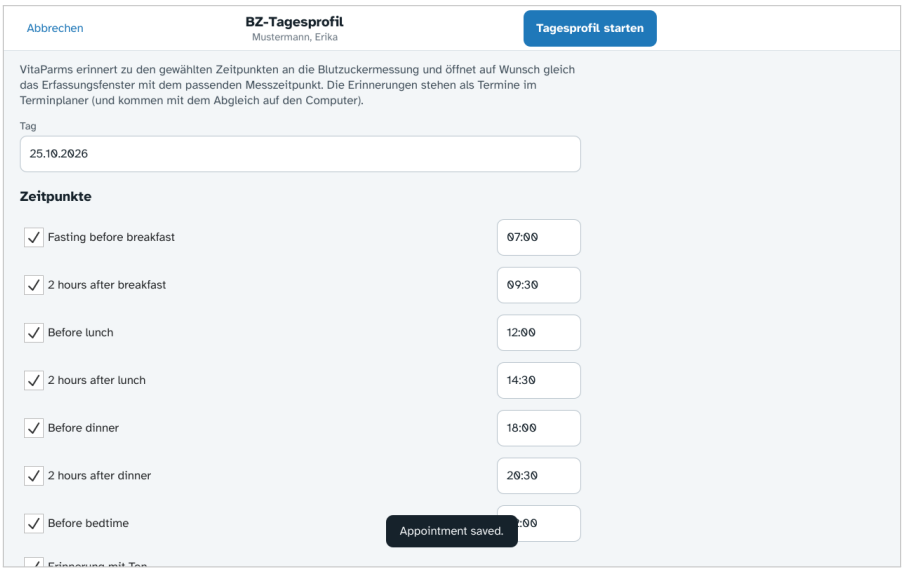


Fig. 15.26: BG day profile on the tablet

15.10 Troubleshooting

api/sync.php missing on the website Upload the website folder from this package again – the synchronisation interface is new.

The folder sync/ is not writable Set the permissions of the `sync` folder to 755 or 775.

Older VitaParms version Both devices need at least version 0.10.

The database exceeds post_max_size With many documents in the medical record the database becomes large. Increase `upload_max_filesize` and `post_max_size` with your host.

USB: no device found Is USB debugging switched on on the tablet? Try another cable (some cables only charge). On Windows, some tablets need the manufacturer’s USB driver or the ‘Google USB Driver’.

USB: confirm the prompt If the device list says so, the prompt ‘Allow USB debugging?’ on the tablet has not been answered yet.

USB sync is not open on the tablet Open the page ‘USB sync with the computer’ on the tablet and keep it open.

USB: different databases The devices were set up independently. Replace one with Computer → tablet or Tablet → computer; afterwards they can be synchronised.

Login on the tablet fails The login accounts of the computer apply. A password changed on the computer applies on the tablet only after the next synchronisation.

A Keyboard shortcuts

On macOS use Cmd instead of Ctrl.

Main window

Action	Shortcut
Person	
New person	Ctrl + Shift + N
Master record	Ctrl + E
Main person 1 / 2	Ctrl + 1 / Ctrl + 2
Prepare medication (assistant)	Ctrl + Shift + M
Medication stock and orders	Ctrl + Shift + L
Search controlled narcotics in the medication plan	Ctrl + Shift + B
Readings	
New measurement	Ctrl + M or Ctrl + N
Edit measurement	F2
Delete measurement	Del
Medical record and appointments	
Show medical record	Ctrl + 3
Add documents	Ctrl + D
Pack record / send by e-mail	Ctrl + Shift + S
Show appointment planner	Ctrl + 4
New appointment	Ctrl + T
Output	
Print	Ctrl + P
Print preview	Ctrl + Shift + P
Export to Excel	Ctrl + Shift + E
Session	
Update website	Ctrl + Shift + U
Synchronise database (tablet)	Ctrl + Shift + A
Log out / lock	Ctrl + L
Help	
Open the user manual	F1

‘Prepare medication’ assistant

Action	Key
Next medicine	Space, Enter or →
Repeat the same medicine	R
Back to the previous medicine	←

Command line

`VitaParms --database <file>`	opens a specific database (short form `-d`)
`VitaParms --demo`	starts the demo database (Section 3.5 (p. 12))
`VitaParms --version`	shows the program version
`VitaParms --help`	shows help on the parameters

B Legal background

This appendix explains the legal framework under which VitaParms is provided and can be used. VitaParms is developed in Germany; the appendix therefore refers to German and European law as of October 2026. Where the law of another country applies to you, the rules may differ – some pointers are given at the relevant places.

Not legal advice: The following explanations are general information. They do not replace legal advice in an individual case. Anyone who uses VitaParms professionally – for example in a care service or a practice – should agree its use with their data protection officer.

B.1 VitaParms is not a medical device

In the EU, whether software is a medical device is determined by Regulation (EU) 2017/745 on medical devices (MDR). The decisive factor is the *intended purpose* defined by the manufacturer (Art. 2(1) and (12) MDR). Software is a medical device if it is intended for a medical purpose, such as the diagnosis, monitoring, treatment or prediction of diseases in individual patients. Guidance on this distinction is given in the guideline MDCG 2019-11 of the Medical Device Coordination Group. Other jurisdictions, such as the United Kingdom or the United States, have their own regulations that follow a comparable approach based on intended purpose.

The intended purpose of VitaParms is:

> *VitaParms is a personal diary for recording, filing, displaying and outputting data that users enter themselves – readings, medication plans, documents and appointments. The program is intended for documentation and organisation. It is not intended to assess readings, to make or support diagnoses, to recommend treatments or doses, to monitor diseases or to warn of health risks.*

The program's calculations – BMI, daily amounts, active amounts – are simple arithmetic with the data entered, as a pocket calculator would perform them. The classification of medicines into groups is a search aid within your own data, not drug information. Anyone who uses VitaParms contrary to this intended purpose, for example to control a therapy, does so on their own responsibility.

B.2 Licence

VitaParms is open-source software under the MIT licence. You may use, copy, modify, merge, publish, distribute, sublicense and sell the program free of charge. The only condition: the copyright notice and the licence text must be retained in all copies or substantial portions. The licence text reads:

```
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```

B.3 Warranty and liability

The MIT licence excludes warranty and liability comprehensively (‘as is’). Such exclusions do not apply without limit in every legal system. Under German law, software that – like VitaParms – is provided free of charge is widely considered to be a gift, for which the following applies:

- The donor is only liable for **intent and gross negligence** (§ 521 of the German Civil Code, BGB).
- The donor is only liable for defects if they **fraudulently concealed** them (§ 524 BGB).

These statutory minimum standards, and any mandatory rights under the law applicable to you, remain unaffected by the licence. Beyond that, no liability is accepted – in particular not for loss of data, for incorrect or incomplete entries, for the automatic classification of medicines, for reminders that do not appear or appear late, or for content from the internet.

Product liability

Under the new EU Product Liability Directive (EU) 2024/2853, software is now considered a product. However, the directive expressly does not apply to free and open-source software developed or supplied *outside the course of a commercial activity* (Art. 2(2) of the directive). In Germany, the new Product Liability Act implements the directive; the Bundestag passed it in October 2026, and it is due to enter into force on 9 December 2026. VitaParms is developed without any intention to make a profit and is provided free of charge.

Cyber resilience

The same applies to the Cyber Resilience Act, Regulation (EU) 2024/2847: free software that is not made available in the course of a commercial activity is outside its scope. However, anyone who distributes VitaParms commercially or builds it into a commercial product takes on the obligations of a manufacturer – under these rules as well as under product liability law.

B.4 Third-party components

Component	Use	Licence
Qt 6 (Core, GUI, Widgets, PrintSupport, UiTools, Network)	program framework, user interface, printing	LGPL v3
Qt PDF (with PDFium)	preview, text and printing of PDF documents	LGPL v3 (PDFium: BSD-3-Clause and others)
Qt TextToSpeech	reading aloud in the assistant	LGPL v3
Qt Multimedia	signal tones	LGPL v3
SQLite3 Multiple Ciphers	encrypted database	MIT (Ulrich Telle)
SQLite	database engine	public domain
Themes ‘Dark’ and ‘VS Code Dark’	colour schemes from AmigaED 4.0	by the author

The Qt libraries are linked *dynamically*. This means VitaParms meets the conditions of the LGPL v3: users can replace the Qt libraries with their own modified versions. The source code of Qt is available at https://download.qt.io; the licence text is supplied with the Qt libraries and can be displayed via Help » About Qt.

The modules *Qt Charts* and *Qt Graphs* are deliberately *not* used because in the open-source edition they are only available under the GPL v3. Charts, Excel and ZIP generation as well as the AES encryption of ZIP archives are therefore implemented by VitaParms itself.

B.5 Data protection

What VitaParms itself does with your data

VitaParms stores all data exclusively locally in the encrypted database on your computer. There is no user account with the author, no cloud, no advertising and no usage statistics (telemetry). Connections to the internet are only made when you yourself

- look up a package leaflet – only name, strength, active substance or PZN of the medicine are transmitted, no personal data;
- have reminders sent by e-mail or send an export by e-mail – here the terms of your e-mail provider apply.

Private use

Health data is specially protected under Art. 9 of the General Data Protection Regulation (GDPR). However, the GDPR does not apply to processing by a natural person *in the course of a purely personal or household activity* (Art. 2(2)(c) GDPR, the so-called household exemption); the UK GDPR contains a comparable exemption. Anyone who records their own readings or those of their mother in need of care with VitaParms is therefore not subject to the obligations of the GDPR in this respect. Nevertheless: please treat other people's data with care and, where possible, ask the person concerned for their consent.

Professional use

If an institution, a care service or a practice uses VitaParms, it is the *controller* within the meaning of Art. 4(7) GDPR. In particular it then needs

- a legal basis for processing health data (Art. 9(2) GDPR, e.g. point (h) for health or social care),
- appropriate technical and organisational measures (Art. 32 GDPR), such as personal login accounts, secure passwords, backups, screen locking,
- an entry in the record of processing activities (Art. 30 GDPR) and, if applicable, a data protection impact assessment (Art. 35 GDPR).

Members of the healthcare professions – including doctors, pharmacists and nurses – are also bound by professional secrecy (in Germany under § 203 of the Criminal Code). Outside the EU and the UK, other rules apply, for example HIPAA in the United States for healthcare providers. The encryption of VitaParms supports these obligations but does not replace a data protection concept.

B.6 Medicines and controlled narcotics

What counts as a controlled narcotic in Germany

Controlled narcotics (Betäubungsmittel, BtM) within the meaning of the German Narcotics Act (BtMG) are the substances and preparations listed in its Schedules I to III. Those that can be prescribed – and are therefore relevant for a medication plan – are the narcotics of **Schedule III**, such as morphine, oxycodone, fentanyl, hydromorphone, tapentadol, buprenorphine, methylphenidate or nabilone. They are prescribed on a special narcotics prescription form.

Schedule III expressly exempts certain preparations (*exempted preparations*). These are prescription-only, but not controlled narcotics within the meaning of the Act. Examples that VitaParms takes into account:

Active substance	not a controlled narcotic if …
Codeine, dihydrocodeine	up to 100 mg per divided form (e.g. tablet) or up to 2.5 % in undivided preparations
Tilidine	as a prolonged-release tablet in a fixed combination with naloxone; tilidine drops have been controlled narcotics since 2013
Diazepam	up to 10 mg per divided form
Lorazepam	up to 2.5 mg per divided form
Oxazepam	up to 50 mg per divided form
Bromazepam	up to 6 mg per divided form
Zolpidem	up to 8.5 mg per divided form
Phenobarbital	up to 300 mg per divided form

Some of these exemptions are subject to restrictions, for example when prescribed for people with an addiction. Since the Medical Cannabis Act (MedCanG) of 1 April 2024, medical cannabis and dronabinol are no longer controlled narcotics in Germany; they are prescribed on a normal prescription. Nabilone remains a controlled narcotic.

Other countries classify controlled drugs differently. In the UK, for example, the Misuse of Drugs Regulations apply, in the USA the Controlled Substances Act. Adjust the classification in VitaParms to your local rules where necessary (Section 8.5 (p. 57)).

Travelling with controlled narcotics

For journeys of up to 30 days to a Schengen state, anyone carrying medically prescribed controlled narcotics needs a **certificate under Article 75 of the Convention implementing the Schengen Agreement**. It is completed by the prescribing doctor and certified by the competent authority; in Germany the form is provided by the Federal Institute for Drugs and Medical Devices (BfArM). For journeys to other countries, a multilingual medical certificate is recommended; the embassy of the destination country can provide information. The narcotics search in VitaParms (Section 8.5 (p. 57)) produces an overview with daily amounts that helps with the preparation – it does not replace the certificate.

Medication plan

In Germany, patients with statutory health insurance who take at least three prescribed, systemically acting medicines long-term are entitled to a standardised federal medication plan (§ 31a of Book V of the Social Code, SGB V). The printout from VitaParms is not such a plan, but it can help to keep it up to date.

B.7 Content from the internet

When looking up package leaflets, VitaParms displays content from other providers or opens it in the browser. The respective providers are solely responsible for this content; their terms of use apply.

- **openFDA** (U.S. Food and Drug Administration): the data is in the public domain. The FDA gives no warranty; it is US drug information.
- **European Medicines Agency (EMA):** the product information may be reused provided the source is acknowledged.
- **Apotheken Umschau, Gelbe Liste, search engines and custom sources:** VitaParms merely opens their websites in the browser. No content is taken over automatically.

A package leaflet that you store in the medical record is a copy for private use (in Germany under § 53 of the Copyright Act). Do not pass it on publicly.

B.8 Trademarks and names

All product, company and medicine names mentioned are trademarks or registered trademarks of their respective owners. They are mentioned for descriptive purposes only and do not imply any recommendation. The recognition of brand names in the medication search is purely a search function; VitaParms has no connection with the manufacturers.

B.9 In an emergency

VitaParms is not suitable for emergencies. If life is at risk, call the **emergency number**: 112 throughout Europe, 999 in the UK, 911 in the USA and Canada. Outside surgery hours, contact your out-of-hours medical service (in Germany 116 117, in England NHS 111) and, in case of poisoning, your poison control centre.

C Troubleshooting

Login and database

I have forgotten my password.

An administrator can set a new password under Tools » Login accounts... . If all administrators have forgotten their passwords, the recovery key helps (Section 3.3 (p. 12)).

VitaParms reports that the key file is missing.

The file `....sqlite.keys` must be in the same folder as the database. If only the database was copied, fetch the `.keys` file as well or open the database with the recovery key.

VitaParms reports that the list of accounts has been changed.

The key file was edited outside VitaParms or is damaged. You are logged in with user rights for the time being. Restore a backup or set up the accounts again with the recovery key.

I can no longer see a person's data.

Check the person selection and the period filter in the journal. Check in the status bar that the right database is open.

Readings and chart

The BMI is missing.

For the BMI, VitaParms needs weight and height. If no height is entered for a measurement, the last recorded one is used. If a height has never been recorded, the BMI stays empty.

The chart shows 'No readings in this period'.

Use the arrow buttons to move to a period with measurements, or select under *Panels* the readings you have actually recorded.

The blood glucose values are suddenly much smaller.

The unit has probably been switched to mmol/l (Section 13.3 (p. 94)). The stored values are unchanged.

Medication and assistant

A medicine does not appear in the assistant.

The assistant only shows regular medication that is due on the selected day. Check type, schedule, start and end of the medicine as well as the *Time of day* filter. Is the medicine archived?

The assistant does not speak, or speaks with a foreign accent.

Check that *read aloud automatically* is ticked and the computer is not muted. For natural pronunciation, a voice for the language of the user interface must be installed in the operating system (Section 9.3 (p. 69)).

A medicine is wrongly classified as controlled narcotic, painkiller, blood thinner, blood pressure medicine or cortisone.

Correct the classification in the medication dialog under *Classification* (Section 8.5 (p. 57)).

F1 reports that the user manual was not found.

The manuals belong in the `docs` subfolder of the program folder (Section 4.2 (p. 16)). Unpack the VitaParms ZIP archive completely, including this folder.

Windows Help shows 'Navigation to the webpage was canceled' instead of the manual.

Windows blocks help files that come from the internet or are located on a network drive. Right-click `docs\VitaParms-Manual.chm`, choose *Properties* and tick *Unblock* at the bottom – or unblock the downloaded ZIP archive in the same way before unpacking it. If VitaParms is on a network drive, copy it to the local hard disk.

The package leaflet cannot be found.

For German packs, enter the PZN from the pack and use Apotheken Umschau (PZN). For medicines authorised throughout the EU, try EMA product information; otherwise use Web search or add the website of your national medicines agency as a custom source. If the lookup is switched off in the settings, the buttons are disabled.

Medical record, appointments and sending

The full-text search does not find a document.

Scanned documents contain no text, only an image (the *Text* column is empty). Add keywords in the note or use a scanning app with text recognition.

The recipient cannot open the ZIP archive.

AES-encrypted ZIP archives cannot be opened with Windows Explorer alone. The recipient needs, for example, the free 7-Zip (Section 10.4 (p. 75)).

A reminder did not appear.

Reminders only appear while VitaParms is running and the responsible account is logged in. Check *Responsible* and *Advance notice* in the appointment and whether the appointment has been marked as done.

The test e-mail fails.

Check server, port and encryption (587/STARTTLS or 465/SSL). Many providers require an app password instead of the normal password.

Reporting bugs

Have you found a bug? Describe as precisely as possible what you did, what happened and what you expected, and state the program version (Help › About VitaParms…) and operating system. *Never* send real health data or your database – a screenshot with sample data is enough.

D Glossary

ACE inhibitors
Blood pressure medicines that inhibit the angiotensin-converting enzyme; the substances end in '-pril', e.g. ramipril, enalapril.

AES-256
Widely used encryption method considered secure, with a 256-bit key; used in VitaParms for the key file and ZIP archives.

Anticoagulants
Medicines that inhibit blood clotting ('blood thinners'), e.g. phenprocoumon, apixaban, heparin.

Antihypertensives
Medicines that lower blood pressure, e.g. ACE inhibitors, sartans, calcium channel blockers, beta blockers, diuretics.

Antiplatelet agents
Medicines that inhibit platelet aggregation, e.g. low-dose aspirin, clopidogrel.

As-needed medication
Medicines that are only given on a particular occasion.

BG
Blood glucose.

BMI
Body mass index: weight in kg divided by the square of height in m.

BP
Blood pressure, given as systolic/diastolic.

BtM, BtMG
Betäubungsmittel – controlled narcotic under the German Narcotics Act (Betäubungsmittelgesetz).

CHM
Compiled HTML Help: the file format of Windows Help; on Windows, VitaParms also ships the manual in this format.

Daily profile
Several measurements on one day at fixed times, e.g. blood glucose fasting, after breakfast, before lunch.

DOAC
Direct oral anticoagulants: apixaban, dabigatran, edoxaban, rivaroxaban.

Emergency medication
Medicines for a specific emergency, e.g. nitro spray for angina pectoris.

GDPR
General Data Protection Regulation of the EU.

Glucocorticoids
Cortisone and related substances, e.g. prednisolone, dexamethasone, budesonide; depending on the product systemic (tablet, injection), inhaled, nasal or topical.

iCalendar (.ics)
Standard format (RFC 5545) for exchanging appointments between calendar programs.

ICD-10
International Classification of Diseases; diagnosis codes such as 'I10.90' for hypertension.

IU International unit, a quantity used e.g. for insulin or vitamin D.

Journal
The history table of all measurements of a person.

Login account
Access for a person who operates VitaParms – as opposed to the *person* whose data is recorded.

mg/dl, mmol/l
Units for blood glucose; 1 mmol/l corresponds to about 18 mg/dl.

mmHg
Millimetres of mercury, the unit of blood pressure.

NRS
Numeric rating scale for pain, from 0 (no pain) to 10 (worst pain imaginable).

NSAIDs
Non-steroidal anti-inflammatory drugs, e.g. ibuprofen, diclofenac, naproxen.

Peak flow (PEF)
Peak expiratory flow: the highest air flow when breathing out forcefully, in litres per minute, measured with a peak flow meter.

PZN
Pharmazentralnummer: eight-digit number that uniquely identifies a medicine pack in Germany; the last digit is a check digit.

Recovery key
Emergency key that opens the database without password and without key file.

Regular medication
Medicines taken according to a fixed plan.

s.c.
Subcutaneous, under the skin (e.g. insulin, heparin).

Sartans
AT1 blockers, blood pressure medicines whose substances end in '-sartan', e.g. candesartan, valsartan.

Schedule III BtMG
List of the narcotics that may be traded and prescribed in Germany.

Schengen certificate
Certificate under Art. 75 of the Convention implementing the Schengen Agreement for carrying medically prescribed controlled narcotics when travelling.

SMTP
Protocol for sending e-mails; the outgoing mail server of your e-mail provider.

SpO_2
Peripheral oxygen saturation of the blood in per cent, measured with a pulse oximeter on the finger.

E About the author



Michael Bergmann

Michael Bergmann develops software and writes technical books – with a particular fondness for the Amiga. He contributed to the development of AmigaOS 3.2 and is the author of the (German-language) books *MUI-Programmierung* and *Das Amiga-C-Kompendium*. On his website https://mbergmann-sh.de he publishes tutorials and the online course 'BlitzBasic für Einsteiger'.

Michael Bergmann has been a state-registered nurse (*staatlich examinierter Krankenpfleger*) since 1991 and has worked on many interdisciplinary wards over the course of his career. He is also a certified practical instructor (*Praxisanleiter*) for internal emergency and disaster management in hospitals and care facilities.

His current projects include AmigaED, from which the dark themes of VitaParms originate, and MuiBuilderQt. With VitaParms he has written a program that is not aimed at developers, but at people who look after their own health or care for relatives.

Contact and source code

Web	https://mbergmann-sh.de
GitHub	https://github.com/mbergmann-sh

Bug reports, suggestions for improvement and translations are welcome. Please never send real health data with them.

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VitaParms builds on the work of many others: the Qt project, SQLite and SQLite3 Multiple Ciphers by Ulrich Telle, PDFium and the speech engines of the operating systems. My thanks go to everyone who writes and maintains free software.

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Back cover blurb

to be added – file `cover/backcover.tex`

